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Northern Hemisphere Vaccine Recommendation Meeting: February 2010.
Summary of NIMR Results

Influenza activity September 2009 to February 2010.

Influenza activity continued to be dominated by the pandemic A(H1N1) virus. The situation across Europe and the wider WHO European Region was complex and heterogeneous with countries showing distinct patterns of activity.

Figure 1 (page 5) shows the number of viruses characterised through the sentinel and non-sentinel surveillance systems within the WHO European Region (~153,000 samples being reported to WHO EuroFlu). Ninety nine percent of the subtyped influenza A viruses were H1N1pdm with 0.35% being H3N2. The identity of the remaining 0.65% as seasonal H1N1 needs to be verified. Influenza B virus identification was very low and represented only 0.3% of the total number of characterised influenza viruses. The collection of viruses across the European Region is not uniform and the number of viruses identified for the EU/EAA countries (data from ECDC) makes up a large proportion of the characterised isolates within the WHO European Region (**Figure 2**, page 6). The number of deaths from influenza in the WHO European Region was in the order of 3600 whilst in the EU/EAA the number was 1528.

Figures 3 and 4 (pages 7 and 8) illustrate the spread of influenza across the WHO European Region, with high activity first detected in England by July. Activity declined generally before increasing in the autumn and then subsiding in January. Different patterns of infection between countries can be seen: some experienced a significant first wave of the pandemic and a subsequent smaller second wave (e.g. England and The Netherlands and, to a lesser degree, Spain) but others had a high intensity of infection in the second wave (e.g. Germany and Estonia) (**Figure 4** page 8). Even within a small geographic region distinct patterns could be observed. The two patterns of activity are exemplified in **Figure 5** (page 9) in which data from the United Kingdom are presented. These data show a distinct pattern for England and Wales, with a summer peak of infection, contrasting with Scotland and Northern Ireland, with a later autumn peak. (Note that the school terms differ between England and Wales compared with Scotland and Northern Ireland).

To illustrate the kinetics of influenza activity this season, data are shown from EuroFlu to compare the timing of peak influenza activity over recent years. **Figure 6** (page 10) shows that the peak activity at the end of 2009 declined prior to January, when in recent years activity has increased in January. **Figure 6** also shows that influenza B activity has been delayed behind influenza A activity in the seasons 2007/08 and 2008/09 yet marked influenza B activity has yet to be seen in the European Region in 2010. It is notable that this season in the European Region the relative timing of Respiratory Syncytial Virus (RSV) activity differed from the preceding three seasons (**Figure 7**, page 11). In the three years preceding 2009/10 RSV activity was observed prior to influenza but in 2009/10 RSV activity was delayed and seemingly reduced compared to the three preceding years. The cause of this is not clear.

Antigenic and Genetic Characteristics of pandemic A(H1N1) and seasonal influenza viruses isolated between September 2009 through to January 2010.

Pandemic A(H1N1) viruses

From September 2009 through to January 2010 pandemic H1N1 viruses or clinical samples were received from 38 countries in three continents. From these specimens 596 viruses were propagated and 495 were tested for their antigenic profile by haemagglutination inhibition assay (HI); 325 were analysed for their susceptibility to inhibition by oseltamivir and zanamivir using sialidase inhibition assays (**Tables 1, 2 and 17,** pages 12, 13 and 58).

The vast majority were closely related antigenically to the prototype and vaccine viruses (A/California/4/2009 and A/California/7/2009) as analysed by HI using a panel of reference post-infection ferret antisera (**Tables 3 to 5**, pages 14-16).

A small proportion (46 out of a total of 495 viruses tested, 9.3%) showed low reactivity against one or more members of the antiserum panels. These viruses are shown in **Table 6**, page 17, and were received from a variety of countries in Europe.

Genetic analysis of a representative set of viruses has been conducted either through full genome sequencing or focussed analysis of the HA and NA gene sequences. Like the antigenic analysis, genetic analysis of the HA and NA genes showed only minor differences from the sequences of the prototype and vaccine viruses. Phylogenetic analyses are shown in **Figures 8 and 11** (pages 18 and 22) with amino acid alignments of the HA and NA protein sequences shown in **Figures 9 and 12** (pages 19 and 23) respectively. The viruses showing reduced reactivity with members of the antiserum panels are dispersed within both phylogenetic trees and do not form any distinct groupings.

Some amino acid residue substitutions within the HA (primarily residues 203 and 222) are associated with major genetic clades while others (residues 83 and 297) appear to be associated with sub-clades (**Figure 8**, page 18). The location of these amino acids in the structure of the HA glycoprotein is shown in **Figure 10**, page 21. It is notable that residues 203 and 222 are in the head of the HA and are closest to each other across the HA trimer interface. Some of the viruses that showed low HI titres had amino acid substitutions in the region 154-156 of the HA. In all cases examined, this substitution was not present in the clinical sample.

While the HA D222E substitution is associated with a major genetic clade (**Figure 8**, page 18), other substitutions at this position are not. Many clinical samples were received from fatal cases, notably from Ukraine. In these cases heterogeneity was observed at residue 222 of the HA, with viruses displaying the substitutions either D222G or D222N or mixed populations displaying varying proportions of D/G/N/S. It is currently unclear whether selection of D222 substitutions with uncharged amino acids is associated with more severe infection or is a result of altered sampling in these cases.

We have observed that all viruses show sensitivity to zanamivir but two viruses, A/Belgium/1239/2009 and A/Israel/6290/2009, displayed resistance to oseltamivir (**Table 17**, page 58). Both were associated with the use of oseltamivir. Antiviral resistance to oseltamivir does not associate with any HA genetic clade, but the mutation to induce the resistance-conferring H275Y substitution in NA is sufficient to associate these viruses into a small genetic clade (**Figure 11**, page 22).

Conclusion: there was little evidence of antigenic drift or the development of antiviral resistance amongst the viruses analysed at the NIMR WHO CC. The vast majority reacted well with reference antisera to the prototype (A/California/4/2009) and vaccine (A/California/7/2009) viruses.

Seasonal A(H1N1) viruses

Very few seasonal H1N1 viruses were received and these were isolated from samples collected in September and October from only three countries (**Tables 7 and 8** pages 25 and 26).

Antigenically, all viruses were distinguishable in HI analysis with the panel of post-infection ferret antisera raised against reference viruses. Notably they all showed low reactivity to antiserum against the previous vaccine virus A/Brisbane/59/2007 (**Tables 8 and 9**, pages 26 and 27).

The antigenic differentiation of recent isolates by HI was not replicated in virus neutralisation tests (VNT) using a micro-plaque assay technique (**Table 10**, page 28). In these assays none of the small panel of viruses examined was markedly less well inhibited (>/= 8-fold) than was the homologous virus by post-infection ferret antiserum raised against the vaccine virus A/Brisbane/59/2007. A similar result was seen for sera raised against recent virus isolates.

All isolates of recently circulating virus fall within genetic clade 2B, but several amino acid substitutions have been observed compared with the current vaccine virus A/Brisbane/59/2007. Notably, sub-clades defined by S141N and G185A or N183S and G185S have emerged (**Figures 13 and 14**, pages 29 and 30). An additional substitution, V131A, has been observed in isolates from a number of countries in association with S141N and G185A. The locations of these changes are displayed in **Figure 15**, page 32.

All viruses examined were resistant to oseltamivir but sensitive to zanamivir (**Table 17**, page 58). Resistance to oseltamivir is caused by the amino acid substitution H275Y in the NA (**Figures 16 and 17**, pages 33 and 34).

Conclusion: HI assays and VNT gave contrasting results. In VNTs recent viruses were antigenically indistinguishable from the most recent seasonal H1N1 vaccine virus.

A(H3N2) viruses

Tables 7 and 11 (pages 25 and 36) show summaries of the origins and characteristics of the H3N2 viruses received. Eighty-three viruses were propagated and examined for their antigenic characteristics.

HI analysis was carried out using guinea pig red blood cells in either the presence (**Table 12**, page 37) or the absence of oseltamivir (**Table 13**, page 38) as described in the report for WHO in September 2009. The vast majority were shown to be antigenically similar to A/Perth/16/2009, the virus recommended for the Southern Hemisphere 2010 vaccine. Only a few of the viruses characterised were antigenically similar to the 2009-10 Northern Hemisphere vaccine virus, A/Brisbane/10/2007.

Genetic analysis of the HA gene showed that most viruses fell approximately evenly into either the Perth/16 genetic clade or the Victoria/208 genetic clade (**Figure 18**, page 39). The clades can be differentiated by E62K and N144K (Perth/16), and T212A (Victoria/208) substitutions. Both clades share the amino acid substitutions K158N and N189K compared with A/Brisbane/10/2007 (**Figure 19**, page 40). The locations of the amino acid substitutions associated with each clade are shown in **Figure 20**, page 42. Notably, N144K results in the loss of a potential glycosylation site (**Figure 19**, page 40). Viruses of the Perth/16 clade and Victoria/208 clade are antigenically indistinguishable.

Genetic analysis of the NA gene showed that recently isolated viruses were distributed throughout the phylogenetic tree falling into a number of sub-groups defined by specific amino acid substitutions (**Figures 21 and 22**, pages 43 and 44). The clustering of genes within the NA phylogenetic tree mostly mirrors the corresponding clustering of genes in the HA phylogenetic tree.

Of the 65 H3N2 viruses tested for resistance to oseltamivir and zanamivir none showed resistance to either drug (**Table 17**, page 58).

Conclusion: there was little evidence of antigenic drift in the H3N2 viruses away from the 2010 Southern Hemisphere vaccine virus A/Perth/16/2009.

B viruses

Very few influenza B viruses were received from September 2009 and only ten viruses were propagated. Five each were from the Victoria and the Yamagata lineages (**Table 14**, page 46).

The five viruses of the **B/Victoria-lineage** reacted poorly with post-infection ferret serum raised against the B/Brisbane/60/2008 vaccine virus. However, cross reactivity to antiserum raised against B/Paris/1762/2009 was observed, despite the low homologous potency of the antiserum (**Table 15**, page 47).

Genetic analysis of the influenza B/Victoria-lineage HA genes showed that almost all the viruses belonged to the Brisbane/60 clade. Several sub-clades could be differentiated based on the amino acid substitutions at residues 58, 146, 197, 202 and 225 (**Figures 23 and 24**, pages 48 and 49).

NA gene sequence analysis showed that all the recently isolated viruses defined as belonging to the B/Victoria-lineage retained a B/Victoria-lineage NA gene (**Figures 25 and 26**, pages 51 and 52).

The five viruses of the **B/Yamagata-lineage** reacted well with post-infection ferret antiserum raised against the previous B/Yamagata-lineage vaccine virus B/Florida/4/2006 and all other reference antisera raised against more recently-circulating viruses (**Table 16**, page 54).

Genetic analysis showed that all the recent isolates clustered in the Bangladesh/3333 genetic clade with amino acid substitutions at residues 202 and 251 defining sub-clades (**Figures 27 and 28**, pages 55 and 56).

Similar to viruses of the B/Victoria-lineage, recently isolated viruses of the B/Yamagata-lineage HA retained the B/Yamagata-lineage NA (**Figures 25 and 26**, pages 51 and 52).

Of the 7 B viruses tested for resistance to oseltamivir and zanamivir none showed resistance to either drug (**Table 17**, page 58).

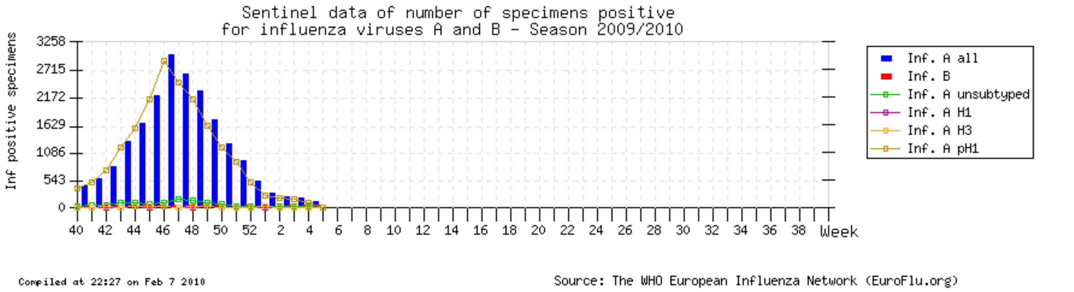
Conclusion: the situation for B viruses is complex and results based on the small numbers of virus isolates received, reflecting the low level of detection in the majority of countries up to the end of January 2010, are thus far inconclusive.

Zoonotic avian influenza virus infections: H5N1 and H9N2.

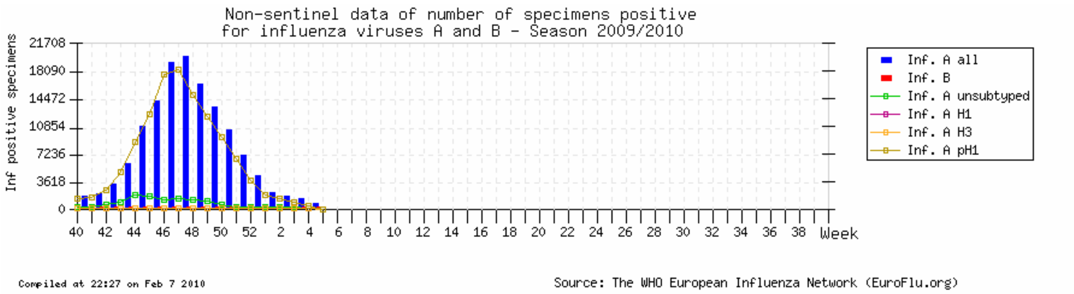
No human isolates were received and we have not analysed any examples of recently isolated H5N1 or H9N2 viruses of avian origin.

Figure 1: Current European Situation (EuroFlu.org: 07.02.2010)

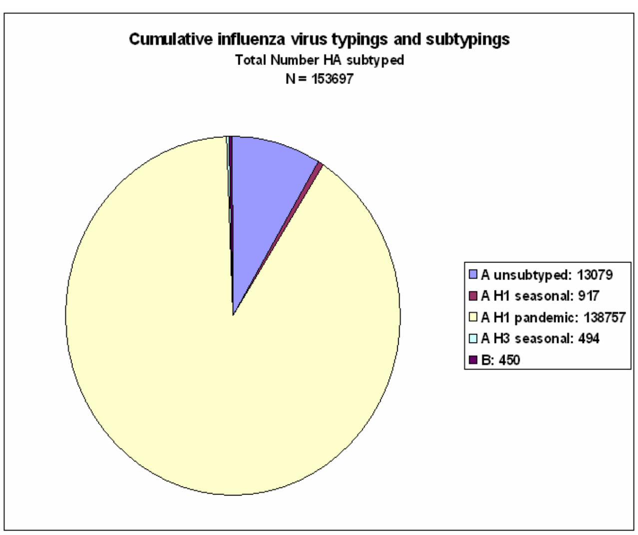
Sentinel Data



Non-sentinel Data



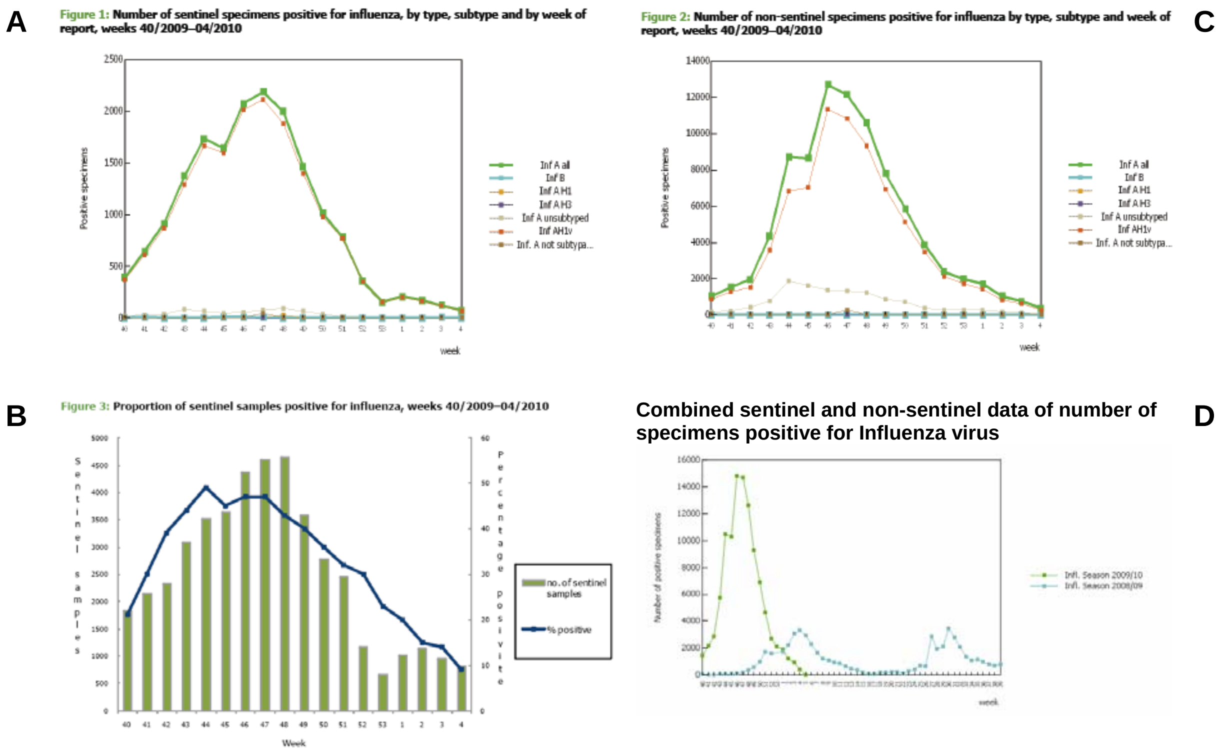
Virus typing/subtyping and H1pdm-associated deaths



Europe: H1pdm-associated deaths since the beginning of the pandemic (confirmed/probable to week 04/2010).

EuroFlu: 3649
WHO/ud86: ~3605
ECDC/WISO: 1528

Figure 2: EU/EAA situation – Figures from ECDC for week 04/2010

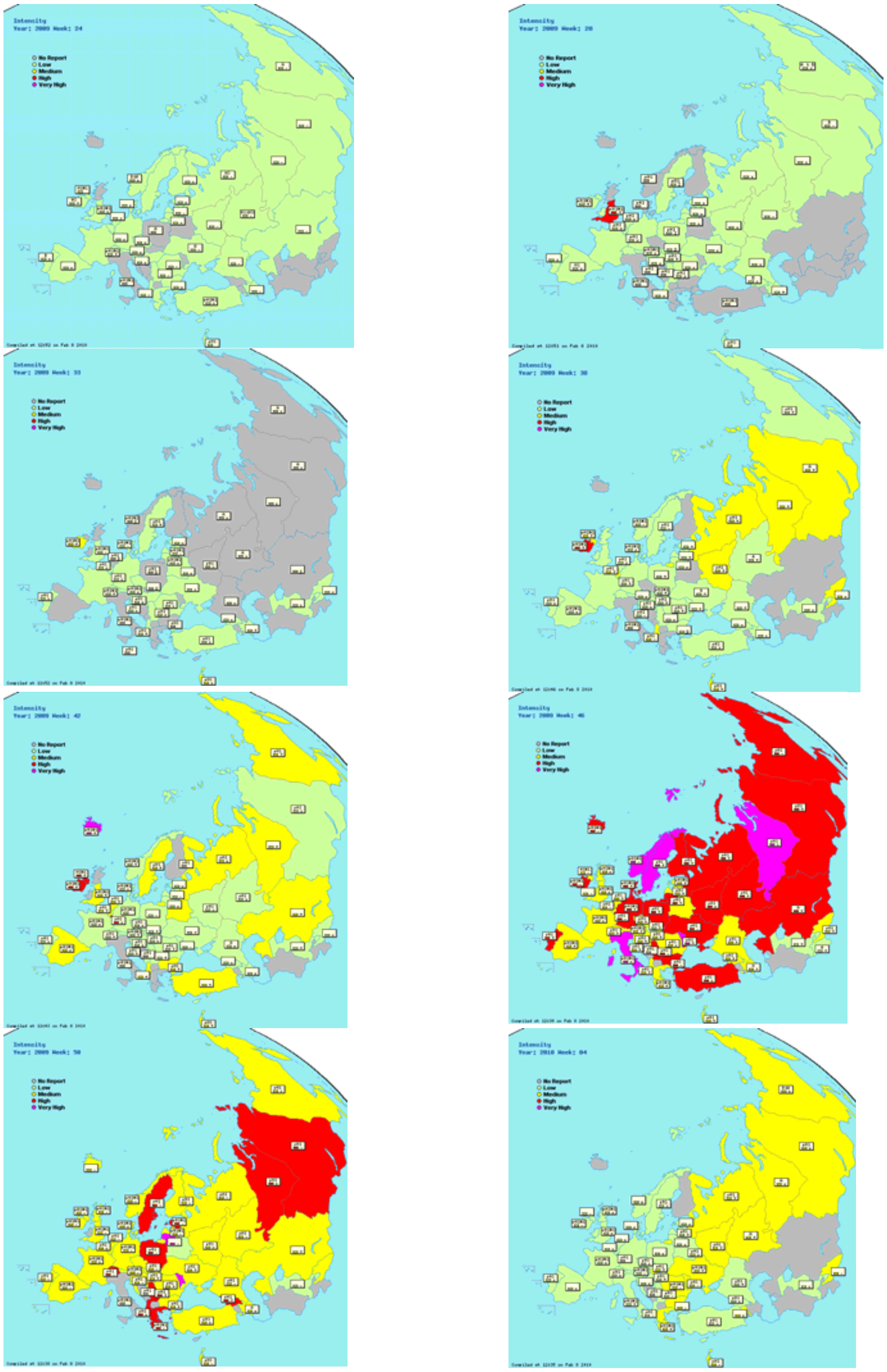


Based on sentinel surveillance, specimen influenza positivity peaked in week 47 (A) while the percentage positive specimens peaked in week 44 (B).

Non-sentinel surveillance yielded a peak in week 46 that was ~6-fold higher than the sentinel peak (C).

Combined figures for sentinel/non-sentinel surveillance showed Europe to have had a Summer 2009 wave of H1pdm activity, comparable to the earlier seasonal influenza epidemic, followed by a second 2009-10 wave of ~6-fold higher intensity (D).

Figure 3: Influenza - Intensity ‘snapshots’ (EuroFlu.org: 07.02.2010)



Wk 24 09 — 8-14 Jun

Wk 28 09 — 6-12 Jul

Wk 33 09 — 10-16 Aug

Wk 38 09 — 14-20 Sep

Wk 42 09 — 12-18 Oct

Wk 46 09 — 9-15 Nov

Wk 50 09 — 7-13 Dec

Wk 04 10 — 25-31 Jan

Figure 4: EU/EAA Country situations – ECDC for week 04/2010

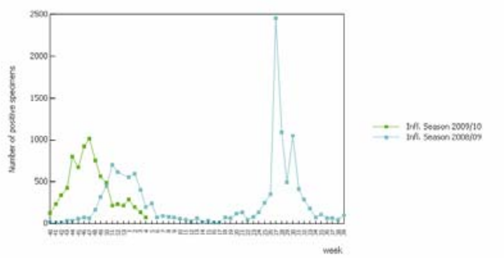
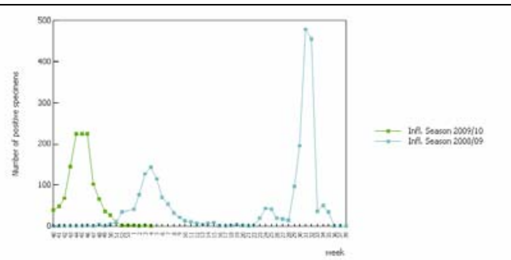
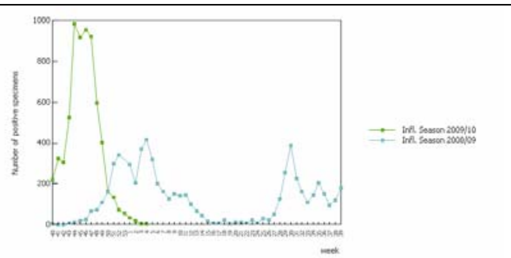
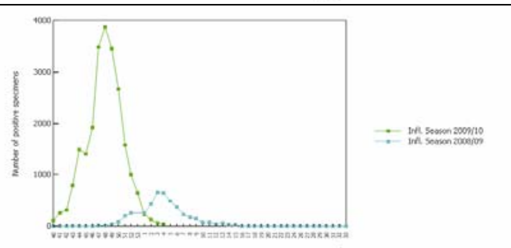
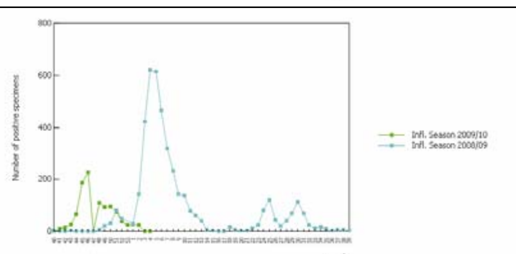
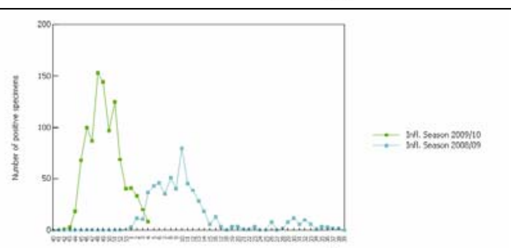
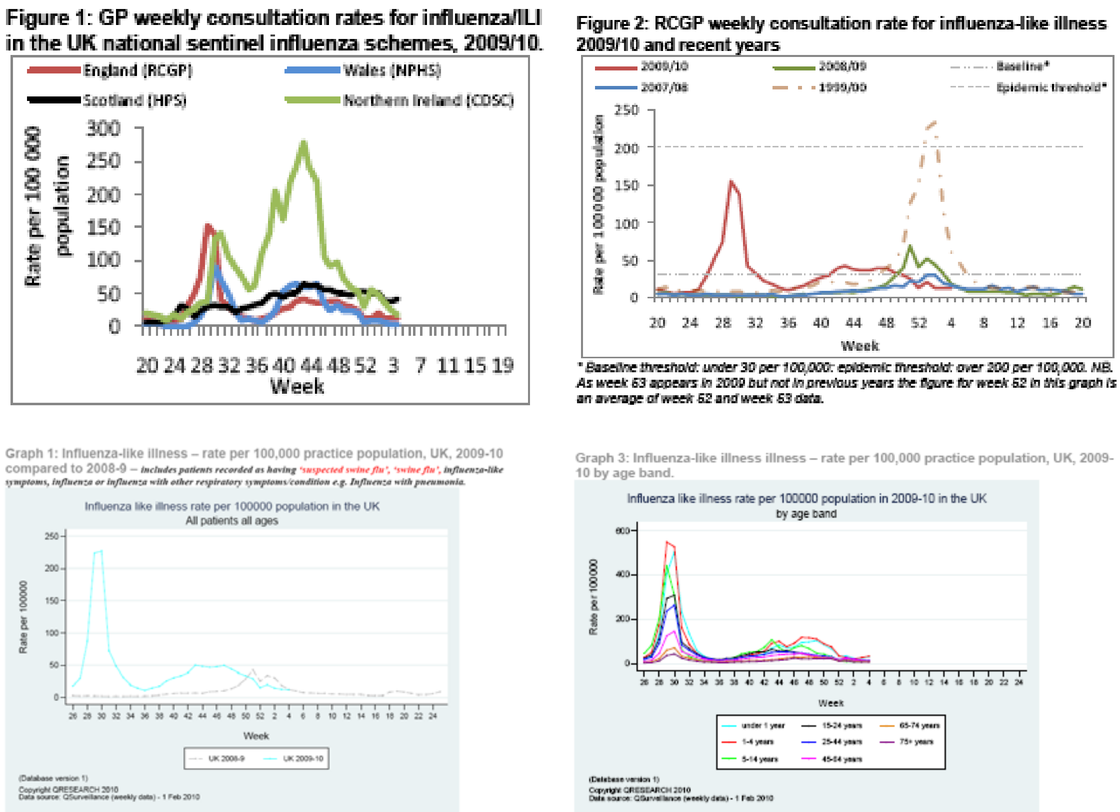
Country	Combined Sentinel/Non-sentinel influenza positive specimens	
UK (England)		**Some countries, e.g. UK (England) and Netherlands, had significant summer 2009 waves of H1pdm activity that were greater than their earlier seasonal influenza epidemic.**
Netherlands		
Spain		
France		**The majority of EU/EAA countries had primary Summer 2009 waves of H1pdm activity that were of lower intensity than their earlier seasonal influenza epidemic. Secondary 2009-10 H1pdm waves in these countries were generally of greater intensity than the first waves.**
Germany		
Estonia		

Figure 5: UK situation – Figures from HPA for week 04/2010

A B C D



Based on the different national sentinel schemes, Northern Ireland and Scotland appear to have secondary waves of H1pdm activity higher than their first (A).

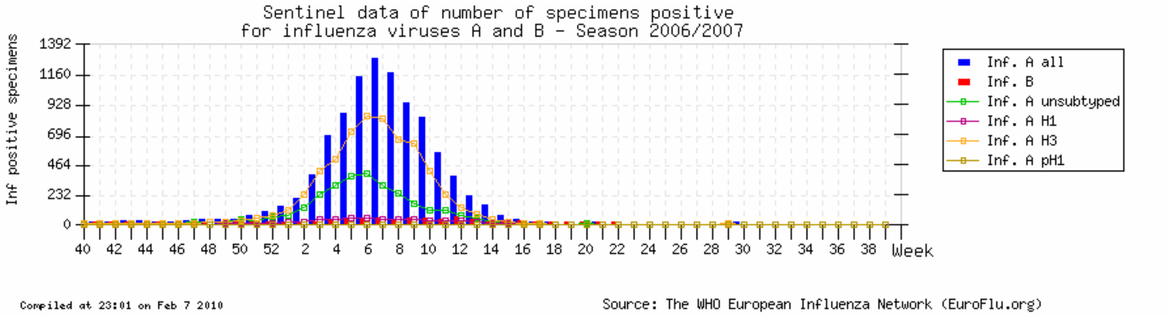
Based on a limited number of sentinel practices within England, at no stage did the incidence of H1pdm activity cross the epidemic threshold (200/100000) (B).

Based on Q-surveillance that covers 19.4 million people in England, Wales, Scotland and Northern Ireland (2996 Medical practices), the epidemic threshold was crossed in the Summer 2009 wave of H1pdm activity (C).

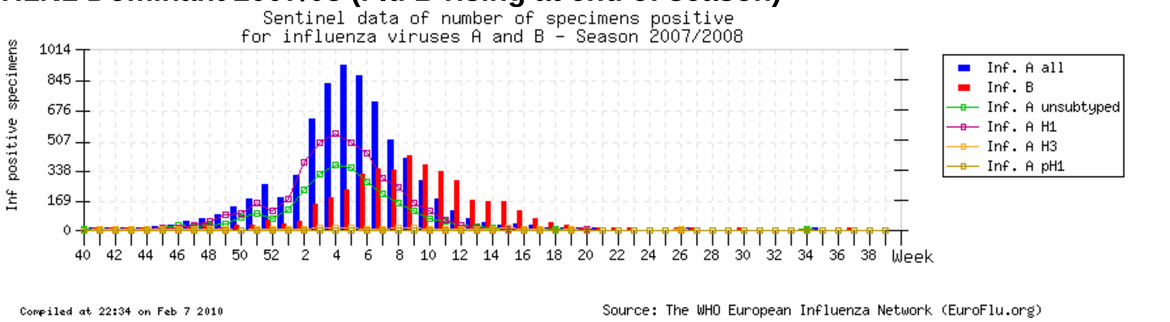
When age-stratified, the Q-surveillance data show that all groups aged <1-44 years presented at epidemic threshold crossing levels in the Summer 2009 wave of H1pdm activity (D).

Figure 6: Influenza Season 'timing' by year (EuroFlu.org: 07.02.2010) Sentinel Data

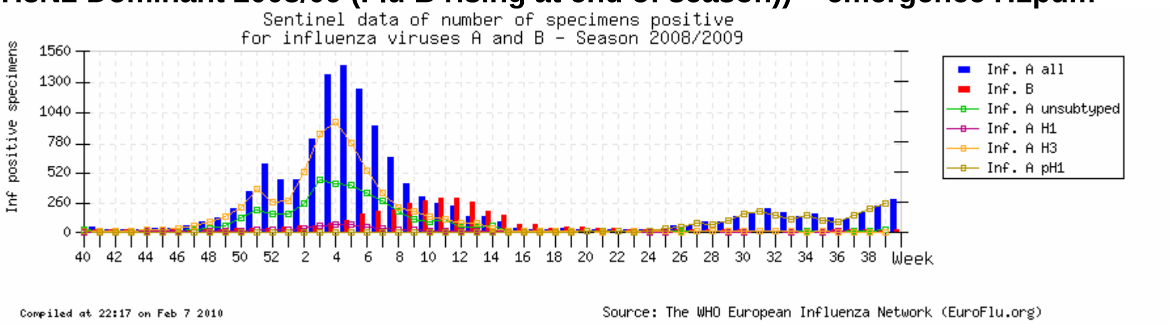
H3N2 Dominant 2006/07



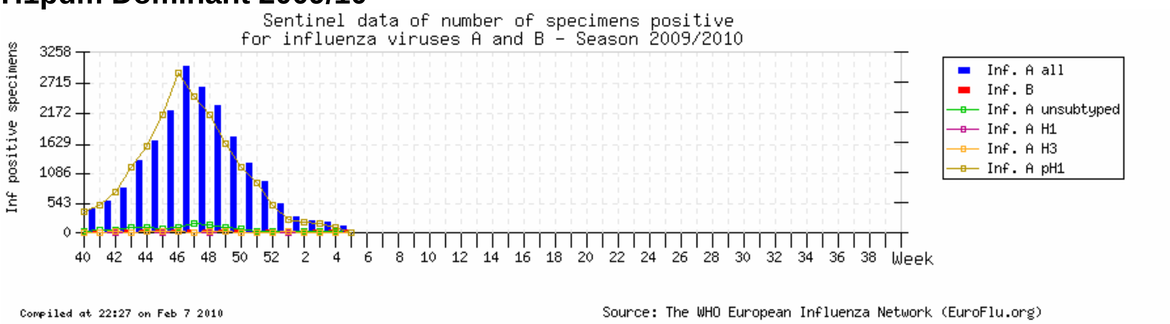
H1N1 Dominant 2007/08 (Flu B rising at end of season)



H3N2 Dominant 2008/09 (Flu B rising at end of season)) + emergence H1pdm



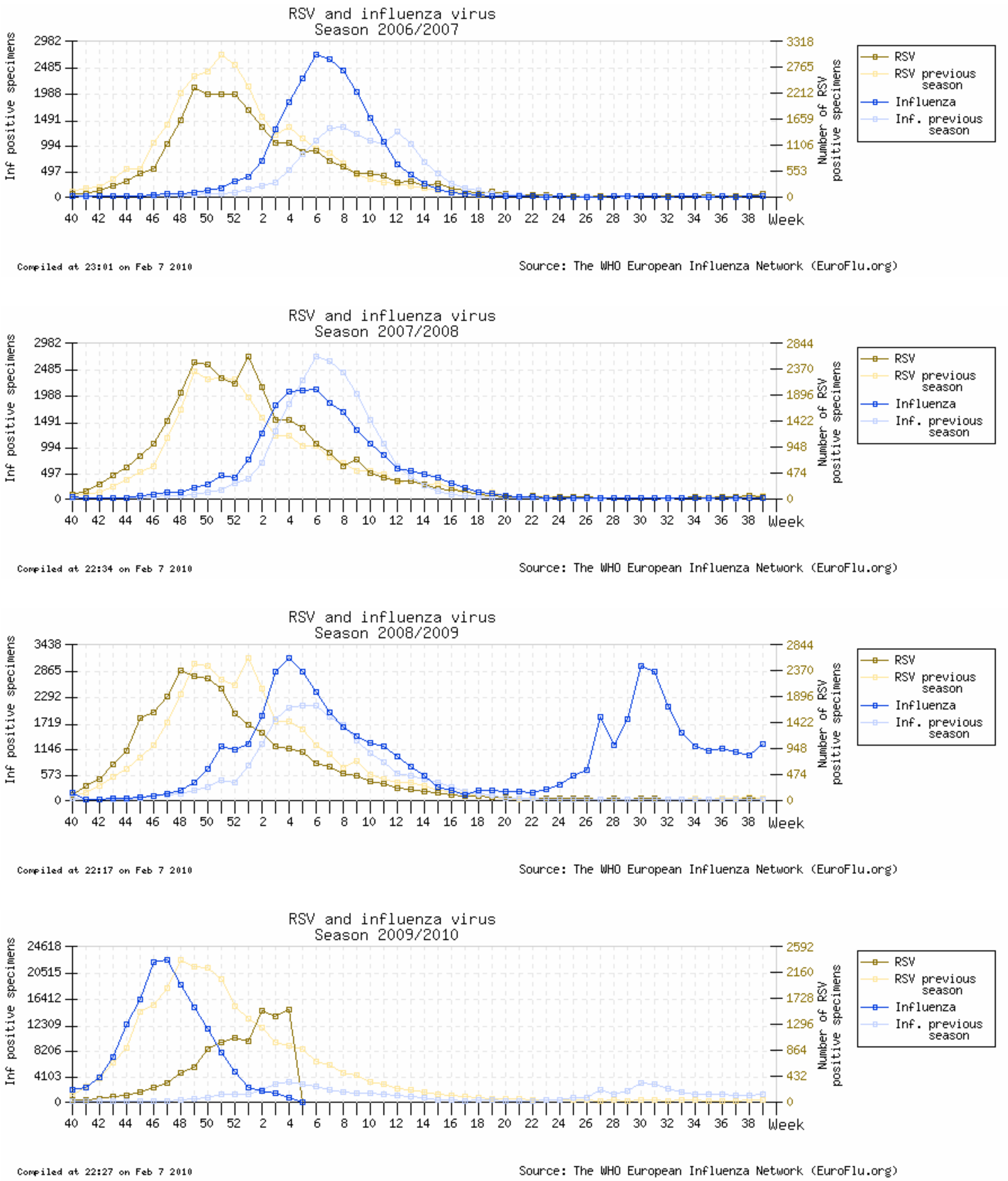
H1pdm Dominant 2009/10



2-3 fold higher number of positive specimens compared to previous three seasons. 'Season' started early, effectively now over, H1pdm suppressed seasonal influenza.

Figure 7: RSV and influenza – activity timings (EuroFlu.org: 07.02.2010)

Combined sentinel and non-sentinel data for numbers of positive specimens.



In the three previous seasons RSV peaks have preceded those for influenza. 'Early' H1pdm in 2009/10 coincided with reduced and later RSV activity.

Table 1. Summary of H1N1 pandemic analyses conducted on samples collected September to December 2009

	Clinical samples received									Isolates received					
Country	Number received	qRT-PCR positive	HA sequence	NA sequence	Number grown	Antigenic characterisation[1]	NI phenotyping[2]	HA sequence	NA sequence	Number received	Number grown	Antigenic characterisation[1]	NI phenotyping[2]	HA sequence	NA sequence
SEPTEMBER															
Afghanistan	1				0	0	0								
Albania										1	1	1	NT		
Austria										1	1	1	NT		
Belarus										1	1	1	1		
Cyprus	6	NT			6	6	6								
Denmark										3	3	3	3		
Finland										1	1	1	NT		
France										5	5	5			
Hong Kong										4	4	4	4	2	2
Ireland										1	0	0	0		
Madagascar	1	1			1	1	NT			3	3	3	3		
Moldova	6	4	4	4	2	2	2								
Norway										6	3	3	3		
Portugal										15	15	15			
Romania										4	4	4	4	1	1
Russia										9	9	9	9	2	2
Senegal	12	NT			4	4	4	1	1						
Slovenia										4	4	4	4		
Sweden										1	1	NT			
Switzerland	4	NT			0	0	0								
OCTOBER															
Afghanistan	1	NT			1	1	NT								
Austria										2	2	2	NT		
Belarus										2	2	2	2		
Belgium	14	NT			13	13	12	1	1						
Cyprus	3				3	3	3	2	2						
Egypt	2	NT			1	1	NT			1	0	0	0		
France										14	14	14	1		
Germany										1	1	1	NT		
Iran										1	1	1	NT		
Ireland										8	1	1	1		
Madagascar	12				11	11	NT	1	1						
Moldova	16	11	10	10	8	8	8	1	1						
Norway										15	10	10	7		
Poland	1	NT			0										
Portugal										14	14	14	8		
Slovenia										6	6	6	6		
Sweden										6	6	6	NT		
Switzerland	13	NT			3	3	3								
Ukraine	46	24	15	9	2	2	2			2	1	1	1		
United Kingdom										2	2	2	NT		
NOVEMBER															
Afghanistan	18	NT			9	9	NT								
Albania										5	5	5	NT		
Algeria										4	4	4	4	2	2
Armenia	5	NT	4	4	4	4	4								
Austria										4	4	4	NT		
Belarus										4	4	4	3		
Belgium	47	NT			39	39	35	4	4						
Egypt	9	NT			5	5	NT			7	0	0	0		
France										20	18	18	12		
Germany	4		4	4						20	20	20	20	3	3
Hong Kong										3	3	3			
Iran										21	20	20	10		
Israel										4	4	4	NT		
Kosovo	83	76			30	30	27								
Latvia	6	NT	3	4	1	1	1			5	4	4	4	4	4
Madagascar	11	NT			6	6				5	4	4	4		
Moldova	4	3	3	3	3	3	3	1	1						
Nepal	58	21	30	19	9	9	8								
Norway										11	11	11	1		
Poland	18	NT			13	13	13								
Portugal	3	NT	2		1	1	1			15	15	15	1	2	2
Russia										12	12	12	10	3	3
Slovenia										12	12	12	12		
Spain										13	13	NT	NT		
Sweden										7	7	7	NT		
Switzerland	16	NT			2	2	2								
Ukraine	244	128	28	1	7	7	7			19	16	16	16	17	11
United Kingdom										9	8	8	NT		
DECEMBER															
Albania										9	9	9	NT		
Algeria	3	NT			1	1	1	1	1						
Austria										1	1	1	NT		
Belgium	1	NT													
Egypt	33				23	23	12	5	5						
Germany										12	12	12	NT		
Greece	28	NT			17	17	10	3	3						
Iran										4	4	4	1	1	1
Israel										7	7	7			
Kosovo	9	0													
Madagascar	18	NT			14	14	NT	2	2						
Nepal	36	21			7	7	3								
Poland	4	NT			4	4	3								
Portugal										7	7	7	NT		
Spain										7	7	NT	NT		
UNKNOWN															
Bosnia	73	72	6	6											
Lithuania	105	48													
TOTAL	974	409	109	64	250	250	170	22	22	380	346	325	155	37	31

1. By HI all classified as A/California/7/2009-like (< 4-fold reduction in titre)
2. NI assay performed with Oseltamivir and Zanamivir. All viruses tested were sensitive to both drugs except A/Belgium/1239/2009 resistant to Oselatamivir.
NT = not tested (qRT-PCR performed by NIC)

Table 2. Summary of H1N1 pandemic analyses conducted on samples collected September to December 2009

		Clinical samples received			Isolates received	
MONTH Continent	Number of Countries	Number received	Number propagated		Number received	Number propagated
SEPTEMBER						
Africa	2	13	5		3	3
Asia	2	1	0		4	4
Europe	14	16	8		52	48
OCTOBER						
Africa	2	14	12		1	0
Asia	2	1	1		1	1
Europe	16	93	29		72	59
NOVEMBER						
Africa	3	20	11		16	8
Asia	4	76	18		24	23
Europe	21	430	100		160	153
DECEMBER						
Africa	3	54	38		0	0
Asia	2	36	7		4	4
Europe	10	42	21		43	43
UNKNOWN						
Europe	2	178				
TOTAL		974	250		380	346

Table 3. Antigenic analyses of pandemic influenza A(H1N1) viruses [showing geographical distribution of homogeneous viruses]

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre[1]						
				Post infection ferret sera					
			A/NJ[2] 8/76 R15/79	A/NJ 8/76 F34/82	A/Cal 4/09 C4/F14/09	A/Cal 7/09 C4/31/09	A/Eng 195/09 NIBSC F18/09	A/Auck 3/09 C4/17/09	
REFERENCE VIRUSES									
A/New Jersey/8/76		Ex	>5120	1280	320	640	640	1280	
A/California/4/2009		C1,E2	5120	320	2560	2560	2560	2560	
A/California/7/2009		E5	2560	320	2560	2560	2560	2560	
A/England/195/2009		MDCK5	2560	320	2560	2560	2560	2560	
A/Auckland/3/2009		Ex+4	5120	640	2560	2560	5120	2560	
A/Bayern/69/2009		MDCK4	640	<	80	320	160	160	
A/Lviv/N6/2009		MDCK4	640	<	320	640	160	160	
TEST VIRUSES									
A/swine/Northern Ireland/998-75/2009[3]	unknown	E2	2560	320	1280	2560	1280	1280	
A/South Africa/2/2009	Jun-09	MDCK4	1280	80	1280	1280	1280	1280	
A/Ghana/FS-09-1306/2009	17/08/2009	MDCK3	2560	160	2560	1280	1280	2560	
A/Dakar/37/2009	11/09/2009	MDCK2	2560	160	2560	1280	2560	2560	
A/Ialomita/3040/2009	12/09/2009	MDCK3	1280	80	1280	1280	1280	1280	
A/St. Petersburg/5/2009	21/09/2009	E4	5120	320	2560	2560	2560	5120	See figures 8 and 11
A/Slovenia/4377/2009	Nov-09	MDCK1/Siat1	1280	80	1280	1280	1280	1280	
A/Moldova/G-186/2009	01/11/2009	MDCK2	2560	160	1280	2560	2560	5120	
A/Dnipropetrovsk/268/2009	04/11/2009	MDCK1/Siat1	2560	160	1280	1280	1280	2560	
A/Switzerland/2471729/2009	04/11/2009	MDCK3	640	80	1280	1280	2560	2560	
A/Madagascar/9567/2009	04/11/2009	MDCK2/Siat1	2560	80	1280	1280	1280	1280	
A/Armenia/N47/2009	08/11/2009	Siat2	5120	160	5120	2560	2560	5120	
A/Latvia/11-686/2009	11/11/2009	MDCK1/Siat3	2560	160	1280	1280	1280	2560	
A/Ireland/M85000/2009	25/11/2009	Siat2	2560	160	1280	1280	1280	2560	
A/Bosnia-Herzegovina/28Sw/2009	Aug-09	MDCK2	2560	160	1280	1280	1280	1280	
A/Mauritius3KP/2009	06/08/2009	MDCK2	2560	160	1280	1280	1280	1280	
A/Lisboa/64/2009	14/09/2009	MDCK2,Siat1	≥5120	160	5120	5120	5120	5120	
A/Hong Kong/26578/2009	16/09/2009	MDCK4	1280	80	1280	1280	1280	2560	
A/Cyprus/S2476/2009	07/10/2009	MDCK2	1280	80	1280	1280	1280	2560	
A/Norway/4158/2009	18/10/2009	MDCK1 / Siat1	1280	80	1280	1280	1280	1280	
A/Moldova/G-170/2009	31/10/2009	MDCK2	5120	320	1280	1280	2560	2560	
A/Slovenia/4221/2009	Nov-09	MDCK1/Siat1	2560	160	1280	2560	2560	2560	
A/Cherkasy/346/2009	02/11/2009	E1/1	2560	160	1280	2560	1280	1280	
A/Madagascar/9567/2009	04/11/2009	MDCK2/Siat1	2560	80	1280	1280	1280	1280	
A/Switzerland/2471729/2009	04/11/2009	MDCK3	640	80	1280	1280	2560	2560	
A/Kosovo/578/2009	06/11/2009	Siat2	2560	160	5120	5120	5120	5120	
A/Belarus/226/2009	07/11/2009	MDCK1,Siat1	5120	160	2560	2560	2560	2560	
A/Lviv/411/2009	08/11/2009	Siat6	1280	160	2560	2560	2560	2560	
A/Armenia/N53/2009	10/11/2009	Siat4	5120	320	2560	2560	2560	2560	
A/Latvia/11-686/2009	11/11/2009	MDCK1/Siat3	2560	160	1280	1280	1280	2560	
A/Nepal/SR-11/2009	13/11/2009	MDCK3	2560	160	1280	1280	2560	2560	
A/Lyon/2490/2009	16/11/2009	Siat1	2560	80	2560	2560	2560	2560	
A/Algeria/G35/2009	18/11/2009	MDCK1 / Siat1	2560	80	2560	2560	2560	2560	
A/Belgium/G4363/2009	19/11/2009	Siat2	2560	80	1280	2560	2560	2560	
A/Hong Kong/34428/2009	24/11/2009	MDCK2 / Siat1	2560	160	2560	2560	2560	2560	
A/Ireland/M85000/2009	25/11/2009	Siat2	2560	160	1280	1280	1280	2560	
A/Iran/18101/2009	Dec-09	MDCK2,Siat1	≥5120	160	2560	2560	2560	5120	
A/Nepal/RB-6/2009	02/12/2009	Siat2	2560	80	1280	1280	2560	2560	
A/Egypt/114/2009	06/12/2009	Siat2	5120	320	5120	5120	5120	5120	

1. < = 40; 2. hyperimmune rabbit serum; 3. case of human to swine transmission

Vaccine strain

Table 4. Antigenic analyses of pandemic influenza A(H1N1) viruses

			Haemagglutination inhibition titre[1]					
			Post infection ferret sera					
Viruses	Collection date	Passage History	A/Cal 4/09 C4/F14/09	A/Cal 7/09 C4/31/09	A/Eng 195/09 NIBSC F18/09	A/Auck 3/09 C4/17/09	A/Bayern 69/09 C4/33/09	A/Lviv N6/2009 C4/34/09
REFERENCE VIRUSES								
A/New Jersey/8/76		Ex	320	640	640	1280	320	640
A/California/4/2009		C1,E2	2560	2560	2560	2560	1280	2560
A/California/7/2009		E5	2560	2560	2560	2560	1280	5120
A/England/195/2009		MDCK5	2560	2560	2560	2560	1280	2560
A/Auckland/3/2009		Ex+4	2560	2560	5120	2560	2560	5120
A/Bayern/69/2009		MDCK4	80	320	160	160	640	640
A/Lviv/N6/2009		MDCK4	320	640	160	160	1280	1280
TEST VIRUSES								
A/Bosnia/1250/2009	unknown	SIAT2	2560	1280	1280	2560	1280	1280
A/Glasgow/7-056E/2009	Nov-09	SIAT2	640	2560	2560	2560	1280	2560
A/Lisboa/89/2009	12/11/2009	MDCK2 / SIAT1	2560	5120	2560	5120	1280	1280
A/Belgium/1239/2009	20/11/2009	Siat4	2560	2560	2560	5120	1280	2560
A/Stockholm/106/2009	24/11/2009	p1 / SIAT1	1280	1280	2560	2560	1280	1280
A/Egypt-Vacsera/162/2009	Dec-09	E2	2560	1280	2560	5120	2560	2560
A/Israel/S2942/2009	Dec-09	SIAT1	2560	5120	5120	2560	1280	2560
A/Egypt/114/2009	06/12/2009	Siat2	5120	5120	5120	5120	2560	2560
A/Belarus/94/2009	02/09/2009	E1 / E1	5120	5120	5120	5120	2560	5120
A/Iran/16743/2009	Nov-09	E2/ E1	1280	1280	1280	2560	1280	2560
A/Iran/16677/2009	Nov-09	E2/ E1	1280	1280	1280	2560	1280	1280
A/Aswan/2288/2009	02/11/2009	SIAT2	1280	1280	2560	2560	640	1280
A/Kaliningrad/2/2009	02/11/2009	E2/ E1	2560	2560	2560	5120	2560	2560
A/Lisboa/98/2009	19/11/2009	MDCK2 / SIAT1	1280	1280	1280	1280	640	1280
A/Helwan/742/2009	19/11/2009	SIAT2	2560	2560	2560	2560	1280	2560
A/Kosovo/348/2009	23/11/2009	SIAT3	2560	2560	2560	5120	1280	2560
A/Belgium/G4444/2009	23/11/2009	SIAT3	2560	1280	2560	5120	1280	1280
A/St-Petersburg/61/2009	23/11/2009	C1 / SIAT1	2560	1280	2560	2560	640	1280
A/St-Petersburg/76/2009	23/11/2009	C1 / SIAT1	5120	2560	5120	5120	1280	2560
A/Belgorod/2/2009	23/11/2009	E2/ E1	2560	2560	2560	2560	1280	2560
A/Petrozavodsk/1/2009	23/11/2009	E2/ E1	1280	1280	1280	2560	1280	1280
A/Voronezh/1/2009	30/11/2009	E1 / E1	2560	2560	2560	5120	1280	2560
A/Egypt-Vacsera/159/2009	Dec-09	E2	2560	1280	5120	5120	2560	5120
A/Poland/922/2009	02/12/2009	SIAT2	1280	1280	2560	2560	1280	1280
A/Poland/923/2009	02/12/2009	SIAT2	2560	2560	2560	5120	1280	2560
A/Berlin/164/2009	07/12/2009	Siat3/SIAT1	1280	1280	2560	2560	1280	2560
A/Madagascar/10136/2009	09/12/2009	SIAT2	2560	2560	2560	2560	640	1280
A/Madagascar/10144/2009	10/12/2009	SIAT2	1280	1280	2560	2560	1280	2560
A/Mecklenburg-Vorp/10/2009	14/12/2009	Siat4/SIAT1	2560	2560	2560	5120	1280	1280
A/Berlin/187/2009	14/12/2009	Siat4/SIAT1	2560	2560	5120	5120	1280	2560
A/Athens/16596/2009	26/12/2009	SIAT3	1280	1280	1280	2560	640	1280
A/Athens/16610/2009	26/12/2009	SIAT2	1280	1280	1280	2560	640	1280
A/Leivadeia/16621/2009	26/12/2009	SIAT2	1280	1280	1280	2560	640	1280

See figures 8 and 11

1. < = 40

Vaccine strain

Table 5. Antigenic analyses of pandemic influenza A(H1N1) viruses (09/02/2010)

			Haemagglutination inhibition titre[1]					
			Post infection ferret sera					
Viruses	Collection Date	Passage History	A/Cal 4/09 C4/F14/09	A/Cal 7/09 C4/31/09	A/Eng 195/09 NIBSC F18/09	A/Auck 3/09 C4/17/09	A/Bayern 69/09 C4/33/09	A/Lviv N6/2009 C4/34/09
REFERENCE VIRUSES								
A/California/4/2009		C1,E2	2560	2560	2560	2560	1280	2560
A/California/7/2009		E6	2560	2560	1280	2560	2560	2560
A/England/195/2009		MDCK/Siat1	2560	1280	2560	2560	1280	1280
A/Auckland/3/2009		Ex+3	5120	2560	2560	5120	2560	2560
A/Bayern/69/2009		MDCK4/SIAT1	160	320	160	160	640	640
A/Lviv/N6/2009		MDCK4/SIAT1	640	1280	320	320	2560	1280
TEST VIRUSES								
A/Stockholm/103/2009	01/10/2009	p0/SIAT1	1280	1280	1280	2560	1280	1280
A/Umea/6/2009	02/10/2009	p2/SIAT1	2560	2560	1280	2560	1280	2560
A/Umea/7/2009	07/10/2009	p2/SIAT1	1280	1280	1280	2560	640	1280
A/Umea/8/2009	07/10/2009	p2/SIAT1	1280	1280	1280	2560	640	1280
A/Cameroon/463/2009	09/10/2009	SIAT1	2560	2560	2560	5120	1280	2560
A/Cameroon/566/2009	30/10/2009	SIAT1	1280	1280	1280	2560	640	1280
A/Paris/6232/09	Nov-09	C1/SIAT2	1280	1280	1280	1280	640	640
A/France/6808/09	Nov-09	Cx/SIAT2	2560	2560	2560	2560	1280	2560
A/Austria/523708/2009	05/11/2009	SIAT2/SIAT2	2560	1280	2560	2560	1280	1280
A/Austria/531141/2009	03/12/2009	SIAT1/SIAT2	2560	2560	2560	5120	1280	2560
A/Lisboa/112/2009	06/11/2009	MCDK2/SIAT1	1280	1280	1280	2560	1280	1280
A/Lisboa/116/2009	29/12/2009	MCDK2/SIAT1	2560	1280	1280	2560	1280	1280
A/Israel/MF 14817/2009	Nov-09	SIAT1	2560	2560	2560	5120	1280	2560
A/Israel/MF 17568/2009	Nov-09	SIAT1	2560	2560	2560	5120	1280	2560
A/Israel/S 2838/2009	Nov-09	SIAT1	1280	1280	1280	2560	1280	1280
A/Afghanistan/4236/2009	04/11/2009	SIAT2	1280	1280	1280	1280	640	1280
A/Albania/1828/2009	25/11/2009	P1/SIAT2	2560	2560	2560	2560	2560	2560
A/Albania/2489/2009	22/12/2009	P1/SIAT1	2560	2560	2560	5120	2560	2560
A/Orebro/1/2009	02/11/2009	p2/SIAT1	2560	2560	2560	5120	1280	2560
A/Orebro/2/2009	09/11/2009	p0/SIAT1	1280	1280	1280	2560	1280	1280
A/Uppsala/2/2009	12/11/2009	p1/SIAT1	1280	1280	1280	2560	640	1280
A/Linkoping/1/2009	22/11/2009	p0/SIAT1	1280	1280	1280	5120	1280	1280
A/Stockholm/106/2009	24/11/2009	p1/SIAT1	1280	1280	2560	2560	1280	1280

1. < = 40

Vaccine strain

Table 6. Antigenic analyses of pandemic influenza A(H1N1) viruses [all isolates with ≥ 4-fold reduction against at least one ferret antiserum]

			Haemagglutination inhibition titre[1]								
				Post infection ferret sera							
Viruses	Collection date	Passage History	A/NJ[2] 8/76 R15/79	A/NJ 8/76 F34/82	A/Cal 4/09 C4/F14/09	A/Cal 7/09 C4/31/09	A/Eng 195/09 NIBSC F18/09	A/Auck 3/09 C4/17/09	A/Bayern 69/09 C4/33/09	A/Lviv N6/2009 C4/34/09	
REFERENCE VIRUSES											
A/New Jersey/8/76		Ex	>5120	1280	320	640	640	1280	320	640	
A/California/4/2009		C1,E2	5120	320	2560	2560	2560	2560	1280	2560	
A/California/7/2009		E5	2560	320	2560	2560	2560	2560	1280	5120	
A/England/195/2009		MDCK5	2560	320	2560	2560	2560	2560	1280	2560	
A/Auckland/3/2009		Ex+4	5120	640	2560	2560	5120	2560	2560	5120	
A/Bayern/69/2009		Siat2,MDCK1,Siat1	640	<	80	320	160	160	640	640	
A/Lviv/N6/2009		MDCK3,Siat1	640	<	320	640	160	160	1280	1280	
TEST VIRUSES											
A/Sachsen-Anhalt101/2009	04/05/2009	Siat4	1280	<	320	160	80	160	NT	NT	
A/Hungary/20/2009	15/06/2009	E2/E1	2560	160	640	320	320	640	NT	NT	
A/Bayern/69/2009	Jul-09	Siat2,MDCK1	640	<	80	80	40	80	NT	NT	
A/Sachsen/92/2009	04/07/2009	Siat4	640	<	160	80	80	160	NT	NT	
A/Netherlands/1132/2009	12/08/2009	xMDCK3	1280	80	640	640	1280	1280	NT	NT	
A/Norway/3206-3/2009	01/09/2009	MDCK3	640	<	320	640	80	80	NT	NT	See figures
A/Belarus/24/2009	07/10/2009	MDCK2,Siat1	640	NT	640	640	640	640	640	640	8 and 11
A/Norway/4011/2009	25/10/2009	MDCK1 / Siat1	1280	80	1280	1280	640	320	640	640	
A/Lviv/N6/2009	27/10/2009	MDCK2	640	<	320	640	80	80	NT	NT	
A/Lyon/2288/2009	03/11/2009	MDCK3/Siat1	640	<	320	640	320	320	640	640	
A/Belgium/G4405/2009	23/11/2009	Siat2	2560	40	640	2560	640	640	1280	1280	
A/Azores/13092/2009	12/11/2009	Siat2	1280	80	640	640	640	1280	NT	NT	
A/Athens/16606/2009	26/12/2009	Siat2	NT	NT	320	640	320	320	2560	1280	
A/Brandenburg/19/2009	02/05/2009	Siatx	2560	160	160	320	160	160	NT	NT	
A/Stockholm/63/2009	11/07/2009	p2/SIAT1	NT	NT	640	1280	320	640	1280	1280	
A/Nordrhein-Westfalen/74/2009	20/07/2009	Siat4	2560	<	80	160	80	80	NT	NT	
A/Nordrhein-Westfalen/79/2009	20/07/2009	Siat3	640	<	160	160	160	160	NT	NT	
A/Slovenia/3347/2009	Aug-09	MDCK1/Siat1	640	40	640	640	640	1280	NT	NT	
A/Nordrhein-Westfalen/99/2009	07/08/2009	Siat3+1	2560	NT	640	1280	640	320	2560	1280	
A/Lisboa/51/2009	19/08/2009	MDCK2,Siat1	2560	NT	160	640	160	160	640	640	
A/Switzerland/2243133/2009	27/08/2009	MDCK3	640	80	640	640	1280	1280	NT	NT	
A/Iraq/8529/2009	08/09/2009	E3/E1	NT	NT	320	640	320	320	640	640	
A/St. Etienne/2024/2009	14/09/2009	MDCK4/Siat1	1280	80	640	1280	640	640	640	640	
A/South Carolina/18/2009	16/09/2009	M1/E3/E1	NT	NT	320	1280	320	320	1280	1280	
A/Ontario/RV3226/2009	18/09/2009	C1/E2/E1	NT	NT	320	640	160	320	640	640	
A/Norway/3568/2009	20/09/2009	MDCK6	640	40	640	640	640	1280	NT	NT	
A/Slovenia/4025/2009	Oct-09	MDCK1/Siat1	1280	80	640	640	1280	1280	NT	NT	
A/Norway/3670/200	05/10/2009	MDCK5	640	40	320	640	640	1280	NT	NT	
A/Thuringen/176/2009	17/10/2009	Siat3+1	≥5120	NT	640	1280	640	640	1280	1280	
A/Madagascar/9018/2009	26/10/2009	Siat2	NT	NT	640	1280	640	320	2560	2560	
A/Belgium/1044/2009	28/10/2009	Siat3	640	<	320	1280	320	160	1280	1280	
A/Rivne/214/2009	30/10/2009	Siat4	5120	40	160	1280	160	160	640	1280	
A/Belarus/119/2009	02/11/2009	MDCK2,Siat1	2560	NT	640	640	640	640	640	640	
A/Lisboa/99/2009	02/11/2009	MDCK2/Siat1	NT	NT	1280	640	640	1280	640	640	
A/Dnipropetrovsk/ 267/2009	04/11/2009	MDCK1/Siat1	640	40	640	640	640	640	NT	NT	
A/Volin/226/2009	05/11/2009	Siat3	2560	80	1280	1280	640	640	320	1280	
A/Bremen/20/2009	16/11/2009	Siat3/Siat1	1280	40	640	1280	640	640	1280	1280	
A/Berlin/133/2009	16/11/2009	Siat3/Siat1	5120	40	320	1280	320	320	1280	1280	
A/Sweden/25/2009	16/11/2009	p2/SIAT1	NT	NT	640	640	640	1280	640	1280	
A/Neidersachsen/366/2009	16/11/2009	Siat3/Siat1	640	80	640	1280	640	1280	320	640	
A/Brandenburg/55/2009	17/11/2009	Siat3/Siat1	640	<	320	640	320	320	1280	1280	
A/Kosovo/275/2009	18/11/2009	Siat3	NT	NT	1280	640	640	1280	640	1280	
A/England/869/2009	18/11/2009	Siat1/Siat1	NT	NT	160	320	160	160	1280	640	
A/England/876/2009	20/11/2009	Siat1/Siat1	NT	NT	160	320	160	160	320	320	
A/Sachsen/177/2009	14/12/2009	Siat4/Siat1	NT	NT	320	1280	640	640	1280	640	
A/Albania/2385/2009	18/12/2009	P1/SIAT1	NT	NT	1280	640	640	1280	640	1280	

1. < = 40; 2. hyperimmune rabbit serum

NT = not tested

Vaccine strain

Figure 8. Phylogenetic comparison of pandemic A(H1N1) HA genes

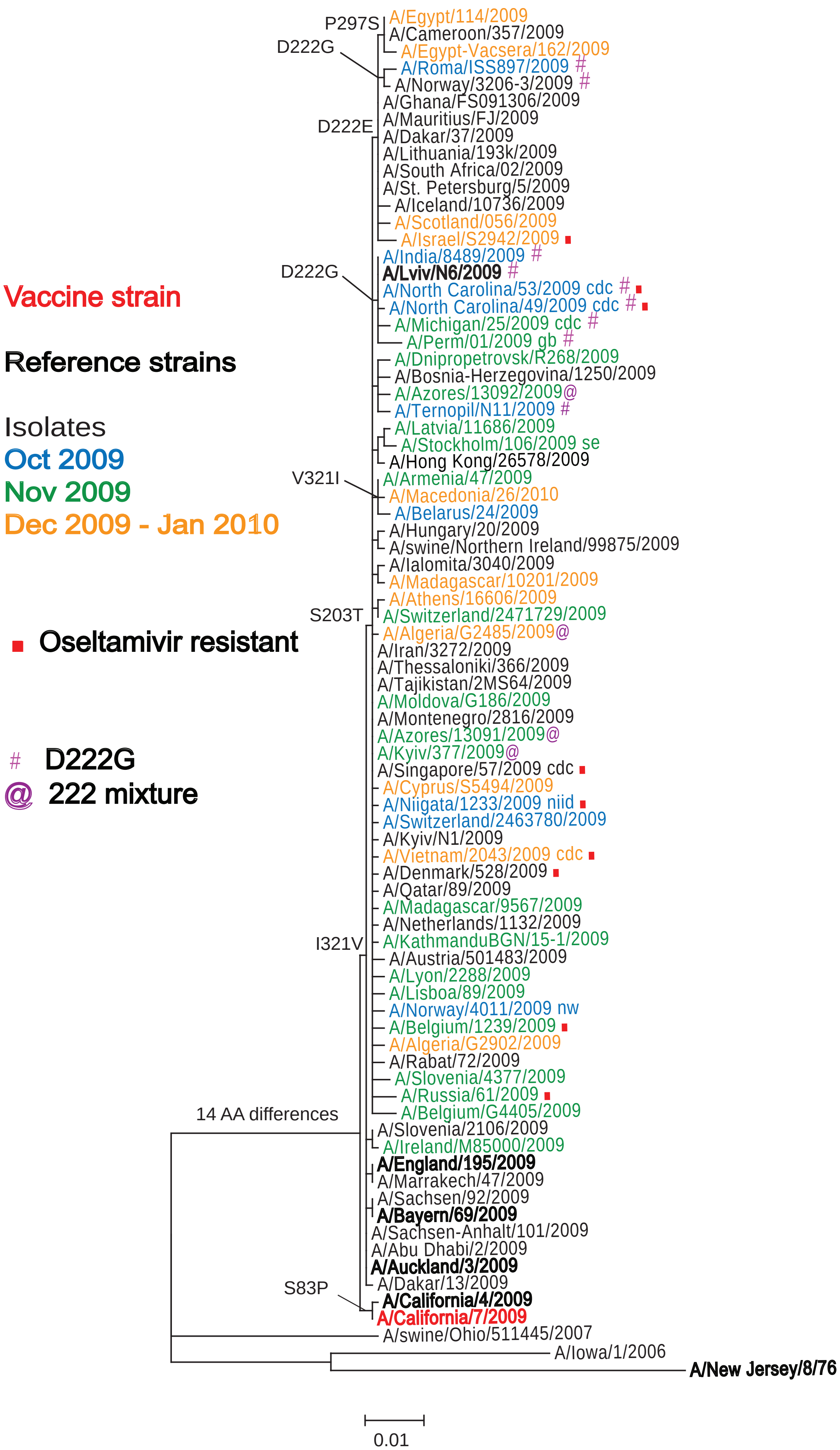


Figure 9: HA1 protein alignment for pandemic A(H1N1) viruses

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                          1                                                                                               83              100
     A/California/7/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- --p------- ----------
        A/Auckland/3/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         A/Bayern/69/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/England/195/2009  ---------- ---------- ---------- -i-------- ---------- ---------- ---------- ---------- ---------- ----------
        A/Russia/61/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Algeria/G2902/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Belgium/1239/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Denmark/528/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Vietnam/2043/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Niigata/1233/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Cyprus/S5494/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Singapore/57/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Azores/13091/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Algeria/G2485/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Athens/16606/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
 A/Madagascar/10201/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Macedonia/26/2010  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Azores/13092/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
          A/Perm/01/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- -----n---- ----------
      A/Michigan/25/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/North Carolina/49/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/North Carolina/53/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
          A/Lviv/N6/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        A/India/8489/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Israel/S2942/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Scotland/056/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Norway/3206-3/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        A/Roma/SS897/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/Egypt-Vacsera/162/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        A/Egypt/114/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
               Consensus  DTLCIGYHAN NSTDTVDTVL EKNVTVTHSV NLLEDKHNGK LCKLRGVAPL HLGKCNIAGW ILGNPECESL STASSWSYIV ETSSSDNGTC YPGDFIDYEE
```

```
                          101                                                                                                             200
     A/California/7/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        A/Auckland/3/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         A/Bayern/69/2009 ---------- ---------- ---------- ---------- ---------- ----e----- ---------- ---------- ---------- ----------
      A/England/195/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        A/Russia/61/2009  ---------- -----m---- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Algeria/G2902/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Belgium/1239/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Denmark/528/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Vietnam/2043/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Niigata/1233/2009  ---------- ---------- ---------- -e-------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Cyprus/S5494/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Singapore/57/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Azores/13091/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Algeria/G2485/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Athens/16606/2009  ---------- ---------- ---------- ---------- ---------- ----e----- ---------- ---------- ---------- ----------
 A/Madagascar/10201/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Macedonia/26/2010  ---------- ---------- ---------- ---------- t--------- ---------- ---------- ---------- ---------- ----------
     A/Azores/13092/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
          A/Perm/01/2009  ---------- ---------- -------p-- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Michigan/25/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/North Carolina/49/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/North Carolina/53/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
          A/Lviv/N6/2009  ---------- ---------- ---------- ---------- ---------- ----e----- ---------- ---------- ---------- ----------
        A/India/8489/2009 ---------- ---------- ----x----- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Israel/S2942/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Scotland/056/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Norway/3206-3/2009  ---------- ---------- ---------- ------s--- ---------- ---------- ---------- ---------- ---------- ----------
        A/Roma/SS897/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/Egypt-Vacsera/162/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        A/Egypt/114/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
               Consensus  LREQLSSVSS FERFEIFPKT SSWPNHDSNK GVTAACPHAG AKSFYKNLIW LVKKGNSYPK LSKSYINDKG KEVLVLWGIH HPSTSADQQS LYQNADAYVF
```

Figure 9: HA1 protein alignment for pandemic A(H1N1) viruses (continued)

```
                     201 203             222                                                                                              297300
   A/California/7/2009  --s------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Auckland/3/2009  --s------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Bayern/69/2009  --s------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/England/195/2009  --s------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Russia/61/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Algeria/G2902/2009  ---------- ---------- ---------- ---------- --------a- ---------- -----v---- ---------- ---------- ----------
   A/Belgium/1239/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Denmark/528/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Vietnam/2043/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Niigata/1233/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Cyprus/S5494/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Singapore/57/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Azores/13091/2009  ---------- ---------- -x-------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Algeria/G2485/2009  ---------- ---------- -x-------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Athens/16606/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/Madagascar/10201/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Macedonia/26/2010  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Azores/13092/2009  ---------- ---------- -x-------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        A/Perm/01/2009  ---------- ---------- -g-------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Michigan/25/2009  ---------- ---------- -g-------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/North Carolina/49/2009 ---------- ---------- -g-------h ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/North Carolina/53/2009 ---------- ---------- -g-------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        A/Lviv/N6/2009  ---------- ---------- -g-------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/India/8489/2009  ---------- ---------- -g-------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Israel/S2942/2009  ---------- ---------- -e-------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Scotland/056/2009  ---------- ---------- -e-------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Norway/3206-3/2009  ---------- ---------- -g-------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Roma/SS897/2009  ---------- ---------- -g-------- ---------- ---------- ---------- ---------- ---------- ---------- ------s---
A/Egypt-Vacsera/162/2009 ---------- ---------- -e-------- ---------- ---------- ---------- ---------- ---------- ---------- ------s---
      A/Egypt/114/2009  ---------- ---------- -e-------- ---------- ---------- ---------- ---------- ---------- ---------- ------s---
             Consensus  VGTSRYSKKF KPEIAIRPKV RDQEGRMNYY WTLVEPGDKI TFEATGNLVV PRYAFAMERN AGSGIIISDT PVHDCNTTCQ TPKGAINTSL PFQNIHPITI
```

```
                        301                   321    328
   A/California/7/2009  ---------- ---------- i-------
     A/Auckland/3/2009  ---------- ---------- --------
      A/Bayern/69/2009  ---------- ---------- --------
    A/England/195/2009  ---------- ---------- --------
      A/Russia/61/2009  ---------- ---------- --------
  A/Algeria/G2902/2009  ---------- ---------- --------
   A/Belgium/1239/2009  ---------- ---------- --------
    A/Denmark/528/2009  ---------- ---------- --------
   A/Vietnam/2043/2009  ---------- ---------- --------
   A/Niigata/1233/2009  ---------- ---------- --------
   A/Cyprus/S5494/2009  ---------- ---------- --------
   A/Singapore/57/2009  ---------- ---------- --------
   A/Azores/13091/2009  ---------- ---------- --------
  A/Algeria/G2485/2009  ---------- ---------- --------
   A/Athens/16606/2009  ---------- ---------- --------
A/Madagascar/10201/2009 ---------- ---------- --------
   A/Macedonia/26/2010  ---------- ---------- i-------
   A/Azores/13092/2009  ---------- ---------- --------
        A/Perm/01/2009  ---------- ---------- i-------
    A/Michigan/25/2009  ---------k ---------- --------
A/North Carolina/49/2009 ---------- ---------- --------
A/North Carolina/53/2009 ---------- ---------- --------
        A/Lviv/N6/2009  ---------- ---------- --------
     A/India/8489/2009  ---------- ---------- --------
   A/Israel/S2942/2009  ---------- ---------- --------
   A/Scotland/056/2009  ---------- ---------- --------
  A/Norway/3206-3/2009  ---------- ---------- --------
     A/Roma/SS897/2009  ---------- ---------- --------
A/Egypt-Vacsera/162/2009 ---------- ---------- --------
      A/Egypt/114/2009  ---------- ---------- --------
             Consensus  GKCPKYVKST KLRLATGLRN VPSIQSRG
```

N-linked glycosylation sites (NXS/T) are highlighted.

Figure 10. Sites of H1N1pdm amino acid substitutions in HA head region.

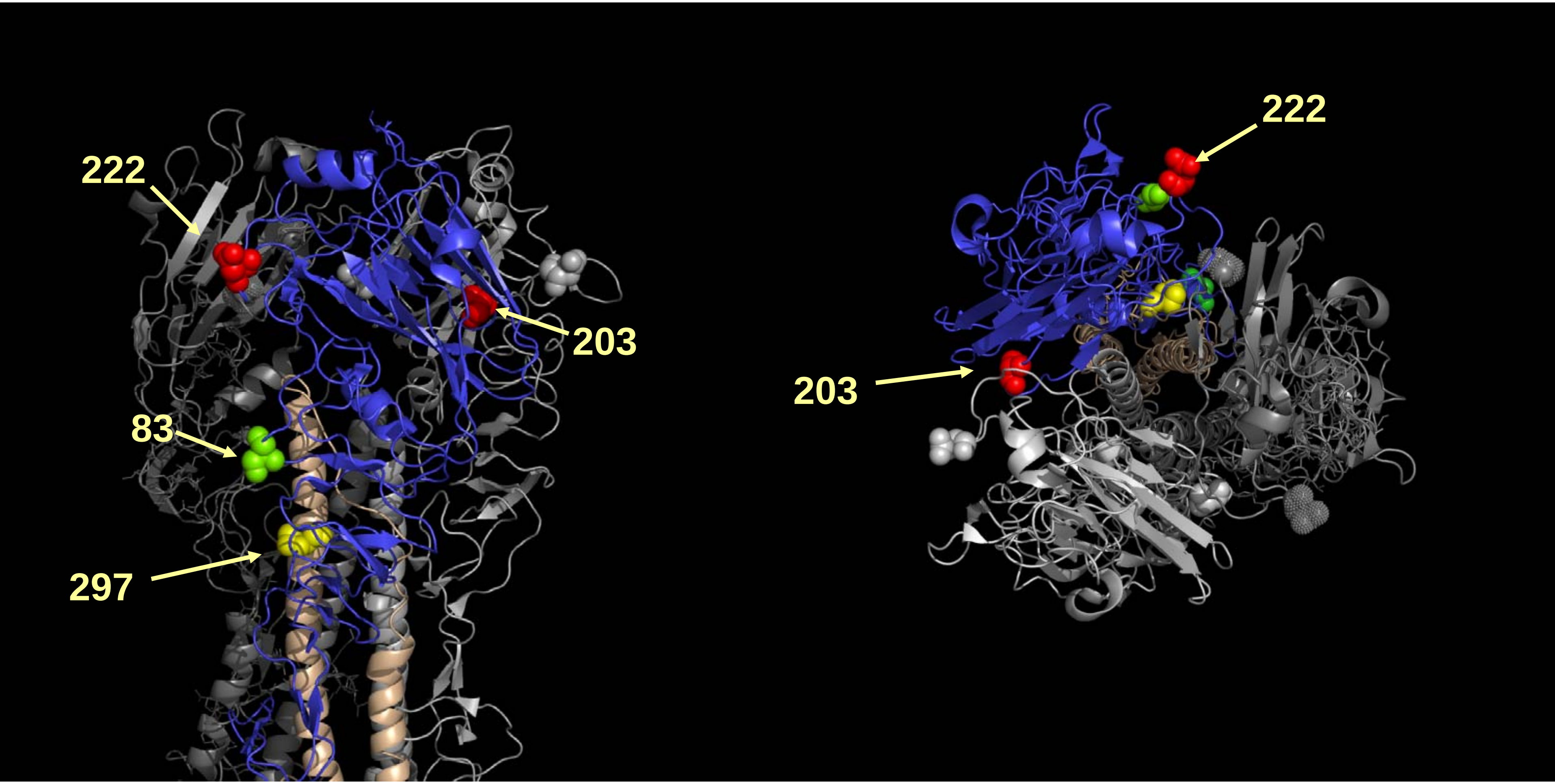

Figure 11. Phylogenetic comparison of pandemic A(H1N1) NA genes

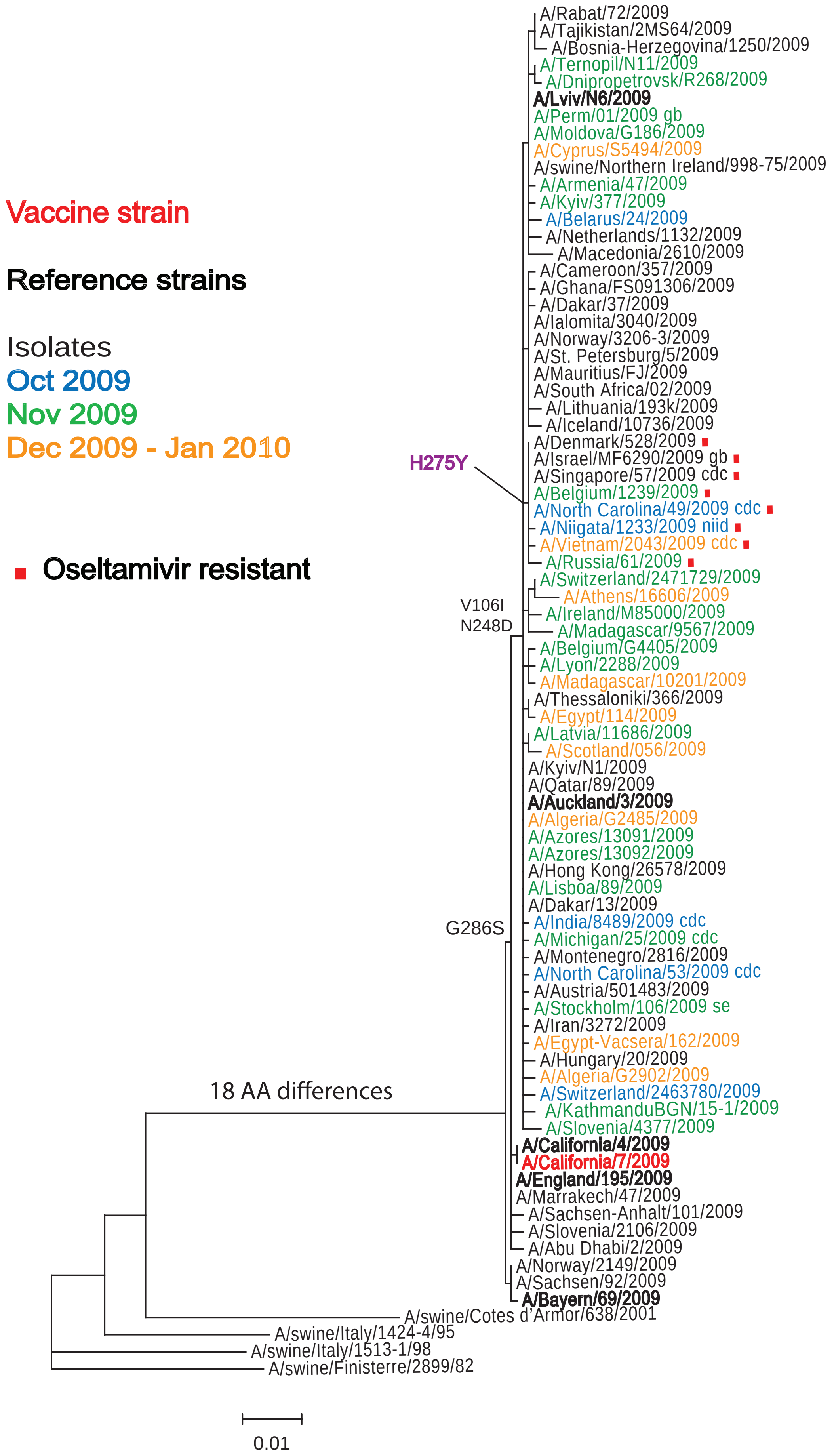

Figure 12: NA protein alignment for pandemic A(H1N1) viruses

```
                          1                                                                                                          100
     A/California/7/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       A/Auckland/3/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         A/Bayern/69/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/England/195/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         A/Russia/61/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Algeria/G2902/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Belgium/1239/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Denmark/528/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Vietnam/2043/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Niigata/1233/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Cyprus/S5494/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Singapore/57/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Azores/13091/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Algeria/G2485/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Athens/16606/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Madagascar/10201/2009 ~~~~~~~~~~ ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Macedonia/26/2010  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Azores/13092/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
          A/Perm/01/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Michigan/25/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/North Carolina/49/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/North Carolina/53/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
          A/Lviv/N6/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       A/India/8489/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Israel/S2942/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ------s--- ----------
     A/Scotland/056/2009  ---------- ---------- ---------- ---------- -----t---- ---------- ---------- ---------- ---------- ----------
    A/Norway/3206-3/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/Egypt-Vacsera/162/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         A/Egypt/114/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
               Consensus  MNPNQKIITI GSVCMTIGMA NLILQIGNII SIWISHSIQL GNQNQIETCN QSVITYENNT WVNQTYVNIS NTNFAAGQSV VSVKLAGNSS LCPVSGWAIY

                          101  106                                                                                                   200
     A/California/7/2009  -----v---- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       A/Auckland/3/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         A/Bayern/69/2009 -----v---- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/England/195/2009  -----v---- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         A/Russia/61/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Algeria/G2902/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Belgium/1239/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Denmark/528/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Vietnam/2043/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Niigata/1233/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Cyprus/S5494/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Singapore/57/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Azores/13091/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Algeria/G2485/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Athens/16606/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Madagascar/10201/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Macedonia/26/2010  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Azores/13092/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
          A/Perm/01/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Michigan/25/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/North Carolina/49/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/North Carolina/53/2009  ---------- ---------- ---------- ---------- --r------- ---------- ---------- ---------- ---------- ----------
          A/Lviv/N6/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       A/India/8489/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Israel/S2942/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Scotland/056/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Norway/3206-3/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/Egypt-Vacsera/162/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         A/Egypt/114/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
               Consensus  SKDNSIRIGS KGDVFVIREP FISCSPLECR TFFLTQGALL NDKHSNGTIK DRSPYRTLMS CPIGEVPSPY NSRFESVAWS ASACHDGINW LTIGISGPDN

                          201                                               248                        275         286           300
     A/California/7/2009  ---------- ---------- ---------- ---------- -------n-- ---------- ---------- ---------- ---------- ----------
       A/Auckland/3/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         A/Bayern/69/2009 ---------- ---------- ---------- ---------- -------n-- ---------- ---------- ---------- -----g---- ----------
      A/England/195/2009  ---------- ---------- ---------- ---------- -------n-- ---------- ---------- ---------- ---------- ----------
         A/Russia/61/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----y----- ---------- ----------
    A/Algeria/G2902/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Belgium/1239/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----y----- ---------- ----------
      A/Denmark/528/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----y----- ---------- ----------
     A/Vietnam/2043/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----y----- ---------- ----------
     A/Niigata/1233/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----y----- ---------- ----------
     A/Cyprus/S5494/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Singapore/57/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----y----- ---------- ----------
     A/Azores/13091/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Algeria/G2485/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Athens/16606/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Madagascar/10201/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Macedonia/26/2010  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Azores/13092/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
          A/Perm/01/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Michigan/25/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/North Carolina/49/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----y----- ---------- ----------
A/North Carolina/53/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
          A/Lviv/N6/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       A/India/8489/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Israel/S2942/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Scotland/056/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Norway/3206-3/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/Egypt-Vacsera/162/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         A/Egypt/114/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
               Consensus  GAVAVLKYNG IITDTIKSWR NNILRTQESE CACVNGSCFT VMTDGPSDGQ ASYKIFRIEK GKIVKSVEMN APNYHYEECS CYPDSSEITC VCRDNWHGSN
```

Figure 12: NA protein alignment for pandemic A(H1N1) viruses (continued)

```
                           301                                                                                                     400
     A/California/7/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       A/Auckland/3/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        A/Bayern/69/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/England/195/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        A/Russia/61/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Algeria/G2902/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Belgium/1239/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Denmark/528/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Vietnam/2043/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Niigata/1233/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Cyprus/S5494/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Singapore/57/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Azores/13091/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Algeria/G2485/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Athens/16606/2009   ---------- ---------- ---------- -k-------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Madagascar/10201/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Macedonia/26/2010   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Azores/13092/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
          A/Perm/01/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Michigan/25/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/North Carolina/49/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/North Carolina/53/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
          A/Lviv/N6/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       A/India/8489/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Israel/S2942/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Scotland/056/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Norway/3206-3/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/Egypt-Vacsera/162/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        A/Egypt/114/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
               Consensus   RPWVSFNQNL EYQIGYICSG IFGDNPRPND KTGSCGPVSS NGANGVKGFS FKYGNGVWIG RTKSISSRNG FEMIWDPNGW TGTDNNFSIK QDIVGINEWS
```

```
                           401                                                                 469
     A/California/7/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------*
       A/Auckland/3/2009   ---------- ---------- ---------- ---------- ---------- ---------- -----~~~~~
        A/Bayern/69/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------*
      A/England/195/2009   ---------- ---------- ---------- ---------- ---------- ---------- -------~~~
        A/Russia/61/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------*
    A/Algeria/G2902/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------*
     A/Belgium/1239/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------*
      A/Denmark/528/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------*
     A/Vietnam/2043/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------*
     A/Niigata/1233/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------*
     A/Cyprus/S5494/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------*
     A/Singapore/57/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------*
     A/Azores/13091/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------*
    A/Algeria/G2485/2009   ---------- ---------- -----x---- ---------- ---------- ---------- ---------*
     A/Athens/16606/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------*
  A/Madagascar/10201/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------*
     A/Macedonia/26/2010   ---------- ---------- ---------- ---------- ---------- ---------- ---------*
     A/Azores/13092/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------*
          A/Perm/01/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------*
      A/Michigan/25/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------*
A/North Carolina/49/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------*
A/North Carolina/53/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------*
          A/Lviv/N6/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------*
       A/India/8489/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------*
     A/Israel/S2942/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------*
     A/Scotland/056/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------*
    A/Norway/3206-3/2009   ---------- ---------- ---------- ----~~~~~~ ~~~~~~~~~~ ~~~~~~~~~~ ~~~~~~~~~~
A/Egypt-Vacsera/162/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------*
        A/Egypt/114/2009   ---------- ---------- -----q---- ---------- ---------- ---------- ---------*
               Consensus   GYSGSFVQHP ELTGLDCIRP CFWVELIRGR PKENTIWTSG SSISFCGVNS DTVGWSWPDG AELPFTIDK-
```

N-linked glycosylation sites (NXS/T) are highlighted.

NA sequence corresponding to the HA1 of A/Roma/SS897/2009, shown in Figure 9, was not available.

Table 7. Summary of seasonal influenza viruses received, collected between September and present

MONTH	H1N1		H3N2		B			
					Yamagata		Victoria	
Country	received	grown	received	grown	received	grown	received	grown
SEPTEMBER								
Cameroon			8	7				
France	2	2	5	5				
Ghana			3	3			1	1
Gibraltar			1	1				
Hong Kong	2	2	3	3				
Sweden			3	3				
Turkey	1	1						
United Kingdom			3	3				
October 2009								
Cameroon			11	11				
Finland			2	2				
France	1	1	1	1				
Ghana			1	1				
Gibraltar			1	1				
Hong Kong	3	3	1	1			1	1
Madagascar							2	0
Norway			2	2			1	0
November 2009								
Algeria			2	1				
Cameroon			15	15				
France			7	4				
Ghana			1	1				
Hong Kong			2	2	1	1		
Israel			6	6				
Senegal			15	3				
Sweden			2	2			1	1
Turkey			7	3	4	4		
December 2009								
United Kingdom							1	1
January 2010								
Germany			1	1				
Sweden							1	1
Turkey			1	1				
Total	9	9	104	83	5	5	8	5

Table 8. Summary of seasonal H1N1 influenza viruses received, collected between September and present

MONTH	H1N1				
Country	received	grown	≤2 fold*	4 fold*	≥8 fold*
SEPTEMBER 2009					
France	2	2			2
Hong Kong	2	2			2
Turkey	1	1			1
OCTOBER 2009					
France	1	1			1
Hong Kong	3	3			3
Total	**9**	**9**			**9**

*** Reduction in HI titre with ferret antiserum raised against vaccine strain (A/Brisbane/59/2007)**

Table 9. Antigenic analyses of seasonal A(H1N1) viruses

Viruses	Collection Date	Passage History	A/NC 20/99 F19/08	A/HK 2652/06 F14/06	A/SI 3/06 F8/07	A/Bris 59/07 F6/08	A/SP 12/08 F17/08	A/Sey 2239/08 F27/08	A/HK 1856/08 F28/08	A/HK 1870/08 F4/09	A/Mos 2/09 F19/09	A/HK 1988/09 F20/09
			Haemagglutination inhibition titre[1] – Post infection ferret sera									
Phylogenetic Clade			1	2C	2A	2B	2B	2B	2C	2C	2B	2B
REFERENCE VIRUSES												
A/New Caledonia/20/99		E2 E3	**160**	40	<	40	<	80	80	<	160	80
A/Hong Kong/2652/2006		E2 E4	40	**320**	320	320	320	160	320	640	80	320
A/Solomon Islands/3/2006		E3 E5	80	320	**640**	160	160	320	640	160	640	320
A/Brisbane/59/2007		E2 E5	40	80	320	**320**	320	160	160	320	80	320
A/St.Petersburg/12/2008		E2 E3	40	160	320	320	**320**	160	160	640	80	320
A/Seychelles/2239/2008		MDCK1 Siat4	<	80	80	40	80	**320**	640	640	640	320
A/Hong Kong/1856/2008		MDCK2 Siat5	40	320	320	40	80	640	**2560**	160	320	640
A/Hong Kong/1870/2008		E2 E5	<	160	160	160	160	80	640	**320**	80	320
A/Moscow/2/2009		MDCK1 Siat6	<	80	40	40	80	160	640	80	**640**	640
A/Hong Kong/1988/2009		MDCK2 Siat4	<	80	40	40	80	160	640	80	320	**640**
TEST VIRUSES												
A/Taiwan/2/2009	unknown	E3+2 E2	160	40	80	160	<	160	320	320	160	320
A/Hubei-Yiling/51/2009	unknown	E3+2 E2	<	40	40	<	<	<	40	80	40	160
A/Wisconsin/13/2009	14/05/2009	E3 E2	80	40	80	80	<	40	160	80	40	80
A/Paraguay/61/2009	05/06/2009	C2 Siat2	80	40	80	<	<	160	160	80	80	80
A/Iceland/8064/2009	09/06/2009	MDCK2 Siat2	160	<	<	<	<	40	40	80	40	80
A/Hong Kong/18106/2009	25/08/2009	MDCK2 Siat2	<	<	<	<	<	<	<	40	<	80
A/Annecy/2013/2009	03/10/2009	MDCK2 Siat1	<	80	40	<	80	640	160	80	320	320
A/Madagascar/6054/2009	17/08/2009	MDCK2 Siat2	<	160	80	<	40	40	<	40	40	160
A/Madagascar/6081/2009	20/08/2009	MDCK2 Siat1	<	40	<	<	40	160	40	40	80	80
A/Caen/5096/09	Sep-09	C1 Siat1	<	<	40	40	80	320	160	80	160	160
A/Turkey/TR-25/2009	01/09/2009	SIAT1 Siat1	<	40	<	40	40	80	40	40	320	160
A/Montpellier/2051/2009	28/09/2009	MDCK2 Siat1	<	80	40	40	160	80	80	160	80	320
A/Hong Kong/25324/2009	16/09/2009	MDCK2 Siat1	<	<	<	40	80	40	320	160	NT	NT
A/Hong Kong/26553/2009	13/09/2009	MDCK2 Siat1	<	<	<	<	80	40	320	160	NT	NT
A/Hong Kong/33925/2009	17/10/2009	MDCK2 Siat1	<	80	<	<	40	640	160	40	640	320
A/Hong Kong/33597/2009	18/10/2009	MDCK2 Siat1	<	40	<	<	40	640	80	40	320	160
A/Hong Kong/34079/2009	25/10/2009	MDCK2 Siat1	<	40	40	<	40	40	<	80	40	80

See figure 10 (Neutralisation assay)

1. < = <40; NT = not tested

Vaccine strain

Table 10. Antigenic analyses of seasonal A(H1N1) viruses by neutralisation assay

Viruses	Collection Date	Passage History	A/SI 3/06 F8/07	A/Bris 59/07 F6/08	A/Sey 2239/08 F27/08	A/Mos 2/09 F19/09	Position in HA1 131	141	185	
			Virus Neutralization titre[1] Post infection ferret sera							
Phylogenetic Clade			2A	2B	2B	2B				
REFERENCE VIRUSES										
A/Solomon Islands/3/2006		E3 E6	**2560**	640	640	640				
A/Brisbane/59/2007		E2 E6	320	**320**	320	320				
A/Seychelles/2239/2008		MDCK1 Siat5	80	80	**320**	320				
A/Moscow/2/2009		MDCK1 Siat7	40	80	160	**640**				
TEST VIRUSES										
A/Taiwan/2/2009	unknown	E3+2 E2	80	160	320	320				
A/Hubei-Yiling/51/2009	unknown	E3+2 E2	80	160	160	160	A	N	A	See figure 13
A/Wisconsin/13/2009	14/05/2009	E3 E2	80	320	320	320	V	S	V	
A/Paraguay/61/2009	05/06/2009	C2 Siat2	80	80	320	320	V	S	S	
A/Iceland/8064/2009	09/06/2009	MDCK 2 Siat2	<	80	160	160	V	N	A	
A/Hong Kong/18106/2009	25/08/2009	MDCK 2 Siat2	80	160	160	320	A	N	A	
A/Annecy/2013/2009	03/10/2009	P2MDCK Siat2	40	160	320	1280	V	N	A	
A/Madagascar/6054/2009	17/08/2009	MDCK 2 Siat3	80	160	320	640	A	N	A	
A/Montpellier/2051/2009	28/09/2009	P2MDCK Siat2	80	160	160	320	V	N	A	
A/Hong Kong/25324/2009	16/09/2009	MDCK2 Siat2	40	80	160	640	A	N	A	
A/Hong Kong/33597/2009	18/10/2009	MDCKx2 Siat2	<	40	80	320	A	N	A	
A/Hong Kong/26553/2009	13/09/2009	MDCK2 Siat2	<	40	80	320				
A/Hong Kong/33925/2009	17/10/2009	MDCKx2 Siat2	<	80	160	640				

1. < = <40; NT = not tested

Vaccine strain

Figure 13. Phylogenetic comparison of seasonal A(H1N1) HA genes

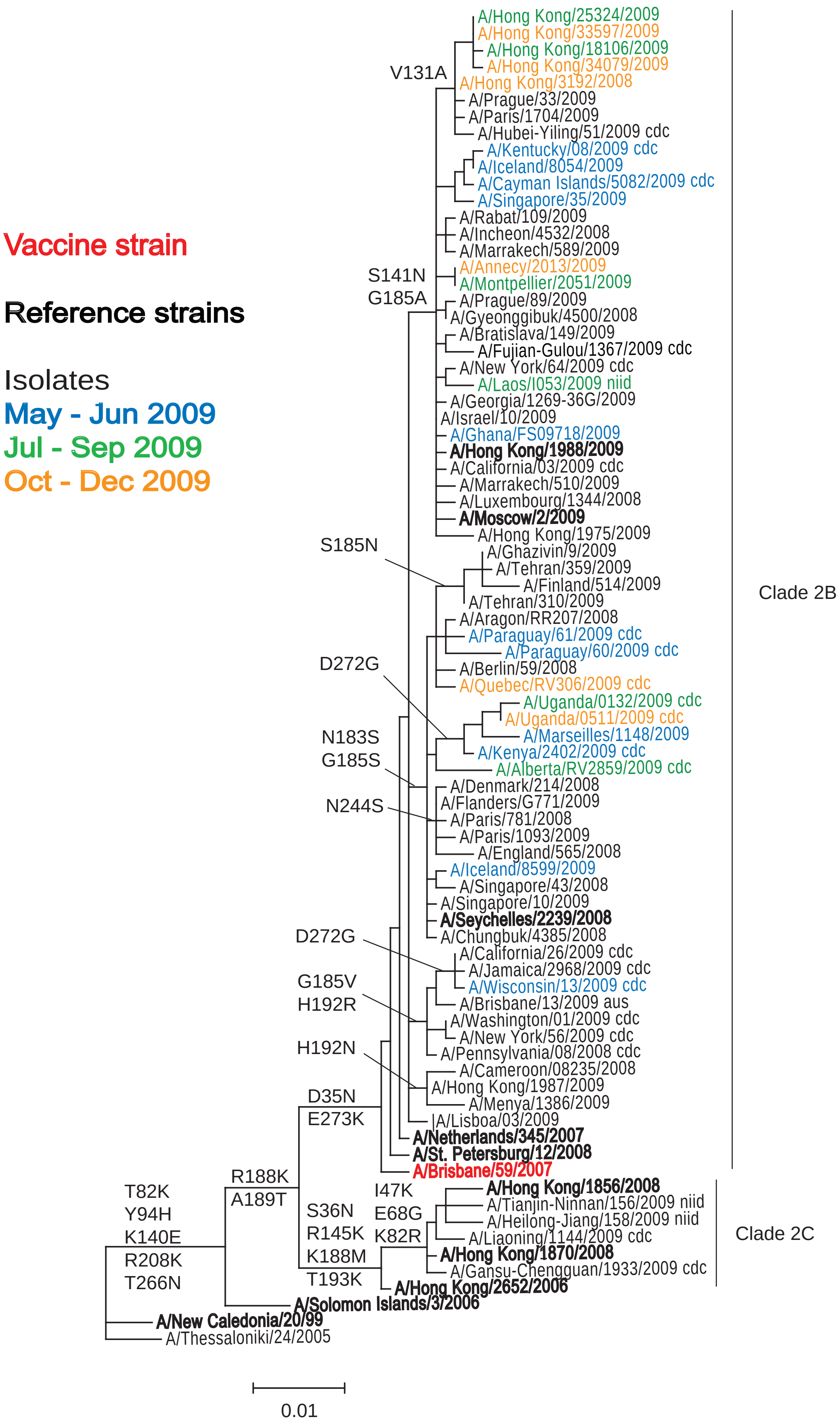

Figure 14: HA1 protein alignment for seasonal A(H1N1) viruses

```
                            1                                                                                                            100
       A/Brisbane/59/2007  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Netherlands/345/2007  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        A/Menya/1386/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       A/New York/56/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Wisconsin/13/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Seychelles/2239/2008  ---------- ---------- ---------- ---------- ---------- --d------- ---------- ---------- ---------- ----------
      A/Iceland/8599/2009  ---------- ---------- -r-------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       A/England/565/2008  ---------- ---------- ---------- ---------- ---------- ---------- ---------- --r------- ---------- ----------
    A/Alberta/RV2859/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        A/Kenya/2402/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Marseille/1148/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       A/Uganda/0511/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Quebec/RV306/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       A/Paraguay/61/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       A/Finland/514/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----t----- ---------- ----------
          A/Moscow/2/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Hong Kong/1988/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Ghana/FS09718/2009  ~~~~~~~--- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         A/Laos/I053/2009  ---------- -----i---- ---------- ---------- ---------- ---------- -----k---- ---------- ---------- ----------
 A/Montpellier/2051/2009  ---------- ---------- ---------- -----n---- ---------- ---------- ---------- ---------- ---------- ----------
       A/Annecy/2013/2009  ---------- ---------- ---------- -----n---- ---------- ---------- ---------- ---------- ---------- ----------
      A/Singapore/35/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/Cayman Islands/5082/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       A/Kentucky/08/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        A/Paris/1704/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Hong Kong/3192/2008  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Hong Kong/34079/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Hong Kong/18106/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Hong Kong/33597/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Hong Kong/25324/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
                Consensus  DTICIGYHAN NSTDTVDTVL EKNVTVTHSV NLLENSHNGK LCLLKGIAPL QLGNCSVAGW ILGNPECELL ISKESWSYIV EKPNPENGTC YPGHFADYEE
```

```
                            101                              131        141                                                  183 185     192      200
       A/Brisbane/59/2007  ---------- ---------- ---------- ---------- s--------- ---------- ---------- ---------- ----g---a- ----------
   A/Netherlands/345/2007  ---------- ---------- ---------- ---------- s--------- ---------- ---------- ---------- ----g---a- ----------
        A/Menya/1386/2009  ---------- ---------- ---------- ---------- s--------- -m------s- ---------- ---------- ----g----- -n--------
       A/New York/56/2009  ---------- ---------- ---------- ---------- s--------- ---------- ---------- ---------- ----vn---- -r--------
      A/Wisconsin/13/2009  ---------- ---------- ---------- ---------- s--------- ---------- ---------- ---------- ----v----- -r--------
   A/Seychelles/2239/2008  ---------- ---------- ---------- ---------- s--------- ---------- ---------- ---------- --s-s----- ----------
      A/Iceland/8599/2009  ---------- ---------- ---------- ---------- s--------- ---------- ---------- ---------- --s-sn---- ----------
       A/England/565/2008  ---------- ---------- ---------- ---------- s--------- ---------- ---------- g--------- --s-s----- ----------
    A/Alberta/RV2859/2009  ------x--- ---------d ---------- ---------- s--------- -x-------- --------g- ---------- --s-r----- ----------
        A/Kenya/2402/2009  ---------- ---------- ---------- ---------- s--------- -r-------- ---------- ---------- --s-s----- ----------
    A/Marseille/1148/2009  ---------- ---------- ---------- ---------- s--------- ---------- ---------- ---------- --s-s----- ----------
       A/Uganda/0511/2009  ---------- ---------- ---------- ---------- s--------- ---------- ---------- ---------- --stsx---- ----------
      A/Quebec/RV306/2009  ---------- ---------- ---------- ---------- s--------- ---------- ---------- ---------- --s-sn---- -n--------
       A/Paraguay/61/2009  ---------- ---------g ---------- ---------- s--------- ---------- ---------- ---------- --s-s----- ----------
       A/Finland/514/2009  ---------- ---------- ---------- ---------- s--------- ---------- ---------- ---------- --s-n----- ----------
          A/Moscow/2/2009  ---------- ---------- ---------- ---------- n--------- ---------- ---------- ---------- ---------- ----------
    A/Hong Kong/1988/2009  ---------- ---------- ---------- ---------- n--------- ---------- ---------- ---------- ---------- ----------
     A/Ghana/FS09718/2009  ---------- ---------- ---------- ---------- n--------- ---------- ---------- ---------- ---------- ----------
         A/Laos/I053/2009  ---------- ---------- ---------- ---------- n--------- ---------- ---------- ---------- -----n---- ----------
 A/Montpellier/2051/2009  ---------- ---------- ---------- ---------- n--------- ---------- ---------- ---------- -----x---- ----------
       A/Annecy/2013/2009  ---------- ---------- ---------- ---------- n--------- ---------- ---------- ---------- ---------- ----------
      A/Singapore/35/2009  ---------- ---------- ---------- ---------- n--------- ---------- ---------- ---------- ---m------ ----------
A/Cayman Islands/5082/2009  ---------- ---------- ---------- ---------- n--------- ---------- ---------- ---------- -----x---- ----------
       A/Kentucky/08/2009  ---------- ---------- ---------- ---------- n--------- -r-------- ---------- ---------- ---------- ----------
        A/Paris/1704/2009  ---------- ---------- ---------- a--------- n--------- ---------- ---------- ---------- ---------- ----------
    A/Hong Kong/3192/2008  ---------- ---------- ---------- a--------- n--------- ---------- ---------- ---------- ---------- ----------
   A/Hong Kong/34079/2009  ---------- ---------- ---------- a--------- n--------- ---------- ---------- ---------- -----x---- ----------
   A/Hong Kong/18106/2009  ---------- ---------- ---------- a--------- n--------- ---------- ---------- ---------- -----n---- ----------
   A/Hong Kong/33597/2009  ---------- ---------- ---------- a--------- n--------- ---------- ---------- ---------- ---------- ----------
   A/Hong Kong/25324/2009  ---------- ---------- ---------- a--------- n--------- ---------- ---------- ---------- ---------- ----------
                Consensus  LREQLSSVSS FERFEIFPKE SSWPNHTVTG VSASCSHNGE -SFYRNLLWL TGKNGLYPNL SKSYANNKEK EVLVLWGVHH PPNIADQKTL YHTENAYVSV
```

Figure 14: HA1 protein alignment for seasonal A(H1N1) viruses (continued)

```
                            201                                           244                           272                           300
       A/Brisbane/59/2007   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Netherlands/345/2007  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         A/Menya/1386/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------~
        A/New York/56/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       A/Wisconsin/13/2009  ---------- ---------- g--------- ---------- ---------- ---------- ---------- -g-------- ---------- ----------
    A/Seychelles/2239/2008  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       A/Iceland/8599/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        A/England/565/2008  ---------- ---------- ---------- ---------- ---s------ ---------- ---------- ---------- ---------- ----------
     A/Alberta/RV2859/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         A/Kenya/2402/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- -g-------- ---------- ----------
     A/Marseille/1148/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- -ge------- ---------- -----~~~~~
        A/Uganda/0511/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- -g-------- ---------- ----------
       A/Quebec/RV306/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        A/Paraguay/61/2009  ---------- --v------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        A/Finland/514/2009  ---------- ---------- ---------- -----r---- ---------- ---------- ---------- ---------- ---------- --------~~
          A/Moscow/2/2009   ---------- ---------- ---------- ---------- ---------- ------n--- ---------- ---------- ---------- ----------
     A/Hong Kong/1988/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Ghana/FS09718/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- -------p~~
          A/Laos/I053/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Montpellier/2051/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        A/Annecy/2013/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Singapore/35/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- -e-------- ---------- ----------
A/Cayman Islands/5082/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        A/Kentucky/08/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         A/Paris/1704/2009  ---------- ---------- ---------- -------a-- ---------- ---------- ---------- ---------- ---------- ----------
     A/Hong Kong/3192/2008  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Hong Kong/34079/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Hong Kong/18106/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Hong Kong/33597/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Hong Kong/25324/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
                 Consensus  VSSHYSRKFT PEIAKRPKVR DQEGRINYYW TLLEPGDTII FEANGNLIAP RYAFALSRGF GSGIINSNAP MDKCDAKCQT PQGAINSSLP FQNVHPVTIG
```

```
                            301                      326
       A/Brisbane/59/2007   ---------- ---------- ------
    A/Netherlands/345/2007  ---------- ---------- ------
         A/Menya/1386/2009  ~~~~~~~~~~ ~~~~~~~~~~ ~~~~~~
        A/New York/56/2009  ---------- ---------- ------
       A/Wisconsin/13/2009  ---------- ---------- ------
    A/Seychelles/2239/2008  ---------- -----~~~~~ ~~~~~~
       A/Iceland/8599/2009  -~~~~~~~~~ ~~~~~~~~~~ ~~~~~~
        A/England/565/2008  ---------- --~~~~~~~~ ~~~~~~
     A/Alberta/RV2859/2009  ---------- ---------- ------
         A/Kenya/2402/2009  ---------- ---------- ------
     A/Marseille/1148/2009  ~~~~~~~~~~ ~~~~~~~~~~ ~~~~~~
        A/Uganda/0511/2009  ---------- ---------- ------
       A/Quebec/RV306/2009  ---------- ---------- ------
        A/Paraguay/61/2009  ---------- ---------- ------
        A/Finland/514/2009  ~~~~~~~~~~ ~~~~~~~~~~ ~~~~~~
          A/Moscow/2/2009   ---------- ---------- ------
     A/Hong Kong/1988/2009  ---------- ---------- ------
      A/Ghana/FS09718/2009  ~~~~~~~~~~ ~~~~~~~~~~ ~~~~~~
          A/Laos/I053/2009  ---------- ---------- ------
  A/Montpellier/2051/2009   ---------- ---------- ------
        A/Annecy/2013/2009  ---------- ---------- ------
      A/Singapore/35/2009   ---------- ---------- ------
A/Cayman Islands/5082/2009  ---------- ---------- ------
        A/Kentucky/08/2009  ---------- ---------- ------
         A/Paris/1704/2009  ---------- ---------- ------
     A/Hong Kong/3192/2008  ---------- ---------- ------
    A/Hong Kong/34079/2009  ---------- ---------- ------
    A/Hong Kong/18106/2009  ---------- ---------- ------
    A/Hong Kong/33597/2009  ---------- ---------- ------
    A/Hong Kong/25324/2009  ---------- ---------- ------
                 Consensus  ECPKYVRSAK LRMVTGLRNI PSIQSR
```

N-linked glycosylation sites (NXS/T) are highlighted. Losses of such sites are underlined.

Figure 15. Location of AA changes in HA of recent seasonal A(H1N1) viruses

Sites:
131- red
141- blue
185- magenta

Figure 16. Phylogenetic comparison of seasonal A(H1N1) NA genes

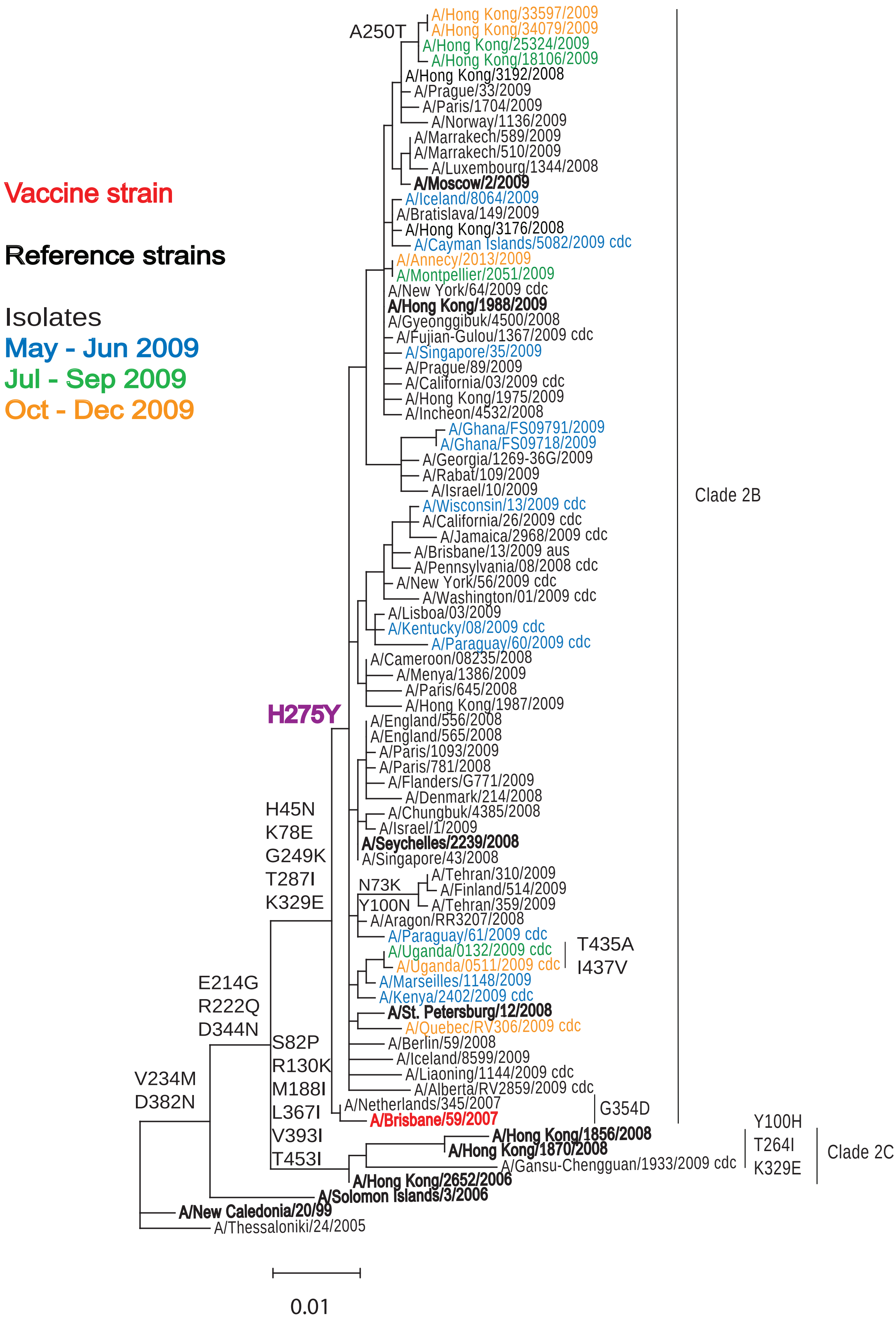

Figure 17: NA protein alignment for seasonal A(H1N1) viruses

```
                         1                                                                                                100
     A/Brisbane/59/2007  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Netherlands/345/2007 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       A/Menya/1368/2009 ~~~~~~~~~~ ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/New York/56/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Wisconsin/13/2009 ---------- ---------- -------s-- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Seychelles/2239/2008 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Iceland/8599/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/England/565/2008 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Alberta/RV2859/2009 ---------- ------m--- ---------- ---------- ---------- ---------- ---------- ---------- --m------- ----------
       A/Kenya/2402/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Marseille/1148/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Uganda/0511/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Quebec/RV306/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- n--------- ----------
      A/Paraguay/61/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Finland/514/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- --k------- ---------- ---------n
         A/Moscow/2/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Hong Kong/1988/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Ghana/FS09718/2009 ---------- ---------- ---------v ---------- ---------- ---------- ---------- ---------- ---------- ----------
 A/Montpellier/2051/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Annecy/2013/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Singapore/35/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/Cayman Islands/5082/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Kentucky/08/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       A/Paris/1704/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Hong Kong/3192/2008 ~~~~------ ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Hong Kong/34079/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Hong Kong/18106/2009 ---------- ---------- ---------- ---------i ---------- ---------- ---------- ---------- ---------- ----------
  A/Hong Kong/33597/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Hong Kong/25324/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
              Consensus  MNPNQKIITI GSISIAIGII SLMLQIGNII SIWASHSIQT GSQNNTGICN QRIITYENST WVNHTYVNIN NTNVVAGEDK TSVTLAGNSS LCSISGWAIY

                         101                                                                                              200
     A/Brisbane/59/2007  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Netherlands/345/2007 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       A/Menya/1368/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/New York/56/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Wisconsin/13/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Seychelles/2239/2008 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Iceland/8599/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/England/565/2008 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Alberta/RV2859/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- -------k-- ----------
       A/Kenya/2402/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Marseille/1148/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Uganda/0511/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Quebec/RV306/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Paraguay/61/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Finland/514/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         A/Moscow/2/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Hong Kong/1988/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Ghana/FS09718/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
 A/Montpellier/2051/2009 ---------- ---------- ---------- ---------- --------i- ---------- ---------- ---------- ---------- ----------
      A/Annecy/2013/2009 ---------- ---------- ---------- ---------- --------i- ---------- ---------- ---------- ---------- ----------
     A/Singapore/35/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/Cayman Islands/5082/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Kentucky/08/2009 ---------- ---------- ---------- ---------- ------r--- ---------- ---------- ---------- ---------- ----------
       A/Paris/1704/2009 ---------- ---------- ---------- ---------- --------i- ---------- ---------- ---------- ---------- ----------
   A/Hong Kong/3192/2008 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Hong Kong/34079/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Hong Kong/18106/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Hong Kong/33597/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Hong Kong/25324/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
              Consensus  TKDNSIRIGS KGDVFVIREP FISCSHLECR TFFLTQGALL NDKHSNGTVK DRSPYRALMS CPLGEAPSPY NSKFESVAWS ASACHDGMGW LTIGISGPDN

                         201                                                 250                      275                     300
     A/Brisbane/59/2007  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----h----- ---------- ----------
  A/Netherlands/345/2007 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----h----- ---------- ----------
       A/Menya/1368/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ------m--- ----------
      A/New York/56/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Wisconsin/13/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Seychelles/2239/2008 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Iceland/8599/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------k ---------- ---------- ----------
      A/England/565/2008 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Alberta/RV2859/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       A/Kenya/2402/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Marseille/1148/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Uganda/0511/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Quebec/RV306/2009 ---------- ---------- ---------- ---------- ---------t ---------- ---------- ---------- ---------- ----------
      A/Paraguay/61/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Finland/514/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         A/Moscow/2/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Hong Kong/1988/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Ghana/FS09718/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
 A/Montpellier/2051/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Annecy/2013/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Singapore/35/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/Cayman Islands/5082/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Kentucky/08/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       A/Paris/1704/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Hong Kong/3192/2008 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Hong Kong/34079/2009 ---------- ---------- ---------- ---------- ---------t ---------- ---------- ---------- ---------- ----------
  A/Hong Kong/18106/2009 ---------- ---------- ---------- ---------- ---------t ---------- ---------- ---------- ---------- ----------
  A/Hong Kong/33597/2009 ---------- ---------- ---------- ---------- ---------t ---------- ---------- ---------- ---------- ----------
  A/Hong Kong/25324/2009 ---------- ---------- ---------- ---------- ---------t ---------- ---------- ---------- ---------- ----------
              Consensus  GAVAVLKYNG IITGTIKSWK KQILRTQESE CVCMNGSCFT IMTDGPSNKA ASYKIFKIEK GKVTKSIELN APNFYYEECS CYPDTGIVMC VCRDNWHGSN
```

Figure 17: NA protein alignment for seasonal A(H1N1) viruses (continued)

```
                            301                                                                                                  400
      A/Brisbane/59/2007    ---------- ---------- ---------- ---------- ---------- ---d------ ---------- ---------- ---------- ----------
  A/Netherlands/345/2007    ---------- ---------- ---------- ---------- ---------- ---d------ ---------- ---------- ---------- ----------
       A/Menya/1368/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/New York/56/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Wisconsin/13/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Seychelles/2239/2008    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Iceland/8599/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/England/565/2008    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Alberta/RV2859/2009    ---------- ---------- --------g- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       A/Kenya/2402/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Marseille/1148/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Uganda/0511/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Quebec/RV306/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Paraguay/61/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Finland/514/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         A/Moscow/2/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Hong Kong/1988/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Ghana/FS09718/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/Montpellier/2051/2009     ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Annecy/2013/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Singapore/35/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/Cayman Islands/5082/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Kentucky/08/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       A/Paris/1704/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Hong Kong/3192/2008    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Hong Kong/34079/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Hong Kong/18106/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Hong Kong/33597/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Hong Kong/25324/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
               Consensus    RPWVSFNQNL DYQIGYICSG VFGDNPRPED GEGSCNPVTV DGANGVKGFS YKYGNGVWIG RTKSNRLRKG FEMIWDPNGW TNTDSDFSVK QDVVAITDWS
```

```
                            401                                                                     470
      A/Brisbane/59/2007    ---------- ---------- ---------- ---------- ---------- ---------- --~~~~~~~~ ~
  A/Netherlands/345/2007    ---------- ---------- ---------- ---------- ---------- ---------- ---------- *
       A/Menya/1368/2009    ---------- --------i- ---------- ---------- ---------- ---------- ---------- *
      A/New York/56/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- *
     A/Wisconsin/13/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- *
  A/Seychelles/2239/2008    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ~
     A/Iceland/8599/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- *
      A/England/565/2008    ---------- ---------- ---------- ---------- ---------- ---------- ---------- *
   A/Alberta/RV2859/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- *
       A/Kenya/2402/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- *
   A/Marseille/1148/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- *
      A/Uganda/0511/2009    ---------- ---------- ---------- ----a-v--- ---------- ---------- ---------- *
     A/Quebec/RV306/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- *
      A/Paraguay/61/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- *
      A/Finland/514/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- *
         A/Moscow/2/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- *
   A/Hong Kong/1988/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- *
    A/Ghana/FS09718/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- *
A/Montpellier/2051/2009     ---------- ---------- ---------- ---------- ---------- ---------- ---------- *
      A/Annecy/2013/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- *
     A/Singapore/35/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- *
A/Cayman Islands/5082/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- *
      A/Kentucky/08/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- *
       A/Paris/1704/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- *
   A/Hong Kong/3192/2008    ---------- ---------- ---------- ---------- ---------- ---------- ---------- *
  A/Hong Kong/34079/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- *
  A/Hong Kong/18106/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- *
  A/Hong Kong/33597/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- *
  A/Hong Kong/25324/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- *
               Consensus    GYSGSFVQHP ELTGLDCIRP CFWVELVRGL PRENTTIWTS GSSISFCGVN SDTANWSWPD GAELPFTIDK –
```

N-linked glycosylation sites (NXS/T) are highlighted.

NA sequence corresponding to the HA1 of A/Laos/I053/2009, shown in Figure 14, was not available.

Table 11. Summary of H3N2 influenza viruses received, collected between September and present

MONTH	H3N2					
Country	received	grown	<2 fold*	4 fold*	>8 fold*	Bris/10/2007
SEPTEMBER 2009						
Cameroon	8	7	4	1	1	1
France	5	5	1	4		
Ghana	3	3	3			
Gibraltar	1	1	1			
Hong Kong	3	3	3			
Sweden	3	3	3			
United Kingdom	3	3	2			1
OCTOBER 2009						
Cameroon	11	11	8	2		1
Finland	2	2	2			
France	1	1		1		
Ghana	1	1	1			
Gibraltar	1	1		1		
Hong Kong	1	1				1
Norway	2	2	2			
NOVEMBER 2009						
Algeria	2	1	1			
Cameroon	15	15	7	7		1
France	7	4	4			
Ghana	1	1	1			
Hong Kong	2	2	2			
Israel	6	6	5	1		
Senegal	15	3	3			
Sweden	2	2	1			1
Turkey	7	3	3			
JANUARY 2010						
Germany	1	1	1			
Turkey	1	1	1			
Total	104	83	59	17	1	6

*** Reduction in HI titre with ferret antiserum raised against vaccine strain (A/Perth/16/2009)**

Table 12. Antigenic analyses of influenza A H3N2 viruses - Guinea Pig RBC - in 20nM Oseltamivir

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret sera A/Wis 67/05 F1/06	A/Bris 10/07 F29/08	A/Uru 716/07 F26/08	A/Fin 9/08 F12/088	A/Jhb 15/08 F22/08	A/HK 1952/09 F22/09	A/HK 1985/09 F21/09	A/Perth 16/09 F25/09	A/Wis 15/09 F24/09
REFERENCE VIRUSES											
A/Wisconsin/67/2005		SpfCk3E3 E7	1280	640	1280	320	320	<	<	<	<
A/Brisbane/10/2007		E2 E3	640	1280	1280	640	320	<	<	<	<
A/Uruguay/716/2007		spfck1, E3 E3	1280	1280	2560	1280	640	<	<	40	<
A/Finland/9/2008		MDCK2 Siat2	1280	1280	1280	640	640	<	80	80	<
A/Johannesburg/15/2008		MDCKx Siat6	1280	1280	1280	1280	2560	40	80	80	40
A/Hong Kong/1952/2009		MDCKx2 Siat7	<	80	80	<	<	160	1280	640	320
A/Hong Kong/1985/2009		MDCKx2 Siat5	<	80	80	<	<	160	1280	640	320
A/Perth/16/2009		E3 E2	<	<	<	<	<	40	320	640	80
A/Wisconsin/15/2009		E2 E2	<	<	<	<	<	40	640	320	160
TEST VIRUSES											
A/Cameroon/445/2009	07/09/2009	Cx Siat1	<	<	80	40	80	<	<	<	<
A/England/700/2009	08/09/2009	MDCK1 Siat1	80	160	320	160	320	<	80	320	<
A/Cameroon/486/2009	13/10/2009	Cx Siat1	<	80	80	40	80	<	<	<	<
A/Hong Kong/33726/2009	18/10/2009	MDCK2 Siat1	80	160	160	80	160	<	40	<	<
A/Paris/3873/09	Sep-09	C1 Siat1	<	80	160	40	40	80	320	160	160
A/Paris/4596/09	Sep-09	C1 Siat1	<	80	160	40	40	80	320	160	80
A/Hong Kong/26554/2009	15/09/2009	MDCK2 Siat1	80	160	160	80	80	640	2560	2560	1280
A/England/815/2009	15/09/2009	Siat1 Siat1	40	160	160	80	80	160	640	320	160
A/Lyon/1715/2009	17/09/2009	MDCK2 Siat2	80	160	80	40	80	40	640	640	80
A/Hong Kong/26227/2009	17/09/2009	MDCK2 Siat1	80	80	80	40	80	320	1280	640	640
A/Hong Kong/26233/2009	17/09/2009	MDCK2 Siat1	80	80	80	40	80	160	1280	640	320
A/Bordeaux/1942/2009	18/09/2009	MDCK2 Siat3	<	160	80	<	40	160	1280	1280	640
A/Gibraltar/SB226/09	22/09/2009	Siat1 Siat1	40	160	160	80	80	80	320	320	160
A/Cameroon/422/2009	28/09/2009	Cx Siat1	<	80	80	<	<	40	80	80	40
A/Paris/5870/09	Oct-09	C1 Siat1	40	<	80	<	<	80	320	160	160
A/Gibraltar/SB252/09	07/10/2009	Siat1 Siat1	40	160	160	40	80	80	320	320	160
A/Paris/6047/2009	Nov-09	C1 Siat2	80	320	320	160	160	320	1280	1280	160
A/Cameroon/663/2009	03/11/2009	Cx Siat1	80	80	320	40	80	80	640	320	160
A/Cameroon/614/2009	05/11/2009	Cx Siat1	<	80	40	<	40	80	320	320	80
A/Cameroon/653/2009	06/11/2009	Cx Siat1	<	80	160	40	40	80	320	320	80
A/Dakar/41/2009	06/11/2009	Siat2	<	40	80	40	40	NT	1280	640	320
A/Dakar/43/2009	06/11/2009	Siat2	<	160	80	40	40	NT	320	320	80
A/Cameroon/658/2009	10/11/2009	Cx Siat1	40	160	160	40	80	80	640	320	160
A/Cameroon/675/2009	11/11/2009	Cx Siat1	<	160	160	40	80	80	640	320	160
A/Dakar/44/2009	11/11/2009	Siat2	<	40	40	<	40	NT	640	640	320
A/Algeria/G46/2009	26/11/2009	MDCK0 Siat2	80	320	160	160	160	320	1280	1280	1280
A/Hong Kong/34194/2009	10/11/2009	MDCK2 Siat1	40	160	80	40	40	160	1280	1280	1280
A/Hong Kong/34430/2009	22/11/2009	MDCK2 Siat1	<	80	80	<	<	160	1280	1280	640
A/Cameroon/598/2009	02/11/2009	Cx Siat1	80	320	320	80	80	160	1280	640	160
A/Cameroon/675/2009	11/11/2009	Cx Siat1	<	160	160	40	80	80	640	320	160
A/Turkey/TR-26/2009	02/12/2009	SIAT1 SIAT:	<	40	80	40	40	NT	2560	1280	640
A/Turkey/TR-28/2009	12/12/2009	SIAT1 SIAT:	<	40	80	<	40	NT	2560	1280	320
A/Turkey/TR-33/2009	05/01/2010	SIAT1 SIAT:	<	40	40	<	<	NT	1280	640	160

1. < = <40

NT = Not tested

Vaccine strain

Table 13. Antigenic analyses of influenza A H3N2 viruses - Guinea Pig RBC

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre[1]										
			Post infection ferret sera										
			A/Wis 67/05 F1/06	A/Bris 10/07 F29/08	A/Uru 716/07 F26/08	A/Fin 9/08 F12/088	A/Jhb 15/08 F22/08	A/HK 1952/09 F22/09	A/HK 1985/09 F21/09	A/Perth 16/09 F25/09	A/Wis 15/09 F24/09	A/HK 26560/09 CDC F38/10	A/Phil 2191/09 CDCF39/10
REFERENCE VIRUSES													
A/Wisconsin/67/2005	31/08/2005	SpfCk3E3 E7	1280	640	1280	640	640	<	<	<	<	160	<
A/Brisbane/10/2007	06/02/2007	E2 E3	2560	2560	2560	1280	1280	40	80	40	40	320	320
A/Uruguay/716/2007	21/06/2007	spfck1, E3 E3	1280	1280	1280	1280	640	<	<	<	<	40	160
A/Finland/9/2008	07/01/2008	MDCK2 Siat2	640	1280	1280	1280	1280	80	320	320	80	320	320
A/Johannesburg/15/2008	25/06/2008	MDCKx Siat6	320	640	640	640	640	80	320	320	80	320	320
A/Hong Kong/1952/2009	24/03/2009	MDCKx2 Siat7	320	320	320	320	320	320	2560	1280	640	2560	2560
A/Hong Kong/1985/2009	01/04/2009	MDCKx2 Siat5	80	160	160	80	80	160	1280	640	320	640	640
A/Perth/16/2009	04/07/2009	E3 E2	<	<	<	<	<	80	640	640	80	640	640
A/Wisconsin/15/2009	06/07/2009	E2 E2	<	40	40	<	<	160	2560	1280	160	1280	1280
A/Hong Kong/26560/2009	18/09/2009	E3 E1	<	<	<	<	<	80	1280	1280	80	1280	1280
A/Philippines/2191/2009	26/05/2009	E3 E1	<	<	<	<	<	80	1280	640	80	640	1280
TEST VIRUSES													
A/Gothenburg/3/2009	27/11/2009	C2 Siat1	160	640	640	320	640	40	80	40	<	160	160
A/Cameroon/699/2009	20/11/2009	Cx Siat1	320	640	640	320	1280	80	160	160	40	320	320
A/Ghana/FS-1551/2009	14/09/2009	Cx Siat1	80	160	160	80	160	320	2560	1280	640	2560	2560
A/Ghana/FS-1589/2009	22/09/2009	Cx Siat1	80	160	160	80	160	640	2560	2560	1280	2560	2560
A/Ghana/FS-1591/2009	22/09/2009	Cx Siat1	<	80	160	40	40	320	1280	1280	640	2560	2560
A/Sweden/2/2009	01/09/2009	C1 Siat1	80	160	160	160	160	320	2560	1280	640	2560	2560
A/Stockholm/89/2009	06/09/2009	C1 Siat1	40	160	160	80	80	80	320	640	160	320	640
A/Umeå/4/2009	02/09/2009	C2 Siat1	40	160	160	80	160	160	640	1280	320	1280	1280
A/Norway/3789/2009	12/10/2009	MDCK1 Siat1	<	40	40	40	40	40	80	160	40	40	80
A/Norway/3790/2009	13/10/2009	MDCK1 Siat1	<	80	80	40	40	40	160	320	40	80	80
A/Finland/638/2009	20/10/2009	MDCK3 Siat1	40	80	80	40	40	40	160	320	80	80	160
A/Finland/640/2009	26/10/2009	MDCK3 Siat1	40	160	160	80	80	40	320	320	80	160	160
A/Cameroon/677/2009	13/11/2009	Cx Siat1	80	320	320	160	160	320	1280	1280	320	1280	1280
A/Cameroon/678/2009	13/11/2009	Cx Siat1	40	160	160	40	80	80	320	320	80	160	160
A/Cameroon/681/2009	13/11/2009	Cx Siat1	160	320	320	160	160	160	1280	640	320	1280	640
A/Cameroon/701/2009	23/11/2009	Cx Siat1	40	160	160	40	40	80	320	320	80	320	320
A/Cameroon/704/2009	24/11/2009	Cx Siat1	80	160	160	80	80	80	640	1280	160	320	320
A/Stockholm/112/2009	28/11/2009	C1 Siat1	80	320	320	160	160	320	1280	1280	320	2560	2560
A/Israel/22/2009	Nov-09	Cx Siat1	40	80	80	40	80	40	160	320	80	160	160
A/Israel/23/2009	Nov-09	Cx Siat1	40	160	80	40	80	40	160	320	80	160	160
A/Israel/24/2009	Nov-09	P2 Siat1	40	160	80	40	80	80	160	320	80	160	160
A/Israel/25/2009	Nov-09	P2 Siat1	<	80	80	40	40	40	160	160	80	80	160
A/Israel/26/2009	Nov-09	P2 Siat1	40	80	80	40	80	80	160	320	80	160	160
A/Israel/27/2009	Nov-09	Cx Siat1	40	160	80	80	80	80	320	320	80	160	160
A/Ghana/FS-2240/2009	25/11/2009	Cx Siat1	40	160	160	80	80	320	2560	2560	640	2560	2560
A/Nordrhein-Westfalen/1/2010	Jan-10	Siat1 Siat1	80	160	160	160	80	640	2560	2560	1280	2560	2560
A/Lyon/Cx-R/3120/2009	Nov-09	p2MDCK Siat1	40	80	80	40	40	40	160	320	80	160	160
A/Lyon/3670/2009	Nov-09	p2MDCK Siat1	80	160	160	80	160	80	640	640	320	640	640
A/Lyon/CHU/52.384/09	Nov-09	p2MDCK Siat1	160	320	320	160	160	160	1280	1280	320	1280	640
A/Ghana/FS-2355/2009	03/12/2009	Cx Siat1	80	160	160	80	80	320	1280	1280	640	2560	2560

1. < = <40

Vaccine strain

Figure 18. Phylogenetic comparison of A(H3N2) HA genes

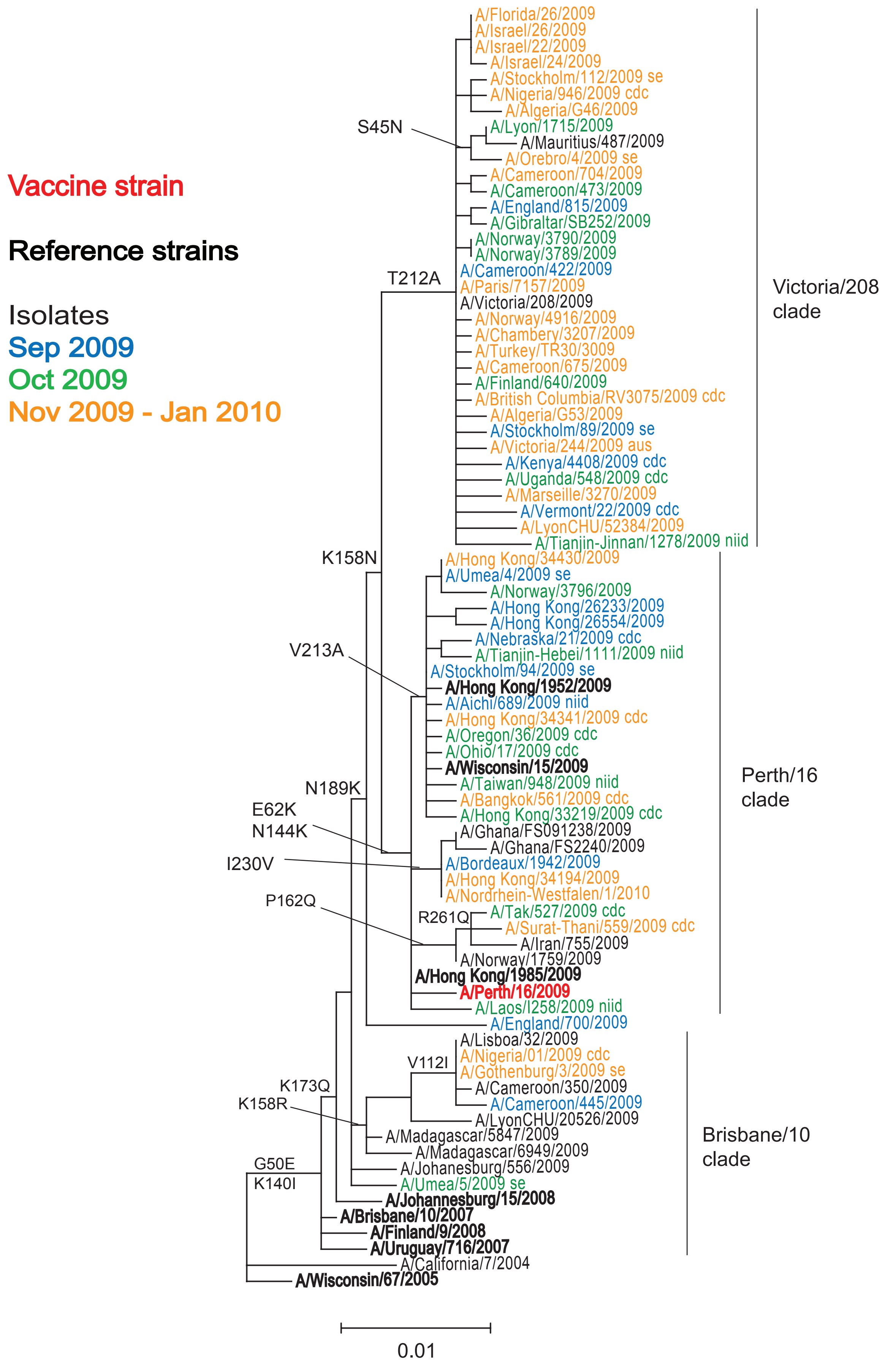

Figure 19: HA1 protein alignment for A(H3N2) viruses

```
                          1                                            45   50            62                                         100
      A/Wisconsin/67/2005 ---------- ---------- ---------- ---------- ---------g ---------- ---------- ---------- ---------- ----------
      A/Uruguay/716/2007  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
           A/Umea/5/2009  ---------m ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Cameroon/445/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Gothenburg/3/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/England/700/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         A/Perth/16/2009  ---------- ---------- ---------- ---------- ---------- ---------- -k-------- ---------- ---------- ----------
         A/Iran/755/2009  ---------- ---------- ---------- ---------- ---------- ---------- -k-------- ---------- ---------- ----------
 A/Surat-Thani/559/2009   ---------- ---------- ---------- ---------- ---------- ---------- -k-------- -------s-- ---------- ----------
  A/Hong Kong/34194/2009  ---------- ---------- ---------- ---------- ---------- ---------- -k-------- ---------- ---------- ----------
    A/Bordeaux/1942/2009  ---------- ---------- ---------- ---------- ---------- ---------- -k-------- ---------- ---------- ----------
     A/Ghana/FS2240/2009  ---------- ---------- ---------- ---------- ---------- ---------- -k-------- ---------- ---------- ----------
       A/Taiwan/948/2009  ---------- ---------- ---------- ---------- ---------- ---------- -k-------- ---------- ---------- ----------
        A/Oregon/36/2009  ---------- ---------- ---------- ---------- ---------- ---------- -k-------- ---------- ---------- ----------
     A/Stockholm/94/2009  ---------- ---------- ---------- ---------- ---------- ---------- -k-------- ---------- ---------- ----------
  A/Hong Kong/26554/2009  ---------- ---------- ---------- ---------- ---------- ---------- -k-------- ---------- ---------- ----------
      A/Norway/3796/2009  ---------- ---------- ---------- ---------- ---------- ---------- -k-------- ---------- -----p---- ----------
  A/Hong Kong/34430/2009  ---------- ---------- ---------- ---------- ---------- ---------- -k-------- ---------- ---------- ----------
A/Tianjin-Jinnan/1278/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/LyonCHU/52384/2009  ---------- ---------- s--------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Marseille/3270/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Victoria/244/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Cameroon/675/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Victoria/208/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Cameroon/422/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/England/815/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        A/Orebro/4/2009   ---------- ---------- ---------- ---------- ----n----- ---------- ---------- ---------- ---------- ----------
      A/Algeria/G46/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        A/Israel/24/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       A/Florida/26/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
               Consensus  QKLPGNDNST ATLCLGHHAV PNGTIVKTIT NDQIEVTNAT ELVQSSSTGE ICDSPHQILD GENCTLIDAL LGDPQCDGFQ NKKWDLFVER SKAYSNCYPY
```

```
                          101         112                              140  144            158  162          173               189         200
      A/Wisconsin/67/2005 ---------- ---------- -d-------- ---------k ---------- -------k-- ---------- --k------- --------n- ----h-----
      A/Uruguay/716/2007  ---------- ---------- ---------- -------s-- ---------- -------k-- ---------- --k------- --------n- ---p------
           A/Umea/5/2009  ---------- ---------- ---------- ---------- ---------- -------k-- ---------- ---------- --------n- ----------
     A/Cameroon/445/2009  ---------- -i-------- ---------- ---------- ---------- -------r-- ---------- ---------- --------n- ----------
     A/Gothenburg/3/2009  ---------- -i-------- ---------- ---------- ---------- -------r-- ---------- ---------- --------n- ----------
      A/England/700/2009  ---------- ---------- x--------- -------s-x -x-------- -------k-- x--------- x--------- --f------- ----------
         A/Perth/16/2009  ---------- ---------- ---------- ---------- ---k------ ---------- ---------- ---------- --l------- ----------
         A/Iran/755/2009  ---------- ---------- ---------- ---------- ---k------ ---------- ---------- ---------- ---------- ----------
 A/Surat-Thani/559/2009   ---------- ---------- ---------- ---------- ---k------ ---------- -q-------- ---------- ---------- ----------
  A/Hong Kong/34194/2009  ---------- ---------- ---------- ---------- ---k------ ---------- ---------- ---------- ---------- ----------
    A/Bordeaux/1942/2009  ---------- ---------- ---------- ---------- ---k------ ---------- ---------- ---------- ---------- ----------
     A/Ghana/FS2240/2009  ---------- ---------- ---------- ---------- ---k------ ---------- ---------- ---------- ---------- ----------
       A/Taiwan/948/2009  ---------- ---------- ---------- ---------- ---k------ ---------- ---------- ---------- ---------- ----------
        A/Oregon/36/2009  ---------- ---------- ---------- ---------- ---k------ ---------- ---------- ---------- ---------- ----------
     A/Stockholm/94/2009  ---------- ---------- ---------- ---------- ---k------ ---------- ---------- ---------- ---------- ----------
  A/Hong Kong/26554/2009  ---------- ---------- ---------- ---------- ---k------ ---------- ---------- ---------- ---------- ----------
      A/Norway/3796/2009  ---------- ---------- ---------- ---------- ---k----k- ---------- ---------- ---------- ---------- ----------
  A/Hong Kong/34430/2009  ---------- ---------- ---------- ---------- ---k------ ---------- ---------- ---------- ---------- ----------
A/Tianjin-Jinnan/1278/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/LyonCHU/52384/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Marseille/3270/2009  ---------- ---------- ---------- ---------- ---------- ------s--- ---------- ---------- ---------- ----------
     A/Victoria/244/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Cameroon/675/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Victoria/208/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Cameroon/422/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/England/815/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- -i-------- ----------
        A/Orebro/4/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Algeria/G46/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        A/Israel/24/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       A/Florida/26/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
               Consensus  DVPDYASLRS LVASSGTLEF NNESFNWTGV TQNGTSSACI RRSNNSFFSR LNWLTHLNFK YPALNVTMPN NEQFDKLYIW GVHHPGTDKD QIFLYAQASG
```

Figure 19: HA1 protein alignment for A(H3N2) viruses (continued)

```
                           201       212213               230                                321                                       300
```

```
                           201       212213               230                                261                                       300
       A/Wisconsin/67/2005 ---------- ---------- --i------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        A/Uruguay/716/2007 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
             A/Umea/5/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       A/Cameroon/445/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----d----- ---------- ----------
       A/Gothenburg/3/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        A/England/700/2009 ---------- ---------- --x------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
           A/Perth/16/2009 ---------- ---s------ ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
           A/Iran/755/2009 ---------- ---------- ---------- ---------- ---------- ---------m q--------- ---------- ---------- ----------
    A/Surat-Thani/559/2009 ---------- ---------- ---------- ---------- ---------- ---------- q--------- ---------- ---------- ----------
    A/Hong Kong/34194/2009 ---------- ---------- ---------v ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Bordeaux/1942/2009 ---------- ---------- ---------v ---------- ---------- ---------- ---------- ---------- ---------- ----------
       A/Ghana/FS2240/2009 ---------- ---------- ---------v ---------- ---------- ---------- ---------- ---------- ---------- ----------
         A/Taiwan/948/2009 ---------- --a------- ---------- ---------- ---------- ---------- ---------- -t-------- ---------- ----------
          A/Oregon/36/2009 ---------- --a-----y- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       A/Stockholm/94/2009 ---------- --a------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Hong Kong/26554/2009 ---------- --a------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        A/Norway/3796/2009 ---------- --a------- ---------- -------e-- ---------- ---------- ---------- ---------- ---------- ----------
    A/Hong Kong/34430/2009 ---------- --a------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/Tianjin-Jinnan/1278/2009 ---------- -a-------- x--------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/LyonCHU/52384/2009 ---------- -a-------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Marseille/3270/2009 ---------- -a-------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       A/Victoria/244/2009 ---------- -a-------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       A/Cameroon/675/2009 ---------- -a-------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       A/Victoria/208/2009 ---------- -a-------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       A/Cameroon/422/2009 ---------- -a-------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        A/England/815/2009 ---------- -a-------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
           A/Orebro/4/2009 ---------- -a-------- x--------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        A/Algeria/G46/2009 ---------- -a-------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
          A/Israel/24/2009 ---------- -a-------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         A/Florida/26/2009 ---------- -a-------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
                 Consensus RITVSTKRSQ QTVIPNIGSR PRVRNIPSRI SIYWTIVKPG DILLINSTGN LIAPRGYFKI RSGKSSIMRS DAPIGKCNSE CITPNGSIPN DKPFQNVNRI
```

```
                           301                            329
       A/Wisconsin/67/2005 ---------- ---------- ---------
        A/Uruguay/716/2007 ---------- ---------- ---------
             A/Umea/5/2009 ---------- ---------- ---------
       A/Cameroon/445/2009 ---------- ---------- ---------
       A/Gothenburg/3/2009 ---------- ---------- ---------
        A/England/700/2009 ---------- ---------- ---------
           A/Perth/16/2009 ---------- ---------- ---------
           A/Iran/755/2009 ---------- ---------- ---------
    A/Surat-Thani/559/2009 ---------- ---------- ---------
    A/Hong Kong/34194/2009 ---------- ---------- ---------
      A/Bordeaux/1942/2009 ---------- ---------- ---------
       A/Ghana/FS2240/2009 ---------- ---------- ---------
         A/Taiwan/948/2009 ---------- ---------- ---------
          A/Oregon/36/2009 ---------- ---------- ---------
       A/Stockholm/94/2009 --------i- ---------- ---------
    A/Hong Kong/26554/2009 ---------- ---------- ----x----
        A/Norway/3796/2009 ---------- ---------- ---------
    A/Hong Kong/34430/2009 ---------- ---------- ---------
A/Tianjin-Jinnan/1278/2009 ---------- ---------- ---------
      A/LyonCHU/52384/2009 ---------- -s-------- ---------
     A/Marseille/3270/2009 ---------- -k-------- ---------
       A/Victoria/244/2009 ---------- ---------- ---------
       A/Cameroon/675/2009 ---------- ---------- ---------
       A/Victoria/208/2009 ---------- ---------- ---------
       A/Cameroon/422/2009 ---------- ---------- ---------
        A/England/815/2009 ---------- ---------- ---------
           A/Orebro/4/2009 ---------- ---------- ---------
        A/Algeria/G46/2009 ---------- ---------- ---------
          A/Israel/24/2009 ---------- ---------- ---------
         A/Florida/26/2009 ---------- ---------- ---------
                 Consensus TYGACPRYVK QNTLKLATGM RNVPEKQTR
```

N-linked glycosylation sites (NXS/T) are highlighted. Losses of such sites are <u>underlined</u>.

Figure 20. Location of AA changes in HA of A(H3N2) that define A/Perth/16/2009-like viruses compared with A/Brisbane/10/2007

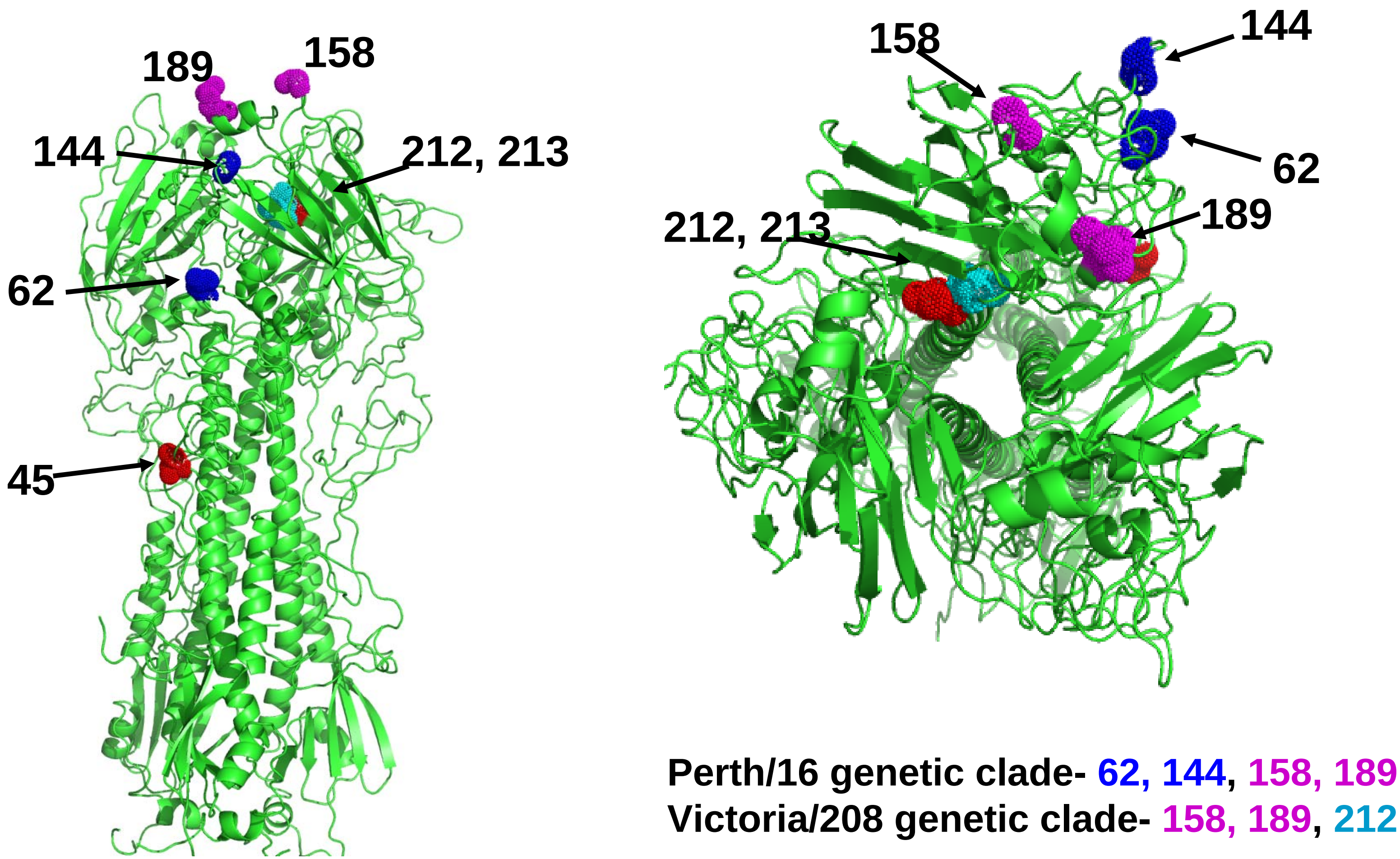

Figure 21. Phylogenetic comparison of H3N2 NA genes

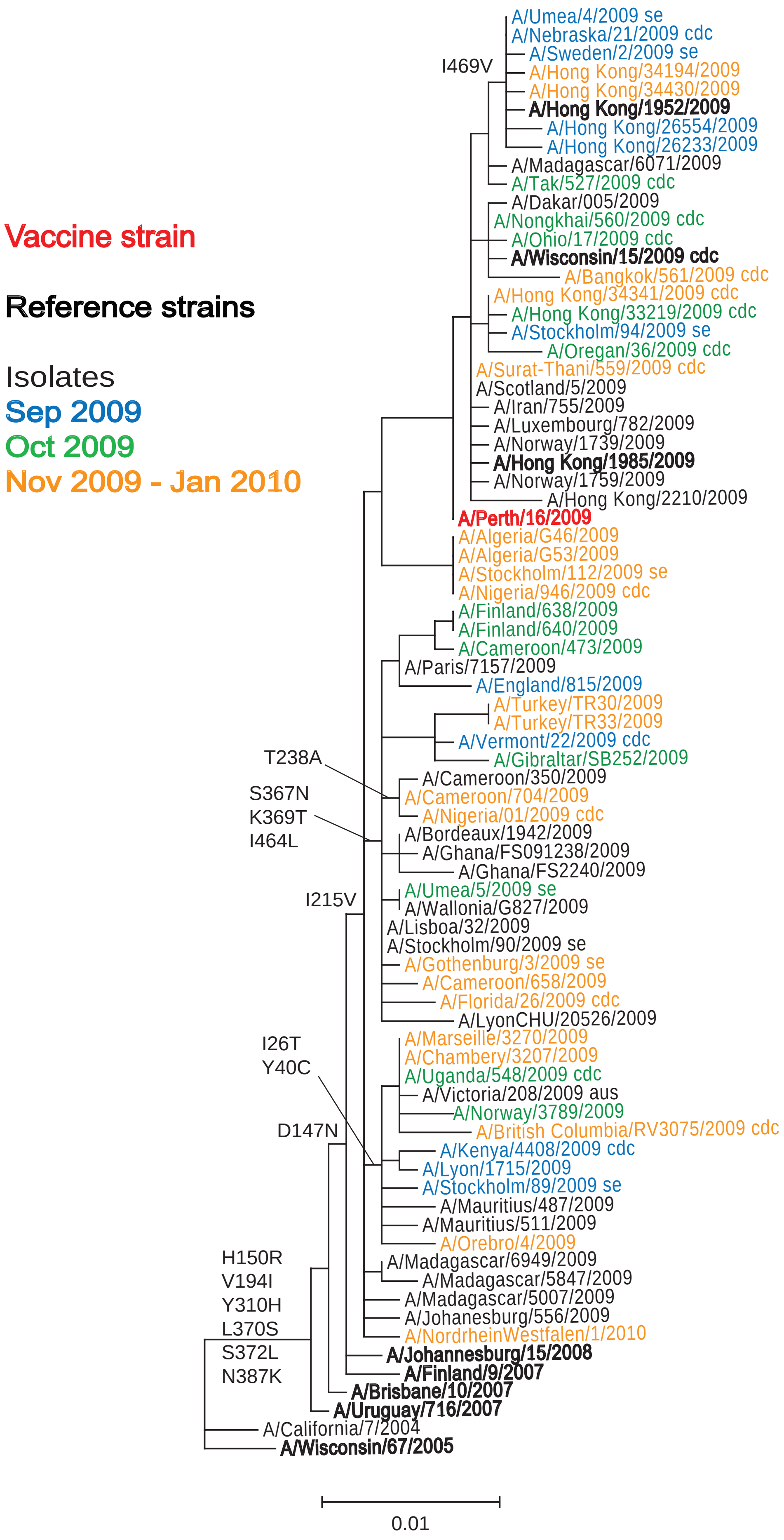

Figure 22: NA protein alignment for A(H3N2) viruses

```
                        1                          26            40                                                                  100
  A/Wisconsin/67/2005   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- --n-------
  A/Uruguay/716/2007    ---------- ---------- -----t---- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Umea/5/2009     ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Gothenburg/3/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Perth/16/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Iran/755/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/Surat-Thani/559/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/Hong Kong/34194/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
 A/Bordeaux/1942/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Ghana/FS2240/2009   ~~~~~~~~~~ ~~~~~----- -----t---- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Oregon/36/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Stockholm/94/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/Hong Kong/26554/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/Hong Kong/34430/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
 A/Marseille/3270/2009  ~~~~~~~~~~ ~~~~~~~~~~ ~~~~~~~~~~ ~~~~~~~~~~ ~~~~~----- ---------- ---------- ---------- ---------- ----------
  A/Cameroon/675/2009   ~~~~~~~~~~ ~~~~~~~~~~ ~~~~~~~~~~ ~~~~~~~~~~ ~~~~~~~~~~ ~~~~~~~~~~ ~~~~~~~~~~ ~~~~~~~~~~ ~~~~~~~~~~ ~~~~~~~~~~
  A/Victoria/208/2009   ---------- ---------- -----t---- ---------c ---------- ---------- ---------- ---------- ---------- ----------
  A/Cameroon/422/2009   ~~~~~~~~~~ ~~~~~~~~~~ ~~~~~~~~~~ ~~~~~~~~~~ ~~~~~~~~~~ ~~~~~~~~~~ ~~~~~~---- ---------- ---------- ----------
   A/England/815/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Orebro/4/2009    ---------- ---------- -----t---- ---------c ---------- ---------- ---------- ---------- ---------- ----------
   A/Algeria/G46/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Florida/26/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
            Consensus   MNPNQKIITI GSVSLTISTI CFFMQIAILI TTVTLHFKQY EFNSPPNNQV MLCEPTIIER NITEIVYLTN TTIEKEICPK LAEYRNWSKP QCDITGFAPF

                        101                                                147  151                                           200
  A/Wisconsin/67/2005   ---------- ---------- ---------- ---------- ------d--h ---------- ---------- ---------- ---------- ---v------
  A/Uruguay/716/2007    ---------- ---------- ---------- ---------- ------d--h ---------- ---------- ---------- ---------- ----------
      A/Umea/5/2009     ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Gothenburg/3/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Perth/16/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Iran/755/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/Surat-Thani/559/2009  ---------- ---------- ---------- ---------- -------x-- ---------- ---------- ---------- ---------- ----------
A/Hong Kong/34194/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
 A/Bordeaux/1942/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Ghana/FS2240/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Oregon/36/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Stockholm/94/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/Hong Kong/26554/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/Hong Kong/34430/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
 A/Marseille/3270/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Cameroon/675/2009   ~~~~~~---- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Victoria/208/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Cameroon/422/2009   ---------- ---------- ---------- ---------- ---------- x--------- ---------- ---------- ---------- ----------
   A/England/815/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Orebro/4/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/Algeria/G46/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Florida/26/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
            Consensus   SKDNSIRLSA GGDIWVTREP YVSCDPDKCY QFALGQGTTL NNVHSNNTVR DRTPYRTLLM NELGVPFHLG TKQVCIAWSS SSCHDGKAWL HVCITGDDKN

                        201            215                                                                                    300
  A/Wisconsin/67/2005   ---------- ----i----- ---------- ---------- ---------- ---------- ---------- ---------- ----l----- ----------
  A/Uruguay/716/2007    ---------- ----i----- ---------- ---------- ---------- ---------- ------i--- ---------- ---------- ----------
      A/Umea/5/2009     ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Gothenburg/3/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Perth/16/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     A/Iran/755/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/Surat-Thani/559/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/Hong Kong/34194/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
 A/Bordeaux/1942/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Ghana/FS2240/2009   ---------- ---------- ---------- ---------- ---------- ------v--- ---------- ---------- ---------- ----------
     A/Oregon/36/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Stockholm/94/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/Hong Kong/26554/2009  ---------- ---------- ---------- ---------- ---------- ---------- -------m-- ---------- ---------- ----------
A/Hong Kong/34430/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
 A/Marseille/3270/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Cameroon/675/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Victoria/208/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
  A/Cameroon/422/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
   A/England/815/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- -----r----
     A/Orebro/4/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- --q------- ----------
   A/Algeria/G46/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Florida/26/2009   ---------- ---------- ---------- ---------- ---------- ---------- ------k--- ---------- ---------- ----------
            Consensus   ATASFIYNGR LVDSVVSWSK EILRTQESEC VCINGTCTVV MTDGSASGKA DTKILFIEEG KIVHTSTLSG SAQHVEECSC YPRYPGVRCV CRDNWKGSNR
```

Figure 22: NA protein alignment for A(H3N2) viruses (continued)

```
                     301                                                                     367 369                                400
 A/Wisconsin/67/2005 ---------y ---------- ---------- ---------- ---------- ---------- ---------l -s-------- ------n--- ----------
 A/Uruguay/716/2007  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      A/Umea/5/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- -----h---- ----------
A/Gothenburg/3/2009  ---------- ---------- ---------- ---------- ---------- ---------- ------n-t- ---------- ---------- ----------
    A/Perth/16/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
    A/Iran/755/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/Surat-Thani/559/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/Hong Kong/34194/2009 ---------- -v-------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/Bordeaux/1942/2009 ---------- ---------- ---------- ---------- ---------- ---------- ------n-t- ---------- ---------- ----------
A/Ghana/FS2240/2009  ---------- ---------- ---------- ---------- ---------- ---------- ------n-t- ---------- ---------- ----------
  A/Oregon/36/2009   ---------- ---------- -------r-- --------g- -----d---- ---------- ---------- ---------- ---------- ----------
A/Stockholm/94/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/Hong Kong/26554/2009 ---------- ---------- ---------- -----n---- ---------- ---------- ---------- ---------- ---------- -----i----
A/Hong Kong/34430/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/Marseille/3270/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/Cameroon/675/2009  ---------- ---------- ---------- ---------- ---------- ---------- ------n-t- ---------- ---------- ----------
A/Victoria/208/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/Cameroon/422/2009  ---------- ---------- ---------- ---------- ---------- ---------- ------n-t- ---------- ---------- ----------
A/England/815/2009   ---------- ---------- ---------- ---------- ---------- ---------- --~~~~~~~~ ~~~~~~~~~~ ~~~~~~~~~~ ~~~~~~~~~~
   A/Orebro/4/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
A/Algeria/G46/2009   ---------- ---------- ---------- ---------- ---------- ---------- ------n-t- ---------- ---------- ----------
A/Florida/26/2009    ---------- ---------- ---------- ---------- ---------- ---------- ------n-t- ---------- ---------- ----------
          Consensus  PIVDINIKDH SIVSSYVCSG LVGDTPRKND SSSSSHCLDP NNEEGGHGVK GWAFDDGNDV WMGRTISEKS RLGYETFKVI EGWSNPKSKL QINRQVIVDR

                     400                                                                    464  469
 A/Wisconsin/67/2005 ---------- ---------- ---------- ---------- ---------- ---------- ---------*
 A/Uruguay/716/2007  ---------- ---------- ---------- ---------- ---------- ---------- ---------*
      A/Umea/5/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---l-----*
A/Gothenburg/3/2009  ----a----- ---------- ---------- ---------- ---------- ---------- ---l-----*
    A/Perth/16/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------*
    A/Iran/755/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------*
A/Surat-Thani/559/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------*
A/Hong Kong/34194/2009 ---------- ---------- ---------- ---------- ---------- ---------- --------v*
A/Bordeaux/1942/2009 ---------- ---s------ ---------- ---------- ---------- ---------- ---l-----*
A/Ghana/FS2240/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---l-----*
  A/Oregon/36/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------*
A/Stockholm/94/2009  ---------- ---------- ---------k ---------- ---------- ---------- ---------*
A/Hong Kong/26554/2009 ---------- ---------- ---------- ---------- ---------- ---------- --------v*
A/Hong Kong/34430/2009 ---------- ---------- ---------- ---------- ---------- ---------- --------v*
A/Marseille/3270/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------*
A/Cameroon/675/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---l----t*
A/Victoria/208/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------*
A/Cameroon/422/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---l-----*
A/England/815/2009   ~~~~~~~~~~ ~~~~~~~~~~ ~~~~~~~~~~ ~~~~~~~~~~ ~~~~~~~~~~ ~~~~~~~~~~ ~~~~~~~~~~
   A/Orebro/4/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------*
A/Algeria/G46/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---l-----*
A/Florida/26/2009    ---------- ---------- ---------- ---------- ---------- ---------- ---l-----*
          Consensus  GNRSGYSGIF SVEGKSCINR CFYVELIRGR KEETEVLWTS NSIVVFCGTS GTYGTGSWPD GADINLMPI-
```

N-linked glycosylation sites (NXS/T) are highlighted. Losses of such sites are underlined.

NA sequences corresponding to eight of the H3 HA1 shown in Figure 19 were not available.

Table 14. Summary of B influenza viruses received, collected between September and present

MONTH	B									
	Yamagata					Victoria				
Country	received	grown	≤2 fold*	4 fold*	≥8 fold*	received	grown	≤2 fold*	4 fold*	≥8 fold*
SEPTEMBER 2009										
Ghana						1	1			1
Norway						1	0			
OCTOBER 2009										
Hong Kong						1	1			1
Madagascar						2	0			
NOVEMBER 2009										
Hong Kong	1	1	1							
Sweden						1	1			1
Turkey	4	4		4						
DECEMBER 2009										
United Kingdom						1	1			1
JANUARY 2010										
Sweden						1	1			1
Total = 7	**5**	**5**	**1**	**4**		**8**	**5**			**5**

*** Reduction in HI titre with ferret antiserum raised against vaccine strain**

Table 15. Antigenic analyses of influenza B viruses (Victoria lineage)

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre[1]								
				Post infection ferret sera							
			B/Shan[2] 7/97	B/Bris 32/02 F2/03	B/HK 45/05 F15/05	B/Mal 2506/04 F28/05	B/Vic 304/06 F16/06	B/Eng 393/08 F31/08	B/Bris 33/08 F1/09	B/Bris 60/08 F2/09	B/Paris 1762/08 F11/09
REFERENCE VIRUSES											
B/Shandong/7/97		Ex E1	1280	160	40	80	80	<	<	40	<
B/Brisbane/32/2002		E2 E7	640	160	80	160	80	20	10	40	<
B/Hong Kong/45/2005		E2 E7	640	160	80	40	40	10	<	20	<
B/Malaysia/2506/2004		E3 E4	640	320	80	160	80	20	10	40	<
B/Victoria/304/2006		E2 E3	1280	320	80	160	160	40	20	80	<
B/England/393/2008		E1 E2	160	80	<	80	160	80	80	320	20
B/Brisbane/33/2008		E3 E5	80	160	<	80	160	80	160	320	20
B/Brisbane/60/2008		E4 E2	320	160	10	80	20	160	160	320	20
B/Paris/1762/2008		C2 Siat2	640	<	<	<	<	80	<	20	40
TEST VIRUSES											
B/Laos/1232/2009	15/07/2009	CK1+1 SIAT1	1280	<	40	<	<	<	<	<	<
B/Madagascar/6945/2009	24/08/2009	MDCK1 SIAT1	160	<	<	<	<	<	<	40	10
B/Madagascar/7002/2009	31/08/2009	MDCK1 SIAT1	160	<	<	<	<	<	<	20	10
B/Ghana/FS-1900/2009	11/09/2009	MDCK1 SIAT1	640	<	<	<	<	<	<	10	40
B/Hong Kong/514/2009	11/10/2009	MDCK2 SIAT1	320	<	<	<	<	10	<	20	40
B/Stockholm/8/2009	27/11/2009	C1 SIAT1	640	<	<	<	<	<	<	20	40
B/England/928/2009	22/12/2009	SIAT1 SIAT1	640	<	<	<	<	<	<	10	20
B/Stockholm/1/2010	01/01/2010	C2 SIAT1	640	<	<	<	<	<	<	20	40

1. < = <10; 2. hyperimmune sheep serum

Vaccine strain

Figure 23. Phylogenetic comparison of influenza B HA genes (Victoria lineage)

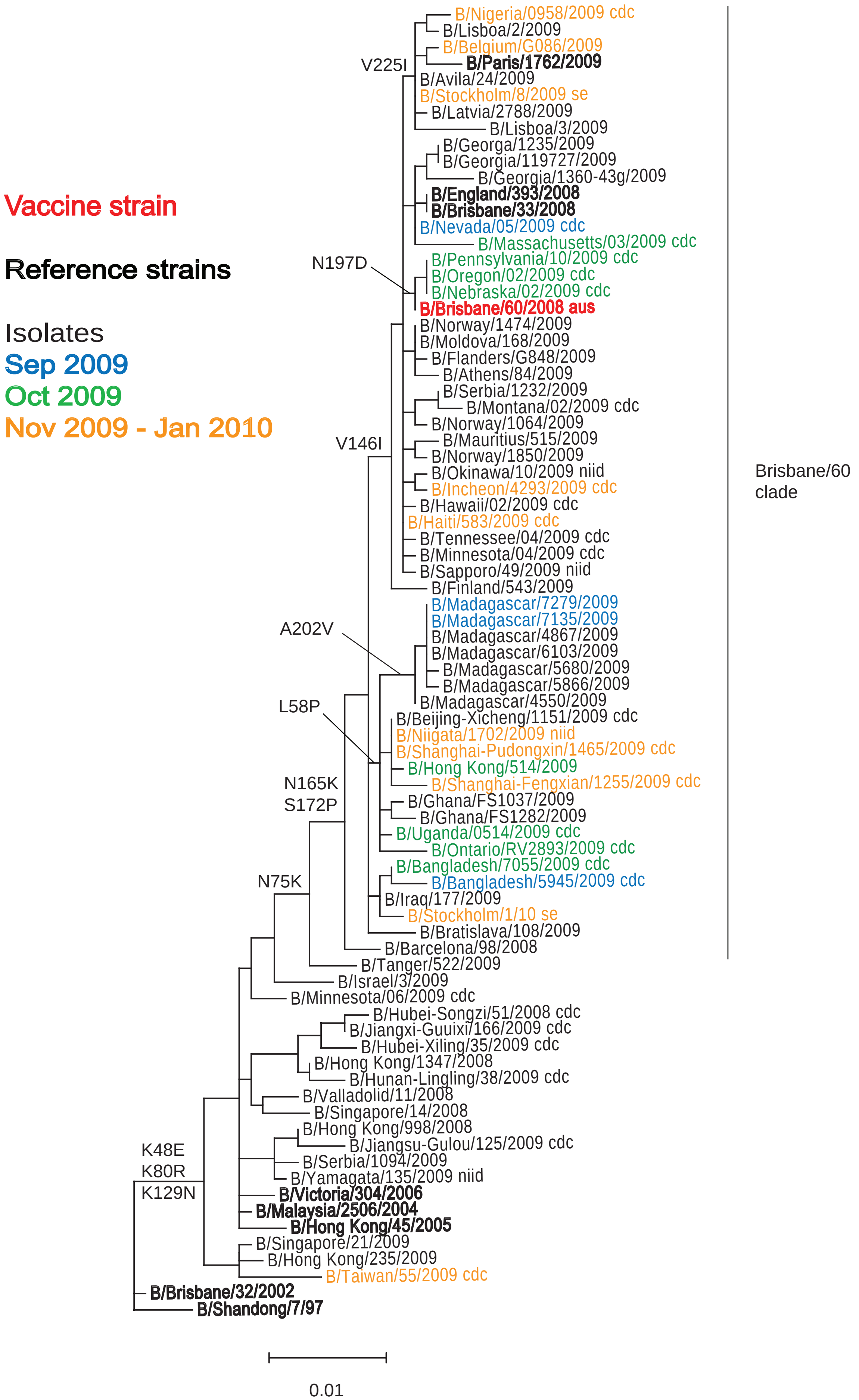

Figure 24: HA1 protein alignment for B-Victoria lineage viruses

```
                                   1                                                              58               75                     100
           B/Malaysia/2506/2004    ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----n----- ---------- ----------
               B/Tanger/522/2009   ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
           B/Barcelona/98/2008     ---------- ---------- ---------- ---------- ---------- -------p-- ---------- ---------- ---------- ----------
            B/Stockholm/1/2010     ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        B/Bangladesh/5945/2009     ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        B/Bangladesh/7055/2009     ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
            B/Uganda/0514/2009     ---------- ---------- ---------- ---------- ---------- -------p-- ---------- ---------- ---------- ----------
          B/Ghana/FS1282/2009      ---------- ---------- ---------- ---------- ---------- -------p-- ---------- ---------- ---------- ----------
 B/Shanghai-Fengxian/1255/2009     ---------- ---------- ---------- ---------- ---------- -------p-- ---------- ---------- ---------- ----------
          B/Hong Kong/514/2009     ---------- ---------- ---------- ---------- ---------- -------p-- ---------- ---------- ---------- ----------
B/Shanghai-Pudongxin/1465/2009     ---------- ---------- ---------- ---------- ---------- -------p-- ---------- ---------- ---------- ----------
           B/Niigata/1702/2009     ---------- ---------- ---------- ---------- ---------- -------p-- ---------- ---------- ---------- ----------
        B/Madagascar/7135/2009     ---------- ---------- ---------- ---------- ---------- -------p-- ---------- ---------- ---------- ----------
        B/Madagascar/7279/2009     ---------- ---------- ---------- ---------- ---------- -------p-- ---------- ---------- ---------- ----------
            B/Sapporo/49/2009      ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
             B/Haiti/583/2009      ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
          B/Incheon/4293/2009      ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
            B/Okinawa/10/2009      ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
          B/Norway/1064/2009       ~--------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
          B/Serbia/1232/2009       ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
             B/Athens/84/2009      ---------- ---------- ---------- ---------- ---------- -------p-- ---------- ---------- ---------- ----------
          B/Moldova/168/2009       ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
          B/Brisbane/60/2008       ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
             B/Oregon/02/2009      ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     B/Massachusetts/03/2009       ---------- ---------- ---------- ---------- ---------- -------f-- ---------- ---------g ---------- ----------
             B/Nevada/05/2009      ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
          B/England/393/2008       ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
          B/Stockholm/8/2009       ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         B/Belgium/G086/2009       ---------- ---------- ---------- --------t- ---------- ---------- ---------- ---------- ---------- ----------
         B/Nigeria/0958/2009       ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
                   Consensus       DRICTGITSS NSPHVVKTAT QGEVNVTGVI PLTTTPTKSH FANLKGTETR GKLCPKCLNC TDLDVALGRP KCTGKIPSAR VSILHEVRPV TSGCFPIMHD
```

```
                                   101                                                 146                   165      172                  197200
           B/Malaysia/2506/2004    --------k- ---------- ---------- ---------- -----v---- ---------- ----n----- -s-------- ---------- ----------
               B/Tanger/522/2009   ---------- ---------- ---------- ---------- -----v---- ---------- ----n----- -s-------- ---------- ----------
           B/Barcelona/98/2008     ---------- ---------- ---------- ---------- -----v---- ---------- ---------- ---------- ---------- ----------
            B/Stockholm/1/2010     ---------- ---------- ---------- ---------- -----v---- ---------- ---------- ---------- ---------- ----------
        B/Bangladesh/5945/2009     ---------- ---------- -n-------- ---------- -----v---- ---------- ---------- ---------- ---------- --------i-
        B/Bangladesh/7055/2009     ---------- ---------- ---------- ---------- -----v---- ---------- ---------- ---------- ---------- ----------
            B/Uganda/0514/2009     ---------- ---------- ---------- ---------- -----v---- ---------- ---------- ---------- ---------- ----------
          B/Ghana/FS1282/2009      ---------- ---------- ---------- ---------- -----v---- ---------- ---------- ---------- ---------- ----------
 B/Shanghai-Fengxian/1255/2009     ---------- ---------- ---------- ---------- -----v---- ---------- ---------- ---------- ---------- --------x-
          B/Hong Kong/514/2009     ---------- ---------- ---------- ---------- -----v---- ---------- ---------- ---------- ---------- ----------
B/Shanghai-Pudongxin/1465/2009     ---------- ---------- ---------- ---------- -----v---- ---------- ---------- ---------- ---------- ----------
           B/Niigata/1702/2009     ---------- ---------- ---------- ---------- -----v---- ---------- ---------- ---------- ---------- ----------
        B/Madagascar/7135/2009     ---------- ---------- ---------- ---------- -----v---- ---------- ---------- ---------- ---------- ----------
        B/Madagascar/7279/2009     ---------- ---------- ---------- ---------- -----v---- ---------- ---------- ---------- ---------- ----------
            B/Sapporo/49/2009      ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
             B/Haiti/583/2009      ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
          B/Incheon/4293/2009      ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
            B/Okinawa/10/2009      ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
          B/Norway/1064/2009       ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
          B/Serbia/1232/2009       ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
             B/Athens/84/2009      ---------- ---------- ---------- ---------- ---------- ---------- ----n----- ---------- ---------- ----------
          B/Moldova/168/2009       ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
          B/Brisbane/60/2008       ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
             B/Oregon/02/2009      ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ------d---
     B/Massachusetts/03/2009       ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
             B/Nevada/05/2009      ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
          B/England/393/2008       ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ------s---
          B/Stockholm/8/2009       ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         B/Belgium/G086/2009       ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         B/Nigeria/0958/2009       ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
                   Consensus       RTKIRQLPNL LRGYEHIRLS THNVINAENA PGGPYKIGTS GSCPNITNGN GFFATMAWAV PKNDKNKTAT NPLTIEVPYI CTEGEDQITV WGFHSDNETQ
```

Figure 24: HA1 protein alignment for B-Victoria lineage viruses (continued)

```
                              201                     225                                                                                  300
          B/Malaysia/2506/2004 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
             B/Tanger/522/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- --------k- ---------- ----------
          B/Barcelona/98/2008 ---------- ---------- ---------- ---------- ---------- ---------- ---------- --------k- ---------- ----------
           B/Stockholm/1/2010 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       B/Bangladesh/5945/2009 --------r- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       B/Bangladesh/7055/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
           B/Uganda/0514/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
          B/Ghana/FS1282/2009 ---------- ---------- ---------- ---------- ---------- ------r--- ---------- ---------- ---------- ----------
B/Shanghai-Fengxian/1255/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         B/Hong Kong/514/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- --------k- ---------- ----------
B/Shanghai-Pudongxin/1465/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
          B/Niigata/1702/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       B/Madagascar/7135/2009 -v-------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       B/Madagascar/7279/2009 -v-------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
            B/Sapporo/49/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
             B/Haiti/583/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
          B/Incheon/4293/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
            B/Okinawa/10/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
           B/Norway/1064/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
           B/Serbia/1232/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
             B/Athens/84/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
           B/Moldova/168/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
           B/Brisbane/60/2008 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
             B/Oregon/02/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
     B/Massachusetts/03/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
             B/Nevada/05/2009 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
           B/England/393/2008 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
           B/Stockholm/8/2009 ---------- ---------- ----i----- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         B/Belgium/G086/2009 ---------- ---------- ----i----- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         B/Nigeria/0958/2009 ---------- ---------- ----i----- ---------- ---------- ---------- ---------- ---------- ---------- ----------
                    Consensus MAKLYGDSKP QKFTSSANGV TTHYVSQIGG FPNQTEDGGL PQSGRIVVDY MVQKSGKTGT ITYQRGILLP QKVWCASGRS KVIKGSLPLI GEADCLHEKY
```

```
                              301                                          347
          B/Malaysia/2506/2004 ---------- ---------- ---------- ---------- -------
             B/Tanger/522/2009 ---------- ---------- ---------- ---------- -------
          B/Barcelona/98/2008 ---------- ---------- ----~~~~~~ ~~~~~~~~~~ ~~~~~~~
           B/Stockholm/1/2010 ---------- ---------- ---------- ---------- -------
       B/Bangladesh/5945/2009 ---------- ---------- ---------- ---------- -------
       B/Bangladesh/7055/2009 ---------- ---------- ---------- ---------- -------
           B/Uganda/0514/2009 ---------- ---------- ---------- ---------- -------
          B/Ghana/FS1282/2009 ---------- ---------- ---------- ---------a -------
B/Shanghai-Fengxian/1255/2009 ---------- ---------- ---------- ---------- -------
         B/Hong Kong/514/2009 ---------- ---------- ---------- ---------- -------
B/Shanghai-Pudongxin/1465/2009 ---------- ---------- ---------- ---------- -------
          B/Niigata/1702/2009 ---------- ---------- ---------- ---------- -------
       B/Madagascar/7135/2009 ---------- ---------- ---------- ---------- ------~
       B/Madagascar/7279/2009 ---------- ---------- ---------- ---------- -------
            B/Sapporo/49/2009 ---------- ---------- ---------- ---------- -------
             B/Haiti/583/2009 ---------- ---------- ---------- ---------- -------
          B/Incheon/4293/2009 ---------- ---------- ---------- ---------- -------
            B/Okinawa/10/2009 ---------- ---------- ---------- ---------- -------
           B/Norway/1064/2009 ---------- ---------- ---------- ---------- -------
           B/Serbia/1232/2009 ---------- ---------- ---------- ---------- -------
             B/Athens/84/2009 ---------- ---------- ---------- ---------~ ~~~~~~~
           B/Moldova/168/2009 ---------- ---------- ---------- ---------- -------
           B/Brisbane/60/2008 ---------- ---------- ---------- ---------- -------
             B/Oregon/02/2009 ---------- ---------- ---------- ---------- -------
     B/Massachusetts/03/2009 ---------- ---------- ---------- ---------- -------
             B/Nevada/05/2009 ---------- ---------- ---------- ---------- -------
           B/England/393/2008 ---------- ---------- ---------- ---------- -------
           B/Stockholm/8/2009 ---------- ---------- ---------- ---------- -------
         B/Belgium/G086/2009 ---------- ---------- ---------- ---------- -------
         B/Nigeria/0958/2009 ---------- ---------- ---------- ---------- -------
                    Consensus GGLNKSKPYY TGEHAKAIGN CPIWVKTPLK LANGTKYRPP AKLLKER
```

N-linked glycosylation sites (NXS/T) are highlighted. Losses of such sites are underlined.

Figure 25. Phylogenetic comparison of influenza B NA genes

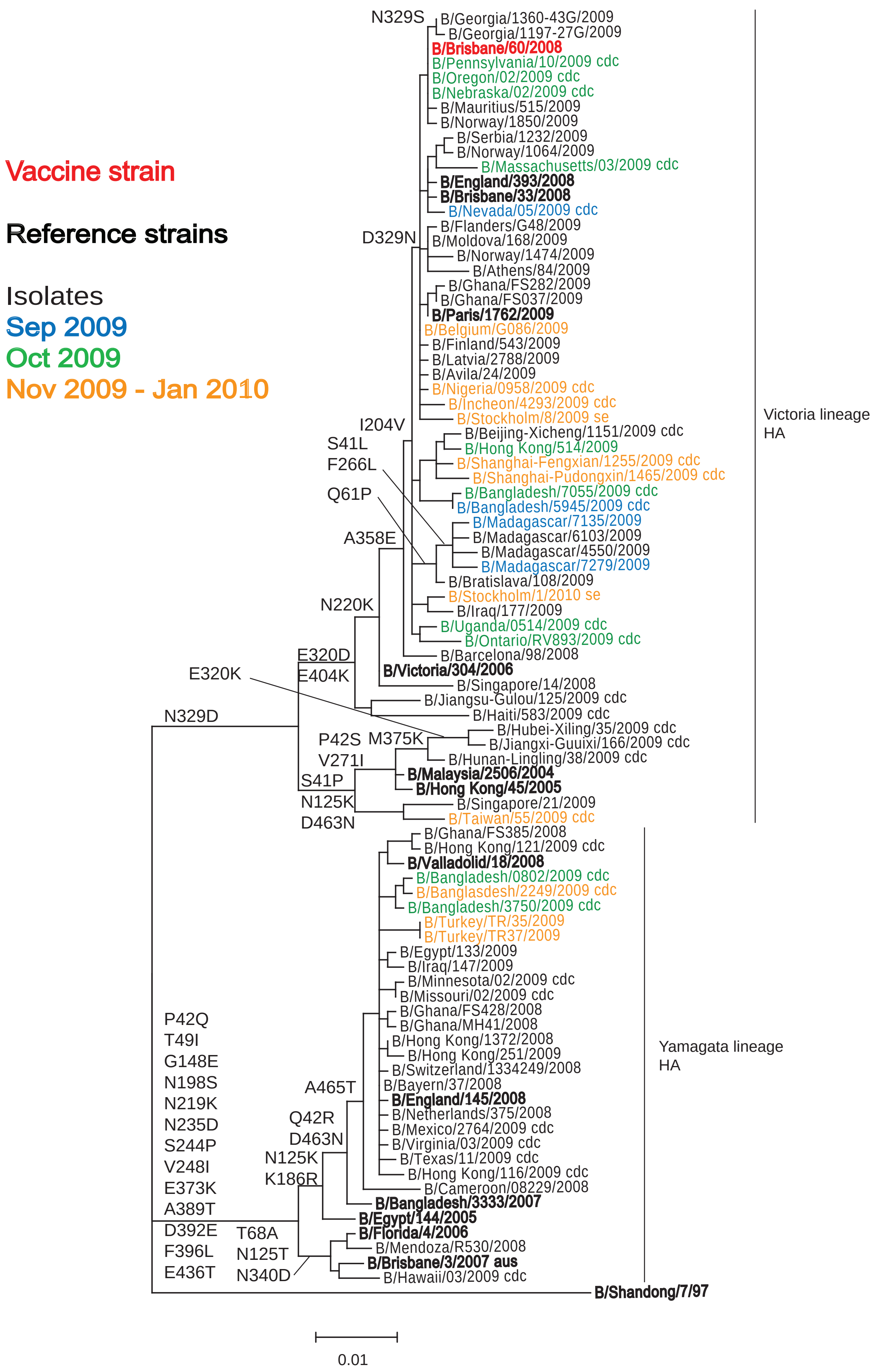

Figure 26: NA protein alignment for B viruses

```
                                1                                        42                       68                              100
          B/Malaysia/2506/2004  ---------- ---------- ---------- ---------- ps------t- ---------- ---------- ---------- ---------- ----------
           B/Brisbane/60/2008  ---------- ---------- ---------- ---------- --------t- ---------- ---------- ---------- ---------- ----------
           B/Barcelona/98/2008  ---------- ---------- ---------- ---------- --------t- ---------- h--------- ---------- ---------- ----------
B/Shanghai-Pudongxin/1465/2009  ---------- ---------- ---------- ---------- --------t- s--------- ---------- --f------- ---------- ----------
 B/Shanghai-Fengxian/1255/2009  ---------- ---------- ---------- ---------- --------t- s--------- ---------- --f------- ---------- ----------
          B/Hong Kong/514/2009  ---------- ---------- ---------- ---------- --------t- s--------- ---------- --s------- ---------- ----------
            B/Stockholm/8/2009  ---------- ---------- ---------- ---------- --------t- ---------- ---------- ---------- ---------- ----------
           B/Incheon/4293/2009  ---------- ---------- ---------- ---------- --------t- ---------- ---------- ---------- ---------- ----------
           B/Nigeria/0958/2009  ---------- ---------- ---------- ---------- --------t- ---------- ---------- ---------- ---------- ----------
           B/Belgium/G086/2009  ---------- ---------- ---------- ---------- --------t- ---------- ---------- ---------- ---------- ----------
            B/Ghana/FS282/2009  ---------- ---------- ---------- ---------- --------tt ---------- ---------- ---------- ---------- i---------
            B/England/393/2008  ---------- ---------- ---------- ---------- --------t- ---------- ---------- ---------- ---------- ----------
            B/Norway/1064/2009  ---------- ---------- ---------- ---------- --------t- ---------- ---------- ---------- ---------- ----------
            B/Serbia/1232/2009  ---------- ---------- ---------- ---------- --------t- ---------- ---------- ---------- ---------- ----------
              B/Oregon/02/2009  ---------- ---------- ---------- ---------- --------t- ---------- ---------- ---------- ---------- ----------
              B/Florida/4/2006  ---------- ---------- ---------- ---------- -q------i- ---------- -------a-- ---------- ---------- ----------
             B/Brisbane/3/2007  -------k-- ---------- ---------- ---------- -q------i- ---------- -------a-- ---------- ---------- ----------
        B/Bangladesh/3333/2007  ---------- ---------- ---------- ---------- -r--t---i- ---------- ---------- ---------- ---------- ----------
          B/Hong Kong/116/2009  ---------- ---------- ---------- -------q-- -r------i- ---------- ---------- ---------- ---------- ----------
            B/England/145/2008  ---------- ---------- ---------- ---------- -r------i- ---------- ---------- ---------- ---------- ----------
          B/Hong Kong/251/2009  ---------- ---------- ---------- ---------- -r----v-i- ---------- ---------- ---------- ---------- ----------
           B/Minnesota/02/2009  ---------- ---------- ---------- ---------- -r------i- ---------- ---------- ---------- ---------- ----------
              B/Egypt/133/2009  ~~~~------ ---------- ---------- ---------- -r------i- ---------- ---------- --------d- ---------- ----------
            B/Turkey/TR37/2009  ---------- ---------- ---------- ---------- -r------i- ---------- ---------- --f------- ---------- ----------
            B/Turkey/TR35/2009  ---------- ---------- ---------- ---------- -r------i- ---------- ---------- --f------- ---------- ----------
        B/Bangladesh/3750/2009  ---------- ---------- ---------- ---------- -r------i- ---------- --m------- ---------- ---------- ----------
        B/Bangladesh/2249/2009  ---------- ---------- ---------- ---------- -r------i- --------d- ---------- --p------- ---------- ----------
        B/Bangladesh/0802/2009  ---------- ---------- ---------- ---------- -r------i- ---------- ---------- --p------- ---------- ----------
          B/Valladolid/18/2008  ---------- ---------- ---------- ---------- -r------i- ---------- ---------- ---------- ---------- ----------
          B/Hong Kong/121/2009  ---------- ---------- ---------- ---------- -r------i- ---------- ---------- ---------- ---------- ----------
                     Consensus  MLPSTIQTLT LFLTSGGVLL SLYVSASLSY LLYSDILLKF SPTEITAP-M PLDCANASNV QAVNRSATKG VTLLLPEPEW TYPRLSCPGS TFQKALLISP

                                101                    125                                                              186          200
          B/Malaysia/2506/2004  ---------- ---------- ---------- ---------- -------g-- ---------- ---------- ---------- ---------- -------n--
            B/Brisbane/60/2008  ---------- ---------- ----n----- ---------- -------g-- ---------- ---------- ---------- ---------- -------n--
           B/Barcelona/98/2008  ---------- ---------- ----n----- ---------- -------g-- ---------- ---------- ---------- ---------- -------n--
B/Shanghai-Pudongxin/1465/2009  ---------- ---------- ----n----- ---------- -------g-- ---------- ---------- ---------- ---------- -------n--
 B/Shanghai-Fengxian/1255/2009  ---------- ---------- ----n----- ---------- -------g-- ---------- ---------- ---------- ---------- -------n--
          B/Hong Kong/514/2009  ---------- ---------- ----n----- ---------- -------g-- ---------- ---------- ---------- ---------- -------nd-
            B/Stockholm/8/2009  ---------- ---------- ----n----- ---------- -------g-- ---------- ---------- ---------- ---------- -------n--
           B/Incheon/4293/2009  ---------- ---------- ----n----- ---------- -------g-- ---------- ---------- ---------- ---------- -------n--
           B/Nigeria/0958/2009  ---------- ---------- ----d----- ---------- -------g-- ---------- ---------- ---------- ---------- -------n--
           B/Belgium/G086/2009  ---------- ---------- ----n----- ---------- -------g-- ---------- ---------- ---------- ---------- -------n--
            B/Ghana/FS282/2009  ---------- ---------- ----n----- ---------- -------g-- ---------- ---------- ---------- ---------- -------n--
            B/England/393/2008  ---------- ---------- ----n----- ---------- -------g-- ---------- ---------- ---------- ---------- -------n--
            B/Norway/1064/2009  ---------- ---------- ----n----- ---------- -------g-- ---------- ---------- ---------- ---------- -------n--
            B/Serbia/1232/2009  ---------- ---------- ----n----- ---------- -------g-- ---------- ---------- ---------- ---------- -------n--
              B/Oregon/02/2009  ---------- ---------- ----n----- ---------- -------g-- ---------- ---------- ---------- ---------- -------n--
              B/Florida/4/2006  ---------- ---------- ----t----- ---------- -------e-- ---------- ---------- ---------- ---------- -------s--
             B/Brisbane/3/2007  ---------- ---------- ----t----- ---------- -------e-- ---------- ---------- ---------- ---------- -------s--
        B/Bangladesh/3333/2007  ---------- ---------- ---------- ---------- -------e-- ---------- ---------- ---------- -----r---- -------s--
          B/Hong Kong/116/2009  ---------- ---------- ---------- ---------- -------e-- ---------- ---------- ---------- -----r---- -------s--
            B/England/145/2008  ---------- ---------- ---------- ---------- -------e-- ---------r ---------- ---------- -----r---- -------s--
          B/Hong Kong/251/2009  ---------- ---------- ---------- ---------- -------e-- ---------- ---------- ---------- -----r---- -------s--
           B/Minnesota/02/2009  ---------- ---------- ---------- ---------- -------e-- ---------- ---------- ---------- -----r---- -------s--
              B/Egypt/133/2009  ---------- ---------- ---------- ---------- -------e-- ---------- ---------- ---------- -----r---- -------s--
            B/Turkey/TR37/2009  ---------- ---------v ---------- ---------- -------e-- ---------- ---------- ---------- -----r---- -------s--
            B/Turkey/TR35/2009  ---------- ---------v ---------- ---------- -------e-- ---------- ---------- ---------- -----r---- -------s--
        B/Bangladesh/3750/2009  ---------- ---------- ---------- ---------- -------e-- ---------- ---------- ---------- -----r---- -------s--
        B/Bangladesh/2249/2009  ---------- ---------- ---------- ---------- -------e-- ---------- ---------- ---------- -----r---- -------s--
        B/Bangladesh/0802/2009  ---------- ---------- ---------- ---------- -------e-- ---------- ---------- ---------- -----r---- -------s--
          B/Valladolid/18/2008  ---------- ---------- ---------- ---------- -------e-- ---------- ---------- ---------- -----r---- -------s--
          B/Hong Kong/121/2009  ---------- ---------- ---------- ---------- -------e-- ---------- ---------- ---------- -----r---- -------s--
                     Consensus  HRFGETKGNS APLIIREPFI ACGPKECKHF ALTHYAAQPG GYYNGTR-DR NKLRHLISVK LGKIPTVENS IFHMAAWSGS ACHDGKEWTY IGVDGPD-NA

                               201 204             220                                                                            300
          B/Malaysia/2506/2004  ---------- --------n- ---------- ----n----- ---s------ ---------- ---------- i--------- ---------- ----------
            B/Brisbane/60/2008  ---v------ --------nk ---------- ----n----- ---s------ ---------- ---------- ---------- ---------- ----------
           B/Barcelona/98/2008  ---------- --------nk ---------- ----n----- ---s------ ---------- ---------- ---------- ---------v ----------
B/Shanghai-Pudongxin/1465/2009  ---v------ --------nk ---------- ----n----- ---s------ ---------- ---------- ---------- ---------- ----------
 B/Shanghai-Fengxian/1255/2009  ---v------ --------nk ---------- ----n----- ---s------ ---------- ---------- ---------- ---------- ----------
          B/Hong Kong/514/2009  ---v------ --------nk ---------- ----n----- ---s------ ---------- ---------- ---------- ---------- ----------
            B/Stockholm/8/2009  ---v------ --------nk ---------- ----n----- ---s------ ---------- ---------- ---------- ---------- ----------
           B/Incheon/4293/2009  ---v------ --------nk ---------- ----n----- ---s------ ---------- ---------- ---------- ---------- ----------
           B/Nigeria/0958/2009  ---v------ --------nk ---------- ----n----- ---s------ ---------- ---------- ---------- ---------- ----------
           B/Belgium/G086/2009  ---v------ --------nk ---------- ----n----- ---s------ ---------- ---------- ---------- ---------- ----------
            B/Ghana/FS282/2009  ---v------ --------nk ---------- ----n----- ---s------ ---------- ---------- ---------- ---------- ----------
            B/England/393/2008  ---v------ --------nk ---------- ----n----- ---s------ ---------- ---------- ---------- ---------- ----------
            B/Norway/1064/2009  ---v------ --------nk ---------- ----n----- ---s------ ---------- ---------- ---------- ---------- ----------
            B/Serbia/1232/2009  ---v------ --------nk ---------- ----n----- ---s------ ---------- ---------- ---------- ---------- ----------
              B/Oregon/02/2009  ---v------ --------nk ---------- ----n----- ---s------ ---------- ---------- ---------- ---------- ----------
              B/Florida/4/2006  ---------- --------k- ---------- ----d----- ---p---i-- ---------- ---------- ---------- ---------- ----------
             B/Brisbane/3/2007  ---------- --------k- ---------- ----d----- ---p------ ---------- ---------- ---------- ---------- ----------
        B/Bangladesh/3333/2007  ---------- --------k- ---------- ----d----- ---p---i-- ---------- ---------- ---------- ---------- ----------
          B/Hong Kong/116/2009  ---------- --------k- ---------- ----d----- ---p---i-- ---------- ---------- ---------- ---------- ----------
            B/England/145/2008  ---------- --------k- ---------- ----d----- ---p---i-- ---------- ---------- ---------- ---------- ----------
          B/Hong Kong/251/2009  ---------- --------k- ---------- ----d----- ---p---i-- ---------- ---------- ---------- ---------- ----------
           B/Minnesota/02/2009  ---------- --------k- ---------- ----d----- ---p---i-- ---------- ---------- ---------- ---------- ----------
              B/Egypt/133/2009  ---------- --------k- ---------- ----d----- ---p---i-- ---------- ---------- ---------- ---------- ----------
            B/Turkey/TR37/2009  ---------- --------k- ---------- ----d----- ---p---i-- ---------- ---------- ---------- ---------- ----------
            B/Turkey/TR35/2009  ---------- --------k- ---------- ----d----- ---p---i-- ---------- ---------- ---------- ---------- ----------
        B/Bangladesh/3750/2009  ---------- --------k- ---------- ----d----- ---p---i-- ---------- ---------- ---------- ---------- ----------
        B/Bangladesh/2249/2009  ---------- --------k- ---------- ----d----- ---p---i-- ---------- ---------- ---------- ---------- ----------
        B/Bangladesh/0802/2009  ---------- --------k- ---------- ----d----- ---p---i-- ---------- ---------- ---------- ---------- ----------
          B/Valladolid/18/2008  ---------- --------k- ---------- ----d----- ---p---i-- ---------- ---------- ---------- ---------- ----------
          B/Hong Kong/121/2009  ---------- --------k- ---------- ----d----- ---p---i-- ---------- ---------- ---------- ---------- ----------
                     Consensus  LLKIKYGEAY TDTYHSYA-N ILRTQESACN CIGG-CYLMI TDG-ASGVSE CRFLKIREGR IIKEIFPTGR VKHTEECTCG FASNKTIECA CRDNSYTAKR
```

Figure 26: NA protein alignment for B viruses (continued)

```
                               301                 320       329         340                358                                         400
B/Malaysia/2506/2004           ---------- ---------- --------d- ---------- ---------- ---------- ---------- ---------- --------a- -d---f----
B/Brisbane/60/2008             ---------- ---------d ---------- ---------- ---------- -------e-- ---------- --e------- --------a- -d---f----
B/Barcelona/98/2008            ---------- ---------d --------d- ---------- ---------- -------e-- ---------- --e------- ---n----a- -d---f----
B/Shanghai-Pudongxin/1465/2009 ---------- ---------d --------d- ---------- ---------- -------e-- ---------- --e------- --------a- -d---f----
B/Shanghai-Fengxian/1255/2009  ---------- ---------d --------d- ---------- ---------- -------e-- ---------- --e------- --------a- -d---f----
B/Hong Kong/514/2009           ---------- ---------d --------d- ---------- ---------- -------e-- ---------- --e------- --------a- -d---f----
B/Stockholm/8/2009             ---------- ---------d ---------- ---------- ---------- -------e-- ---------- --e------- --------a- -d---f----
B/Incheon/4293/2009            ---------- ---------d ---------- ---------- ---------- -------e-- ---------- --e------- --------a- -d---f----
B/Nigeria/0958/2009            ---------- ---------d ---------- ---------- ---------- -------e-- ---------- --e------- --------a- -d---f----
B/Belgium/G086/2009            ---------- ---------d ---------- ---------- ---------- -------e-- ---------- --e------- --------a- -d---f----
B/Ghana/FS282/2009             ---------- ---------d --------d- ---------- ---------- -------e-- ---------- --e------- --------a- -d---f----
B/England/393/2008             ---------- ---------d ---------- ---------- ---------- -------e-- ---------- --e------- --------a- -d---f----
B/Norway/1064/2009             ---------- ---------d ---------- ---------- ---------- -------e-- ---------- --e----e-- --------a- -d---f----
B/Serbia/1232/2009             ---------- ---------d ---------- ---------- ---------- -------e-- ---------- --e----e-- --------a- -d---f----
B/Oregon/02/2009               ---------- ---------d ---------- ---------- ---------- -------e-- ---------- --e------- --------a- -d---f----
B/Florida/4/2006               ---------- ---------- ---------- ---------d ---------- ---------- ---------- ---------- --------t- -e---l----
B/Brisbane/3/2007              ---------- ---------- ---------- ---------d ---------- ---------- ---------- ---------- --------t- -e---l----
B/Bangladesh/3333/2007         ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- --------t- -e---l----
B/Hong Kong/116/2009           ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- --------t- -e---l----
B/England/145/2008             ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- --------t- -e---l----
B/Hong Kong/251/2009           ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- --------t- -e---l----
B/Minnesota/02/2009            ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- --------t- -e---l----
B/Egypt/133/2009               ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- --------t- -e---l----
B/Turkey/TR37/2009             ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- --------t- -e---l----
B/Turkey/TR35/2009             ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- --------t- -e---l----
B/Bangladesh/3750/2009         ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- --------t- -e---l----
B/Bangladesh/2249/2009         ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- --------t- -e---l----
B/Bangladesh/0802/2009         ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- --------t- -e---l----
B/Valladolid/18/2008           ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- --------t- -e---l----
B/Hong Kong/121/2009           ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- --------t- -e---l----
Consensus                      PFVKLNVETD TAEIRLMCTE TYLDTPRPND GSITGPCESN GDKGSGGIKG GFVHQRMASK IGRWYSRTMS KTKRMGMGLY VKYDGDPW-D S-ALA-SGVM
```

```
                              401 404                                                               463 465466
B/Malaysia/2506/2004           ---------- ---------- ---------- -----e---- ---------- ---------- --n---*
B/Brisbane/60/2008             ---k------ ---------- ---------- -----e---- ---------- ---------- --d---*
B/Barcelona/98/2008            ---k------ ---------- ---------- -----e---- ---------- ---------- --d---*
B/Shanghai-Pudongxin/1465/2009 ---k------ ---------- ---------- -----e---- ---------- ---------- --d---*
B/Shanghai-Fengxian/1255/2009  ---k------ ---------- ---------- -----e---- ---------- ---------- --d---*
B/Hong Kong/514/2009           ---k------ ---------- ---------- -----e---- ---------- ---------- --d---*
B/Stockholm/8/2009             ---k------ ---------- ---------- -----e---- ---------- ---------- --d---*
B/Incheon/4293/2009            ---k------ ---------- ---------- -----e---- ---------- ---------- --d---*
B/Nigeria/0958/2009            ---k------ ---------- ---------- -----e---- ---------- ---------- --d---*
B/Belgium/G086/2009            ---k------ ---------- ---------- -----e---- ---------- ---------- --d---*
B/Ghana/FS282/2009             ---k------ ---------- ---------- -----e---- ---------- ---------- --d---*
B/England/393/2008             ---k------ ---------- ---------- -----e---- ---------- ---------- --d---*
B/Norway/1064/2009             ---k------ ---------- ---------- -----e---- ---------- ---------- --n---*
B/Serbia/1232/2009             ---k------ ---------- ---------- -----e---- ---------- ---------- --n---*
B/Oregon/02/2009               ---k------ ---------- ---------- -----e---- ---------- ---------- --d---*
B/Florida/4/2006               ---------- ---------- ---------- -----t---- ---------- ---------- --d---*
B/Brisbane/3/2007              ---------- ---------- ---------- -----t---- ---------- ---------- --d---*
B/Bangladesh/3333/2007         ---------- ---------- ---------- -----t---- ---------- ---------- --d---*
B/Hong Kong/116/2009           ---------- ---------- ---------- -----t---- ---------- ---------- --n-t-*
B/England/145/2008             ---------- ---------- ---------- -----t---- ---------- ---------- --n-t-*
B/Hong Kong/251/2009           ---------- ---------- ---------- -----t---- ---------- ---------- --n-t-*
B/Minnesota/02/2009            ---------- ---------- ---------- -----t---- ---------- ---------- --n-t-*
B/Egypt/133/2009               ---------- ---------- ---------- -----t---- ---------- ---------- --n-t-*
B/Turkey/TR37/2009             ---------- ---------- ---------- -----t---- ---------- ---------- --n-t-*
B/Turkey/TR35/2009             ---------- ---------- ---------- -----t---- ---------- ---------- --n-t-*
B/Bangladesh/3750/2009         ---------- ---------- ---------- -----t---- ---------- ---------- --n-t-*
B/Bangladesh/2249/2009         ---------- ---------- ---------- -----t---- ---------- ---------- --n-t-*
B/Bangladesh/0802/2009         ---------- ---------- ---------- -----t---- ---------- ---------- --n-t-*
B/Valladolid/18/2008           ---------- ---------- ---------- -----t---- ---------- ---------- --n-t-*
B/Hong Kong/121/2009           ---------- ---------- ---------- -----t---- ---------- ---------- --n-t-*
Consensus                      VSMEEPGWYS FGFEIKDKKC DVPCIGIEMV HDGGK-TWHS AATAIYCLMG SGQLLWDTVT GV-MAL-
```

N-linked glycosylation sites (NXS/T) are highlighted.

Table 16. Antigenic analyses of influenza B viruses (Yamagata lineage)

Viruses	Collection date	Passage History	Haemagglutination inhibition titre[1]						
				Post infection ferret sera					
			B/Fl[2] 4/06 Sh 478	B/Eg 144/05 F3/07	B/Fl 4/06 F21/07	B/Bris 3/07 F24/07	B/Eng 145/08 F9/08	B/Vall 18/08 F10/08	B/Ban 3333/07 F21/08
REFERENCE VIRUSES									
B/Egypt/144/2005	01/05/2005	E2 E6	1280	80	320	320	40	10	80
B/Florida/4/2006	15/12/2006	E2 E5	5120	80	160	320	40	<	80
B/Brisbane/3/2007	03/09/2007	E2 E3	2560	160	320	320	40	<	80
B/England/145/2008		Ex E3	160	<	<	<	40	<	10
B/Valladolid/18/2008	18/02/2008	MDCK1 SIAT3	1280	40	40	<	80	160	80
B/Bangladesh/3333/2007	19/08/2007	E4 E3	1280	40	40	80	20	10	80
TEST VIRUSES									
B/Hong Kong/542/2009	19/11/2009	MDCK2 SIAT1	2560	80	160	160	160	160	320
B/Turkey/TR-34/2009	25/12/2009	SIAT1 SIAT1	2560	80	80	40	80	160	80
B/Turkey/TR-35/2009	29/12/2009	SIAT1 SIAT1	1280	40	40	40	80	80	80
B/Turkey/TR-36/2009	30/12/2009	SIAT1 SIAT1	2560	40	40	40	80	80	80
B/Turkey/TR-37/2009	30/12/2009	SIAT1 SIAT1	1280	40	40	40	80	80	80

1. < = <10; 2. hyperimmune sheep serum

Figure 27. Phylogenetic comparison of influenza B HA genes (Yamagata lineage)

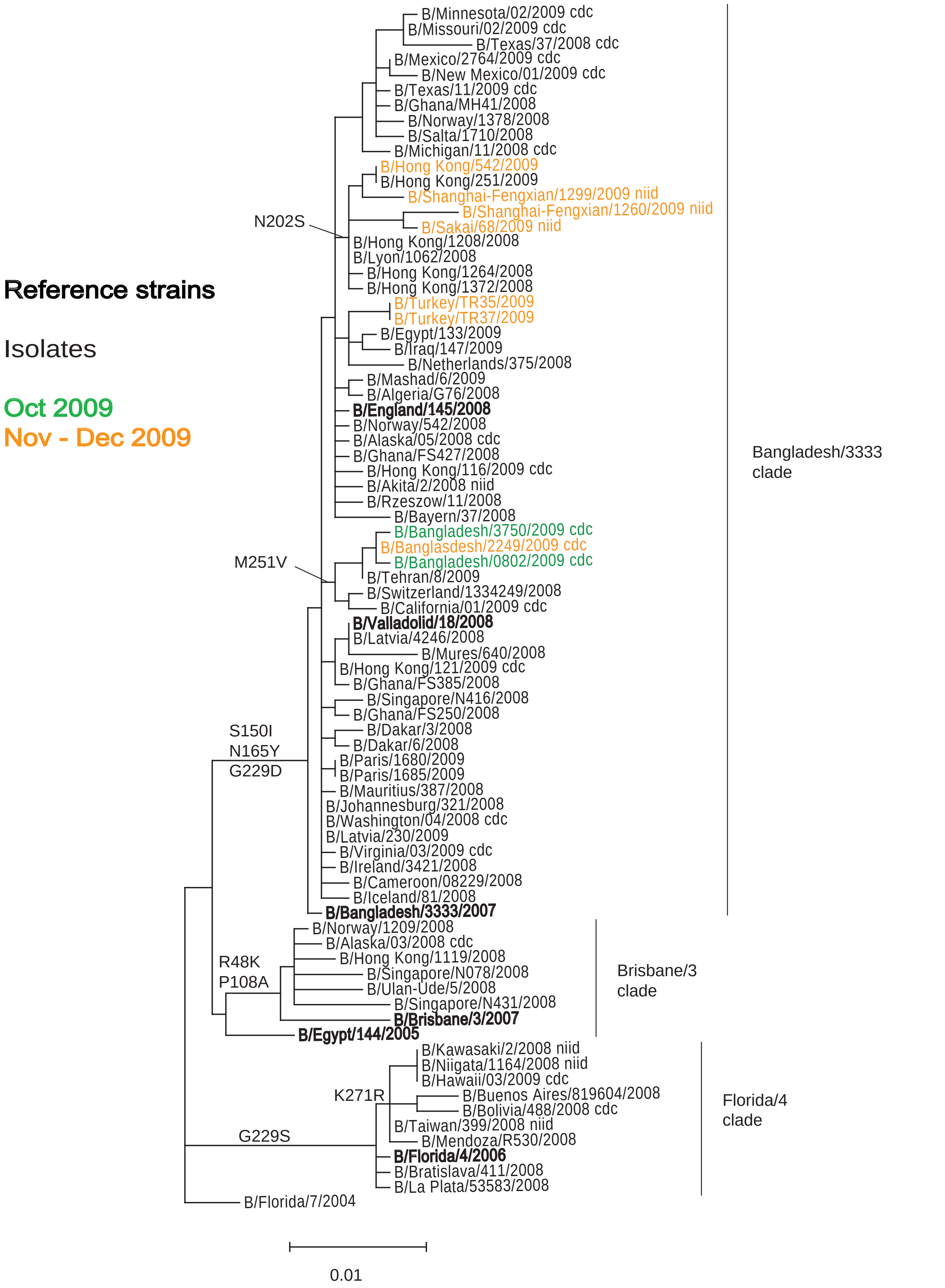

Figure 28: HA1 protein alignment for B-Yamagata lineage viruses

```
                               1                                                  48                                                      100
             B/Florida/4/2006  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- -------k-- ----------
            B/Brisbane/3/2007  ---------- ---------- ---------- ---------- -------k-- ---------- ---------- ---------- ---------- ----------
       B/Bangladesh/3333/2007  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
           B/Virginia/03/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
            B/Latvia/230/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
            B/Paris/1685/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
            B/Paris/1680/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         B/Hong Kong/121/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         B/Valladolid/18/2008  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         B/California/01/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
              B/Tehran/8/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       B/Bangladesh/0802/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       B/Bangladesh/2249/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       B/Bangladesh/3750/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         B/Hong Kong/116/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
           B/England/145/2008  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
              B/Mashad/6/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
              B/Iraq/147/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
             B/Egypt/133/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
           B/Turkey/TR37/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ------i--- ----------
           B/Turkey/TR35/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ------i--- ----------
              B/Sakai/68/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
B/Shanghai-Fengxian/1260/2009  ---------- ---------- ---------- ---------- -------k-- ---------- ---------- ---------- ---------- ----------
B/Shanghai-Fengxian/1299/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         B/Hong Kong/542/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         B/Hong Kong/251/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
              B/Texas/11/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         B/New Mexico/01/2009  ---------- ---------- ---------- ------a--- ---------- ---------- ---------- ---------- ---------- ----------
              B/Texas/37/2008  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------e ---------- ----------
          B/Minnesota/02/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
                    Consensus  DRICTGITSS NSPHVVKTAT QGEVNVTGVI PLTTTPTKSY FANLKGTRTR GKLCPDCLNC TDLDVALGRP MCVGTTPSAK ASILHEVRPV TSGCFPIMHD
```

```
                               101    108                                            150              165                                 200
             B/Florida/4/2006  ---------- ---------- ---------- ---------- ---------s ---------- ----n----- ---------- ---------- -----d----
            B/Brisbane/3/2007  -------a-- ---------- ---------- ---------- ---------s ---------- ----n----- ---------- -q-------- -----d----
       B/Bangladesh/3333/2007  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- -----x----
           B/Virginia/03/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
            B/Latvia/230/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
            B/Paris/1685/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
            B/Paris/1680/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         B/Hong Kong/121/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         B/Valladolid/18/2008  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         B/California/01/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
              B/Tehran/8/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       B/Bangladesh/0802/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       B/Bangladesh/2249/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       B/Bangladesh/3750/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         B/Hong Kong/116/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- -------n--
           B/England/145/2008  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- -------a--
              B/Mashad/6/2009  ---------- ---------- ---------- ---------- -----t---- ---------- ---------- ---------- ---------- ----------
              B/Iraq/147/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
             B/Egypt/133/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- -------i--
           B/Turkey/TR37/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
           B/Turkey/TR35/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
              B/Sakai/68/2009  ---------- ---------- ---------- ---------- -----s---- ---------- ---------- ---------- ---------- ----------
B/Shanghai-Fengxian/1260/2009  ---------- ---------- ---------- ---------- -----s---- ---------- ---------- ---------- ---------- ----------
B/Shanghai-Fengxian/1299/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         B/Hong Kong/542/2009  ---------- ----x----- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         B/Hong Kong/251/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
              B/Texas/11/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         B/New Mexico/01/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
              B/Texas/37/2008  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
          B/Minnesota/02/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
                    Consensus  RTKIRQLPNL LRGYENIRLS TQNVIDAEKA PGGPYRLGTS GSCPNATSKI GFFATMAWAV PKDNYKNATN PLTVEVPYIC TEGEDQITVW GFHSDNKTQM
```

Figure 28: HA1 protein alignment for B-Yamagata lineage viruses (continued)

```
                              201202                          229                       251                                                    300
           B/Florida/4/2006  ---------- ---------- --------s- ---------- ---------- ---------- ---------- ---------- ---------- ----------
           B/Brisbane/3/2007  ---------- ---------- --------g- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      B/Bangladesh/3333/2007  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         B/Virginia/03/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
           B/Latvia/230/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
           B/Paris/1685/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
           B/Paris/1680/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        B/Hong Kong/121/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        B/Valladolid/18/2008  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        B/California/01/2009  ---------- ---------- ---------- ---------- ---------- v--------- ---------- ---------- ---------- ----------
             B/Tehran/8/2009  ---------- ---------- ---------- ---------- ---------- v--------- ---------- ---------- ---------- ----------
      B/Bangladesh/0802/2009  ---------- ---------- ---------- ---------- ---------- v--------- ---------- ---------- ----l----- ----------
      B/Bangladesh/2249/2009  ---------- ---------- ---------- ---------- ---------- v--------- ---------- ---------- ---------- ----------
      B/Bangladesh/3750/2009  ---------- ---------- ---------- ---------- ---------- v--------- ---------- ---------- ---------- ----------
        B/Hong Kong/116/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         B/England/145/2008  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
             B/Mashad/6/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
            B/Iraq/147/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
           B/Egypt/133/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         B/Turkey/TR37/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
         B/Turkey/TR35/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
            B/Sakai/68/2009  -s-------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
B/Shanghai-Fengxian/1260/2009  -s-------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
B/Shanghai-Fengxian/1299/2009  -s-------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        B/Hong Kong/542/2009  -s-------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        B/Hong Kong/251/2009  -s-------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
            B/Texas/11/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
       B/New Mexico/01/2009  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
            B/Texas/37/2008  ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
        B/Minnesota/02/2009  ---------- ---------- ---------- ---------- ---------- ---------- -----i---- ---------- ---------- ----------
                  Consensus  KNLYGDSNPQ KFTSSANGVT THYVSQIGDF PDQTEDGGLP QSGRIVVDYM MQKPGKTGTI VYQRGVLLPQ KVWCASGRSK VIKGSLPLIG EADCLHEKYG

                              301                                            346
           B/Florida/4/2006  ---------- ---------- ---------- ---------- ------
           B/Brisbane/3/2007  ---------- ---------- ---------- pmt------- ------
      B/Bangladesh/3333/2007  ---------- ---------- ---------- ---------- ------
         B/Virginia/03/2009  ---------- ---------- ---------- ---------- ------
           B/Latvia/230/2009  ---------- ---------- ---------- ---------- ------
           B/Paris/1685/2009  ---------- ---------- ---------- ---------- ------
           B/Paris/1680/2009  ---------- ---------- ---------- ---------- ------
        B/Hong Kong/121/2009  ---------- ---------- ---------- ---------- ------
        B/Valladolid/18/2008  ---------- ---------- ---------- ---------- ------
        B/California/01/2009  ---------- ---------- ---------- ---------- ------
             B/Tehran/8/2009  ---------- ---------- ---------- ---~~~~~~~ ~~~~~~
      B/Bangladesh/0802/2009  ---------- ---------- ---------- ---------- ------
      B/Bangladesh/2249/2009  ---------- ---------- ---------- ---------- ------
      B/Bangladesh/3750/2009  ---------- ---------- ---------- ---------- ------
        B/Hong Kong/116/2009  ---------- ---------- ---------- ---------- ------
         B/England/145/2008  ---------- ---------- ---------- ---------- ------
             B/Mashad/6/2009  ---------- ---------- ---------- ---------~ ~~~~~~
            B/Iraq/147/2009  ---------- ---------- ---------- ---------- ------
           B/Egypt/133/2009  ---------- ---------- ---------- ---------- ------
         B/Turkey/TR37/2009  ---------- ---------- ---------- ---------- ------
         B/Turkey/TR35/2009  ---------- ---------- ---------- ---------- ------
            B/Sakai/68/2009  ---------- ---------- ---------- ---------- ------
B/Shanghai-Fengxian/1260/2009  ---------- ---------- ---------- ---------- ------
B/Shanghai-Fengxian/1299/2009  ---------- ---------- ---------- ---------- ------
        B/Hong Kong/542/2009  ---------- ---------- ---------- ---------- ------
        B/Hong Kong/251/2009  ---------- ---------- ---------- ---------- ------
            B/Texas/11/2009  ---------- ---------- ---------- ---------- ------
       B/New Mexico/01/2009  ---------- ---------- ---------- ---------- ------
            B/Texas/37/2008  ---------- ---------- ---------- ---------- ------
        B/Minnesota/02/2009  ---------- ---------- ---------- ---------- ------
                  Consensus  GLNKSKPYYT GEHAKAIGNC PIWVKTPLKL ANGTKYRPPA KLLKER
```

N-linked glycosylation sites (NXS/T) are highlighted. Losses of such sites are <u>underlined</u>.

Table 17. Neuraminidase inhibitor antiviral testing from September 2009 to present

Influenza type/subtype	Number tested*	Number (%) resistant to oseltamivir		Number (%) resistant to zanamivir	
A(H1N1)pdm	635	2	(0.32%)	0	(0%)
A(H1N1)	39	39	(100%)	0	(0%)
A(H3N2)	65	0	(0%)	0	(0%)
B	7	0	(0%)	0	(0%)

*This includes viruses recovered from specimens collected before September 2009